Explaining without predicting: anomaly detection and precedent retrieval for verifiable financial alerts

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I further declare that I have fully acknowledged the Code of Ethical Conduct of P.PORTO.

ISEP, Porto, September 2026

Dedicatory

To my family.

Abstract

When a stock price moves abruptly, retail investors need to determine whether the movement is unusual, whether its origin lies with the company or the market, and whether comparable precedents exist. Free applications display the percentage change, and the few that go further do not expose the evidence behind them; professional terminals do answer them, and are not accessible to this investor profile.

This dissertation presents InvestiGator, a financial alerting system built exclusively on free data sources, which answers the three questions and exposes the evidence supporting each answer. The system integrates four techniques: anomaly detection through a rolling z-score, decomposition of returns into market, sector and company components with beta shrinkage, precedent retrieval by semantic similarity with measurement of the observed outcome, and a supervised triage model trained on a purpose-built dataset.

Three techniques were compared with transparent baselines. Volatility-normalised detection proves consistent across companies; semantic retrieval outperforms the base rate in every sector assessed; the triage model does not outperform a volatility-only baseline, a negative result that is reported and analysed.

By design, the system does not produce price forecasts.

Keywords: Explainable Artificial Intelligence, Financial Alerts, Anomaly Detection, Semantic Retrieval, Retail Investor

Resumo

Quando o preço de uma ação se move de forma abrupta, o investidor particular necessita de determinar se o movimento é invulgar, se a sua origem é a empresa ou o mercado, e se existem precedentes. As aplicações gratuitas apresentam a variação percentual, e as poucas que vão além disso não expõem a evidência que as sustenta; os terminais profissionais respondem, e não estão acessíveis a este investidor.

Esta dissertação apresenta o InvestiGator, um sistema de alertas financeiros assente exclusivamente em fontes de dados gratuitas, que responde às três questões e expõe a evidência de cada resposta. O sistema integra quatro técnicas: deteção de anomalias por z-score deslizante, decomposição do retorno em mercado, setor e empresa com encolhimento do beta, recuperação de precedentes por semelhança semântica com medição do desfecho observado, e um modelo supervisionado de triagem treinado sobre um conjunto de dados próprio.

Três técnicas foram comparadas com linhas de base transparentes. A deteção normalizada pela volatilidade revela-se consistente entre empresas; a recuperação semântica supera a taxa-base nos cinco setores avaliados; o modelo de triagem não supera uma linha de base apenas com volatilidade, resultado negativo reportado e analisado.

Por desenho, o sistema não produz previsões de preço.

Palavras-chave: Inteligência Artificial Explicável, Alertas Financeiros, Deteção de Anomalias, Recuperação Semântica, Investidor Particular

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To Prof. Luís Gomes, supervisor, and to Rafael Silva, co-supervisor, for the conversations that set the direction of this work, for the examples that showed me what already exists in the field, and for the guidance on writing the document. Some of the decisions described here were taken after one of those conversations.

To my family, for the support over these months.

To my colleagues at Sistrade, and to Sistrade, for the flexibility that made it possible to reconcile this master's degree with work, and for the conversations about the application while it was being built.

To those who follow the channel and rate the alerts they receive. That feedback is the only information in this work that did not come from an automatic measurement.

Contents

List of Figures

List of Tables

List of Listings

List of Symbols

P_t	Closing price of session t	Eq. 3.1
r_t	Logarithmic return of day t, $\ln(P_t/P_{t-1})$	Eq. 3.1
μ, σ	Mean and standard deviation of the returns in the window preceding the day	Eq. 3.1
z_t	Rarity of the return of day t against the preceding window	Eq. 3.1
r_m, $\tilde{r}_s$	Daily market return and sector return orthogonalised against the market	Eq. 3.2
β_m, β_s	Historical sensitivity of the stock to the market and to the sector	Eq. 3.2
ε	Part of the return explained neither by the market nor by the sector	Eq. 3.2
$\hat{\beta}$	Beta estimate obtained by regression over the window	Eq. 3.3
β_{ref}	Reference beta towards which the estimate is shrunk	Eq. 3.3
$\text{SE}(\hat{\beta})$	Standard error of the beta estimate	Eq. 3.3
σ^2_{ref}	Reference variance that sets the strength of the shrinkage	Eq. 3.3
w	Weight of the Vasicek shrinkage, between zero and one	Eq. 3.3
cos	Cosine similarity between two sentence vectors	Section 3.6
h	Half-life of the recency decay applied to precedents	Section 3.6
k	Number of precedents retrieved and presented	Section 3.6
p_{bruta}	Triage score before calibration	Eq. 3.4
$p_{\text{calibrada}}$	Triage score after Platt calibration	Eq. 3.4
a, b	Learned parameters of the Platt calibration	Eq. 3.4
P, C	Precision and recall	Section 5.1
F_1	Harmonic mean of precision and recall, $2PC/(P+C)$	Section 5.1

List of Acronyms

AI Artificial Intelligence.

ARCH Autoregressive Conditional Heteroscedasticity.

BERT Bidirectional Encoder Representations from Transformers.

FNSPID Financial News and Stock Price Integration Dataset.

GARCH Generalized Autoregressive Conditional Heteroscedasticity.

HTTP HyperText Transfer Protocol.

LIME Local Interpretable Model-agnostic Explanations.

ONNX Open Neural Network Exchange.

PR-AUC Precision-Recall Area Under the Curve.

PSI Population Stability Index.

ROC-AUC Receiver Operating Characteristic Area Under the Curve.

RQ Research Question.

SBERT Sentence-BERT.

SHAP SHapley Additive exPlanations.

How to read this thesis

This document describes a system that answers three questions a retail investor asks, and measures what each of those answers is worth. It runs to about a hundred pages of body text, and not every reader is after the same thing. This page proposes four paths.

If you want to know what the work concludes read Chapter 1, which states the three questions, and then Section 6.2, where each one receives its answer with the reach delimited. That is about ten pages, and Figure 6.3 summarises the path of each question in a single image.

If you want to know how it was measured read Chapter 3, which describes the four techniques and the temporal discipline that stops each one from using information that did not yet exist, and then Section 5.1, which fixes the metrics and, what matters most, the value each one takes in the absence of information.

If you want to know how the system works read Chapter 4. The path of a news item is in Section 4.3, the policy that decides silence in Section 4.4, and the layer that composes the delivered text in Section 4.5.

If you want to verify go to Appendix A. Table A.1 links every reported result to the section that produces it, and Table A.2 lists every substantive claim with the state it was left in after being measured, including those that were withdrawn.

Three conventions run through the document and are worth stating before you start. No number was written by hand: each one is produced by an automatic procedure and the text cites the file that contains it. No comparison is presented without the value a method with no information would obtain, because a percentage read on its own leads to the wrong conclusion, and the document records three occasions on which that nearly happened. And every result comes with what does *not* follow from it, which is the step that distinguishes an evaluation from a demonstration.

Chapter 1

Introduction

This chapter presents the context of the work described in this dissertation, the problem that motivates it, the research questions and the objectives defined, the scientific contributions and the organisation of the document. Whoever reads it will know what problem the work solves, for whom, and what the system refuses to do by decision rather than by limitation.

1.1 Context

Buying shares has become a low-friction operation, available through mobile applications and without significant brokerage costs. Most adults in the United States now hold shares, directly or through funds and retirement plans (Gallup 2025), in a market whose capitalisation exceeds 62 trillion US dollars (SIFMA 2025), as shown in Figure 1.2.

Ease of access has not extended, however, to the interpretation of information. When the price of an asset moves, the retail investor has limited instruments for determining what that movement means.

The behaviour of this segment is the object of current research, and two recent results delimit the problem addressed in this dissertation. The first is that retail investor attention is today a measured quantity, whose indicators are discussed and compared in the literature (Cahill, Zhangxin Liu, and Smales 2025): what determines what the investor observes is studied as a variable, and not as chance. The second is that retail participation in the market, and the conditions under which it trades, remain the subject of active review (Ernst and Spatt 2026). Chapter 2 develops the behavioural mechanisms that this literature documents, and which explain why the absence of context is more costly to this audience than to a professional.

What matters to retain from this body of work is what delimits the problem. The signals that capture the investor's attention arrive without context, and the decision that follows is taken under conditions the literature documents as unfavourable. Three concrete questions remain unanswered at the moment the investor observes a price movement:

1. **is the movement unusual?** A four per cent change is a rare event in a low volatility company and an ordinary day in a volatile one;
2. **does the movement originate in the company or in the market?** A fall that follows the market and an isolated fall are distinct situations;
3. **are there precedents, and what was their outcome?** The existence of comparable days in the past, and of what followed them, allows the observed movement to be situated.

Current free applications display the percentage change and answer none of the three. There are recent products that claim to go further, and Section 2.2 names them: what their own pages describe is a summary in natural language, and not individual past cases with the outcome measured. Professional financial information terminals do answer all three, and are not available free of charge or at a cost accessible to the investor profile in question. The gap does not follow from the unavailability of data, which exist in quantity and are free, but from the absence of verifiable context built on them.

Figure 1.1 shows the difference on a concrete event.

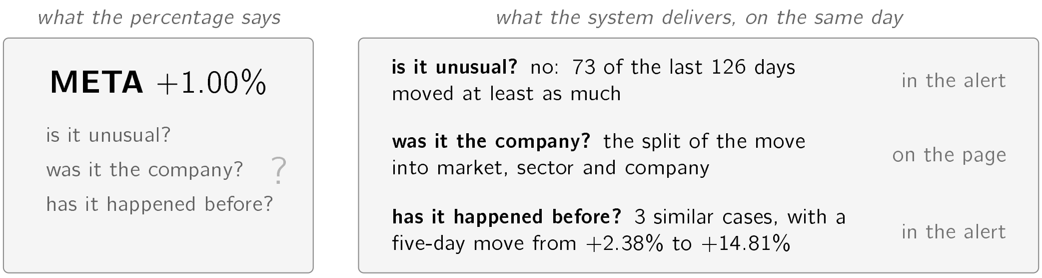

Figure 1.1: The same day of the same company, on the left as a price quote presents it and on the right as the system delivers it. The values on the right are reproduced from the alert sent to the channel on 8 September 2026, and were not constructed for the purpose. The third column indicates where each answer lives, and the distinction is not incidental: the alert answers the first and the third question at the moment it interrupts the reader, whereas the split depends on the day's close and is therefore presented in the application, as Figure 4.8 shows.

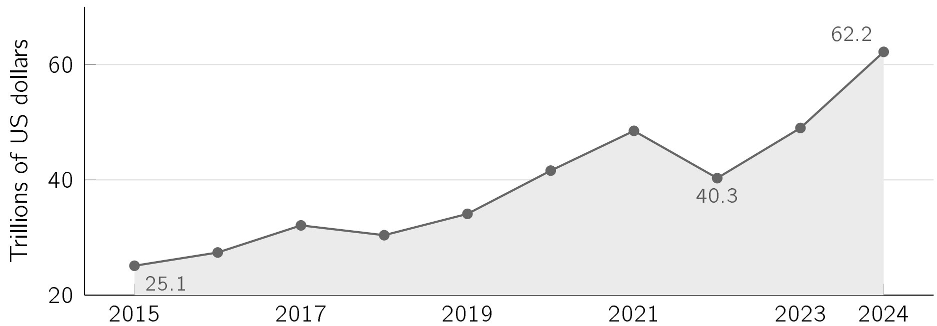

Figure 1.2: Capitalisation of the United States stock market, 2015–2024, in trillions of US dollars. Compiled from the data of the SIFMA Capital Markets Fact Book (SIFMA 2025).

1.2 Problem description

This dissertation presents InvestiGator, a financial alerting system built exclusively on free data sources, which monitors a set of companies, detects events that depart from the usual behaviour of each one, and issues alerts accompanied by the explanation and the evidence that support them.

The system has two triggers, shown in Figure 1.3: a price movement that is unusual for the company in question, or the arrival of a relevant news item about it. In both cases, the message delivered identifies the event, the criterion by which it was considered unusual, the provenance of the data used, and the semantically closest past cases, with the observed behaviour of the price after each of them.

One design decision runs through the system and conditions all the others: **the system does not produce price forecasts**. It does not indicate future direction, does not issue buy or sell recommendations, and does not present price targets. This restriction has two justifications. The first is verifiability: a statement about a past event can be confirmed against the record at the moment it is read, whereas a forecast can only be assessed after the investor's decision has already been taken. The second is theoretical: the efficient market hypothesis, in its semi-strong form, makes the systematic anticipation of movements from publicly available information improbable. Chapter 2 develops it and identifies the corresponding literature. This dissertation does not test that hypothesis, nor does it rely on it to establish any conclusion.

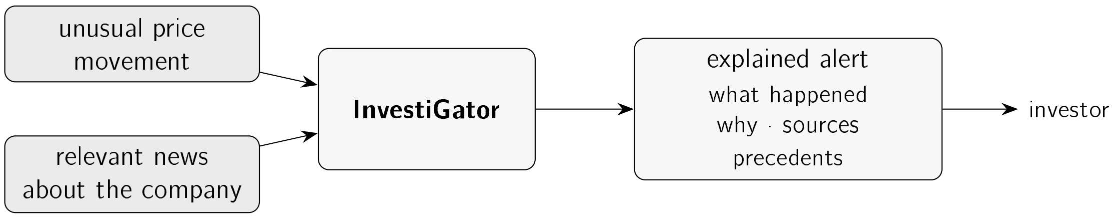

Figure 1.3: The two triggers of the system and the output they produce. The investor receives an alert accompanied by the evidence that allows it to be verified, and not merely the notification that the price has moved.

1.3 Objectives and research questions

The general objective of this dissertation is to design, build and evaluate an explainable financial alerting system for retail investors, using exclusively free data sources.

Four research questions follow from this objective, each associated with a measurement and a success criterion defined before the experiments were carried out:

- **RQ1, detection.** Can a simple and transparent statistical method detect unusual movements consistently across companies with distinct volatilities, and do so better than a fixed threshold rule?
- **RQ2, precedents.** Can the system retrieve news from another company in the same sector better than the evaluated textual and ranking alternatives, and retain an advantage over chance when only news prior to the query is allowed?
- **RQ3, triage.** Can a trained model decide which news items justify an alert better than a baseline built only on volatility?
- **RQ4, comparability.** Can contrastive fine-tuning of the encoder, trained to bring together news items whose subsequent movement has similar *magnitude*, improve the materiality comparability of the precedents retrieved without a cost in thematic relevance?

The fourth question calls for a justification, because it touches the founding constraint of this work. The system does not predict, and training an encoder on observed outcomes might appear to contradict that constraint. The decisive distinction is between *direction* and *magnitude*: what the constraint forbids is asserting where the price will go, and the training objective of this question is the magnitude of the movement, taken in absolute value. What an encoder adjusted in this way produces is not an assertion about the news item under analysis, but an organisation of the case base that brings together news items of comparable materiality. So that the distinction is verifiable and not merely declared, the experiment includes a second arm, trained on the *signed* return: were direction as learnable as magnitude, the two arms would be indistinguishable and the argument would fall.

The investor's second question, concerning the separation between the market component and the company-specific component, is addressed by a technique described in Chapter 3 but does not give rise to a research question. The reason is methodological: the split of a movement has no observable reference value against which it can be compared, since that quantity exists only relative to a model. Section 5.6 establishes what, in spite of that, remains measurable in the technique.

These questions take the form of four objectives:

1. to build the complete system, from data acquisition to the delivery of the alert;
2. to evaluate the components, with transparent baselines for detection, retrieval and triage;
3. to ensure that the explanations presented are faithful to what the system computed;
4. to report the results obtained, including those that do not confirm the initial expectations.

The techniques these objectives call for come from different families, and the question of where machine learning sits in the work has a distinct answer for each of the five decisions the system takes before an alert reaches anyone. One of them is a rule with no estimated parameters. One uses no parameters beyond the mean and the deviation of the preceding window. One has its parameters estimated by regression, with shrinkage. One relies on a model pre-trained by others and used without adjustment. Only the last has parameters estimated in this work, on the historical set. The label of artificial intelligence conflates these situations, and Section 3.3 separates them decision by decision, together with what learning each one from data would require.

1.4 Scientific contributions

This being a dissertation in Artificial Intelligence (AI) Engineering, the contribution does not lie in the formulation of a new algorithm, but in the selection, integration and critical evaluation of established techniques until they constitute a working, explainable and reproducible system. Five contributions are identified:

1. **A complete system in operation**, with both triggers, built on free sources and actually deployed, and not a laboratory demonstration.
2. **A reproducible method for measuring the behaviour of the price following semantically close news items**, combining retrieval by sentence similarity with event study windows, without using information posterior to the moment of the decision. Chapter 2 identifies the two lines of work from which they come.

3. **A purpose-built dataset and a model trained on it**, with 79,753 examples constructed from the public Financial News and Stock Price Integration Dataset (FNSPID) corpus (Dong, Fan, and Peng 2024) and from price series, with the model, the calibration and the metrics kept as versioned artefacts.
4. **An evaluation of the components, with baselines for three of the four techniques**, including the result that contradicts the initial hypothesis, reported as such.
5. **A negative result on learning materiality from text**, obtained with a protocol fixed before execution and with a diagnostic that rejects degenerate representations before any metric is interpreted. The first attempt produced exactly that degeneration, and the diagnostic caught it; Chapter 5 reports both.

1.5 Structure of the document

Chapter 2 presents the state of the art, characterising the target audience, the tools currently available and the technical areas from which the components used are drawn. Chapter 3 describes the data and the four techniques that constitute the system. Chapter 4 describes the architecture of InvestiGator and the path of a news item from collection to delivery. Chapter 5 presents the experimental design and the results obtained for each research question. Chapter 6 synthesises the answers, identifies the limitations of the work and sets out the lines of future development.

Each mechanism described is illustrated on a real case, with the values actually computed by the system, and not on examples constructed for the purpose. One of the twelve companies, AMD, returns in three distinct roles across the document, and serves as a thread for anyone who wants to follow a single name: it is the movement that Section 3.5 splits into its three components, it is the company the decision gate let through in every logged assessment, and it is one of those the model scores without ever having observed them in training. These are three moments of the same company, not of the same event.

Chapter 2

State of the art

This chapter presents the scientific and technological framing of the work. It first characterises the audience the system addresses and the limitations that define the problem; it then examines the solutions currently available; and it finally surveys the technical areas from which the components used are drawn, situating the contribution of this dissertation in relation to each of them. Whoever reads it will know that the retail investor's three questions already have answers in the literature, and why no free tool brings them together.

2.1 The retail investor and the attention problem

This section characterises the documented behaviour of the audience the system addresses, since it is the limitations of that behaviour, and not the size of the market, that delimit the problem treated here.

Chapter 1 established the scale of the universe in question. The characterisation that is missing is behavioural, and it begins with an observation that runs counter to the current portrayal of the retail investor as an agent who only reacts in panic. During the fall of March 2020, retail investors on a commission-free platform increased their aggregate positions rather than reducing exposure (Welch 2022). This is an audience that decides regularly, and that therefore benefits from better information at the moment of the decision.

The behavioural finance literature establishes that those decisions depart systematically from the rational ideal (Barberis and Thaler 2003). Two regularities bear directly on the design of an alerting system. The first is loss aversion (Kahneman and Tversky 1979), already stated in the previous chapter, whose consequence for this work is that the cost of the absence of context is asymmetric: it is greater precisely when the information is unfavourable.

The second is the attention effect, and it is the one that grounds the problem. Barber and Odean (2008) establish that retail investors are net buyers of the stocks that capture their attention, identified by three observable indicators, namely presence in the news, abnormal trading volume and an extreme return on a single day. The mechanism the authors propose is that of the search problem: faced with a very large number of alternatives, the investor does not assess them all and the choice falls on the subset that has been made visible.

The authors delimit the reach of the claim, which concerns an aggregate imbalance between purchases and sales, without extending to a universal behaviour, and report the conclusion that justifies this work: the buying pattern so formed does not generate superior returns. The empirical basis is records from United States brokerages between 1991 and 1999, prior to trading by mobile phone and to the absence of commissions, so what is assumed transferable is the mechanism and not the measured values. The centrality of attention in this process is such that it became measurable through the volume of internet searches (Da, Engelberg, and Gao 2011).

The three indicators identified by the literature are all computed by this system, but the status of each differs. News and the extreme movement are the two triggers described in Chapter 1 and can, on their own, give rise to an alert. Volume is computed and presented as context, and does not interrupt the user. The reason for this asymmetry is empirical and is reported in Section 5.8: combining volume with the remaining signals improved triage in only one of three budgets considered, a result that does not support introducing a third route of interruption. The idea of fusing signals of distinct origins into a single indicator comes from a real-time information panel with that architecture (*World Monitor* 2026), suggested by the co-supervisor of this work, and enters the dissertation in the condition in which it was evaluated, that is, as a measured and not adopted alternative.

There is, on this point, field evidence that is more recent, closer to the object of this work, and that points in an unfavourable direction. C.-W. Liu, Chen, and Wen (2023) followed 20 904 retail investors on a mutual fund platform in Taiwan, between January 2017 and June 2020, of whom 932 activated the platform's own price alerts. Estimating the effect by difference in differences, they quantify it as a deterioration in portfolio performance of the order of one percentage point over six months. The two mechanisms they identify are an increase of about 7% in the number of transactions, and the fact that the funds the investor adjusted after adoption performed worse than those left untouched. The authors read this as the alert directing attention towards short-term volatility and inducing excessive trading.

The result is uncomfortable for a dissertation that builds an alerting system, which is why it appears in the body of the text and not in a footnote. It is worth being exact, even so, about what was measured. The alert studied fires when the price of a fund crosses a threshold set by the investor and reports the current price, with no indication of cause; it is the threshold alert that Section 2.2 encounters again in brokerage applications. The mechanism of harm the authors describe, which is feedback trading on price rather than the use of information about the value of the firm, is precisely what takes hold when the recipient is handed a number and nothing else. The authors themselves conclude that improving access to information is not enough, and that the capacity to process it must also be addressed.

This dissertation takes that result as a reason for the design rather than as validation of the design it already had. Two decisions follow from it. The first is that the alert carries the evidence that justifies it, instead of reporting only that a threshold was crossed. The second is that the number of alerts is capped by explicit decision, and not by whatever the data produce, as described in Section 4.4. Whether these two decisions suffice to avoid the effect C.-W. Liu, Chen, and Wen (2023) measured remains to be demonstrated, and Chapter 6 takes it up in those terms.

On the side of financial institutions, the adoption of artificial intelligence has ceased to be experimental, as documented by the survey of firms in the sector conducted by the Cambridge Centre for Alternative Finance (2026). That survey does not measure the population of products aimed at the retail investor, and therefore does not support claims about it; the characterisation of that population rests on what the providers themselves declare, and is the object of the following section.

2.2 Financial information tools for retail

This section examines the solutions currently available, assessing each one against a fixed criterion, namely the three questions stated in Chapter 1. Table 2.1 presents the result of that assessment.

Table 2.1: The categories of existing tool assessed against the three questions of Chapter 1. The classification *partial* indicates that the tool supplies raw material for the answer and leaves the interpretation to the user.

Tool	**Is it unusual?**	**Company or market?**	**Has it happened?**	**Cost**
Brokerage application	no	no	no	included
News aggregator	no	no	no	free
Sentiment application	partial	no	no	free or low
Automated portfolio manager	no	no	no	percentage of the portfolio
Professional terminal	yes	yes	yes	not published
This work	**yes**	**yes**	**yes**	**free**

The table makes the essential point. The only existing tool that answers all three questions is also the most expensive of all. The problem is therefore not scientific in nature, but one of access: the answers are known and are conditional on a subscription incompatible with the investor profile in question.

Three categories deserve individual examination, since each fails for a distinct reason and those reasons inform decisions taken in the following chapters.

The brokerage application is the most immediate instrument of access to the price, and it is the one whose public documentation describes the smallest set of indicators. It displays the day's percentage change and a price chart. It does not indicate whether that change is large for that particular stock, does not separate the company component from the market component, and has no memory of comparable events.

The price alerts of these applications share the limitation in a more pronounced form. They fire when a fixed threshold is crossed, a criterion that ignores the volatility proper to each stock and therefore produces frequent alerts in agitated companies and rare ones in calm companies. They further communicate that a level was crossed, and never why. This observation determines the methodological option described in Section 3.4, which replaces the absolute threshold with a measure relative to the recent norm of each company, and whose difference is quantified in Section 5.2.

Sentiment applications classify news as positive or negative and present the result of that classification. What they supply is a reading of the news item and not of the movement: a strongly negative classification indicates that something has happened, and does not establish whether the price change is large for that particular stock, which is the first question. It is in this restricted sense that Table 2.1 credits them with a partial answer, and they go no further. Section 5.3 additionally presents a measurement of our own according to which the topic of a news item and the direction of the movement that follows it coincide less often than intuition suggests, which limits the reach of a sentiment score as an indicator of impact.

Automated portfolio managers answer above all a different question, that of the allocation of capital across a set of assets. The literature recognises genuine, though not uniform, benefits: D'Acunto, Prabhala, and Rossi (2019) measure an improvement in diversification and a reduction in volatility among investors who held fewer than five stocks before joining, whereas for those who already held more than ten the effect on diversification is close to nil. The containment of behavioural biases is the most consistently reported benefit (Cardillo and Chiappini 2024; D'Acunto, Prabhala, and Rossi 2019). The difference with respect to this work is one of object and not of transparency: an automated manager recommends a portfolio and executes it, whereas this system makes no recommendations and delivers evidence about a concrete event. Moreover, providing investment advice is a regulated activity, a boundary that the present work avoids by design, as discussed in Section 3.9.3.

There are, finally, recent products that claim to answer exactly the question treated in this dissertation. Robinhood Cortex, announced in March 2025, states on the provider's own page that it is intended to answer the question of why a stock is going up or down on a given day (Robinhood Markets 2025). Google Finance's key moments, presented in June 2026, propose to explain why a stock moved (Google 2026). This is the same question, posed by organisations with incomparably greater resources and user bases.

The difference lies in what the respective pages declare to be delivered. Both describe a summary in natural language, and neither describes the presentation of individual past cases, with dates, measured similarities and observed behaviour of the price after each of them, which is what this work delivers. The distinction claimed is not one of fluency, but of verifiability, and it concerns what the pages declare: the products were not tested, so nothing is asserted about what they actually deliver.

Up to this point the section has examined products. The academic literature is missing, and leaving it out would rest the claim of a gap on the inspection of vendor pages, which is a fragile basis. The class of system at issue here, which is watching for events and notifying the retail investor, has been treated in published research for some two decades, and the most developed example is the *MoFiN DSS* of Muntermann (2009). The parallel is close to the point of being uncomfortable: it likewise addresses the retail investor, likewise starts from the observation that this investor reacts to news more strongly and less quickly than the institutional one, is likewise built and evaluated under the design science research paradigm that Section 3.1 adopts, and likewise publishes its evaluation in a journal article.

It is worth being precise about what that system does, because the positioning of this work follows from that precision. Over 425 announcements by German firms published between 2003 and 2005, and minute-by-minute intraday price series, the author fits two regression models: one estimates the price effect that will follow the announcement, and the other estimates how long that effect will last. The announcement is pushed to the investor when the estimated effect exceeds the average effect observed. The evaluation is an *ex ante* simulation of consumer surplus, that is, it measures the economic opportunity the notification opens, and not the quality of what is communicated. The user is told that an announcement is relevant and is not told why it is.

Three consequences for this dissertation. The first is that the gap cannot be stated as an absence of alerting systems for the retail investor, because they exist and are published; relative to this system, it lies in epistemic stance, between estimating what will happen and explaining what already happened, and as a difference in what is delivered alongside the alert. The second is that the relevance measure of Muntermann (2009) is an abnormal return *in absolute value*, adjusted for the market and corrected for the usual behaviour of the stock itself, that is, magnitude and not direction. It is in essence the same quantity that Section 3.4 measures, with two differences that matter: here it is measured after the fact, and it serves to explain rather than to anticipate. That a work published in 2009 arrived at magnitude, and not direction, as the operational definition of a relevant event for this audience is independent support for a choice this dissertation reached by another route. The third is that that work evaluated its artefact by simulation and explicitly deferred the study with real users to future research, exactly the limitation Chapter 6 states about this one. This is no excuse, but it locates the problem: it is a difficulty of the field and not an isolated omission.

Other works move towards explaining observed price movements. The *Squawk Bot* of Dang, Shah, and Zerfos (2020) jointly learns price series and text to select news associated with price behaviour. The *DeepTrust* of Chan (2022) combines anomaly detection, retrieval of Twitter posts and assessment of their reliability. The latter is a dissertation made available on arXiv, whose evaluation concentrates on two episodes; its existence precludes presenting the explanation of anomalies through information retrieval as a novelty of this work.

The *Stock Pattern Assistant* of Neela (2025), also made available on arXiv, segments prices into monotonic runs, associates events by temporal proximity and proposes a narrative generated from those objects, without prediction or causal attribution. Its demonstration uses events from a constructed sample, so it does not amount to continuous evaluation on captured news. Explaining without predicting is therefore a shared choice. This dissertation is positioned through the combination of the three questions, the evidence delivered and the evaluation of components in operation, without claiming priority for that choice.

The most reasonable objection to the work consists in asking why a general-purpose artificial intelligence assistant would not solve the same problem. No comparison with any such assistant was carried out, so what follows is a design argument and not a result. An assistant without external access to market data lacks three elements. The first is the recent volatility of the stock, which allows one to determine whether a change is large. The second is the separation between the market component and the company-specific component. The third is a base of past cases over which to perform a retrieval.

On this last point, a language model recalls cases seen during training, which does not constitute retrieval and is not verifiable by the reader. There are large models specialised in financial text, whose capability is documented (Li et al. 2023). The output of any of them remains a statement that is read with ease and not checked.

The explainability literature gives reasons to distrust it. Lipton (2018) argues that interpretability does not designate a single property and that a system should declare which of them it offers. Rudin (2019) argues that an explanation produced over an opaque model is an approximation that may mislead. Neither deals with text generated by language models, so the transposition is the author's and what follows from it is a design reason and not a result of the literature. This is why language never produces facts in this system: the numbers come from the components that computed them, and the text is composed from those, as described in Section 4.5.

2.3 Anomaly detection in financial time series

This section situates the first decision of the system in the field of anomaly detection and states the families of method considered.

Anomaly detection is an established area, with a consolidated taxonomy that separates methods by the way they define normality (Chandola, Banerjee, and Kumar 2009; Pang et al. 2021). Three families are relevant for financial time series.

The statistical family defines normality from the local distribution of the data themselves, estimated in a rolling window, and flags as anomalous the observations that depart from it beyond a threshold. The explanation of a detection coincides with the computation that produced it, which makes it fully communicable to the user. In exchange, it assumes that the distribution of returns is locally stable.

The unsupervised learned family defines normality from the geometric structure of the data. Isolation Forest (F. T. Liu, Ting, and Zhou 2008) identifies as anomalous the points that a random partition isolates with few cuts; the Local Outlier Factor (Breunig et al. 2000) identifies them by their reduced density relative to that of their neighbours. Both dispense with labels and capture multivariate patterns that a rule over a single series does not capture, at the cost of the decision ceasing to be statable in one sentence.

The deep family learns a model of normality through a network (Pang et al. 2021). It offers greater representational capacity, requires larger volumes of data, and departs from the explainability requirement that structures this work.

A practical constraint conditions this whole choice. In neighbouring financial domains, such as fraud detection, unsupervised methods occupy a central position precisely because labelled anomalies are scarce and the patterns change over time (Ahmed, Mahmood, and Islam 2016). The same condition holds here: there is no dataset in which it has been recorded, day by day, which movements would justify an alert. This absence determines the way in which Chapter 5 can evaluate the component, and is the reason why the evaluation rests on a criterion derived from the data themselves rather than on human judgement.

The assumption of local stability on which the statistical family rests is confronted and not hidden. Conditional volatility models, introduced by Engle (1982) and generalised by Bollerslev (1986), formalise volatility clustering, that is, the tendency of calm and agitated periods to occur in runs. Both papers formalise the phenomenon over inflation series, so its transposition to stock returns is not assumed: Section 5.2 measures it on the data of this work, comparing the equally weighted window with a variant that assigns greater weight to recent observations. A rolling window follows the phenomenon in part, since the norm against which each day is compared is re-estimated daily from the near past. The choice does not eliminate the problem, and the measurement referred to quantifies what it costs to keep it.

The comparison between the statistical family and the learned family does not remain at the level of argument. Chapter 5 runs both on the same data and reports the result obtained.

2.4 Text representation and semantic retrieval

This section surveys the ways of representing text that make it possible to compare news by meaning, and states the criterion for evaluating retrieval that follows from them.

Comparing news by meaning depends entirely on the representation adopted, and that representation went through three generations, sketched in Figure 2.1 and detailed in Table 2.2.

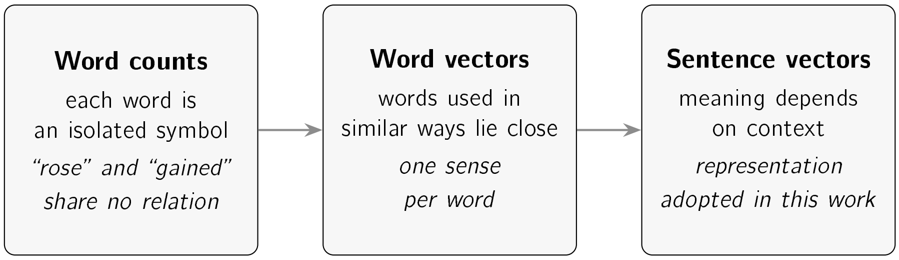

Figure 2.1: The three generations of text representation. Each one resolves a limitation identified in the previous one, and the third is the one that allows two sentences to be compared directly by the position of their vectors.

The first generation represents a document by the frequency of the terms it contains. The vector space model (Salton, Wong, and C. S. Yang 1975) is transparent and computationally cheap, and it is blind to synonyms: the terms "rose" and "gained" are, for this representation, symbols without relation. Within the same generation there is a second line, coming from a distinct probabilistic framework, that of ranking functions of which BM25 (Robertson and Zaragoza 2009) is the current representative; both remain strong baselines in information retrieval and share that blindness. In financial text there is also a specialised variant, the domain lexicons, which correct the fact that generic sentiment dictionaries misread words that are common in financial language (Loughran and McDonald 2011). They are entirely transparent, and are limited to the vocabulary that was compiled.

The second generation represents each word as a vector in a space where proximity reflects similar use, learned from local context (Mikolov et al. 2013) or from global co-occurrence (Pennington, Socher, and Manning 2014). This representation brings synonyms together, resolving the previous limitation, and introduces a new one: each word receives a single vector, regardless of the sentence in which it occurs.

The third generation removes that limitation. The Transformer (Vaswani et al. 2017) made it feasible to process the complete sentence with long-range dependencies, and Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al. 2019) produced contextual representations, in which the vector of a word depends on those surrounding it. An additional step is needed for the problem treated here: comparing two sentences with a model of that nature would require processing them jointly, once per pair, which is not practicable over a base with tens of thousands of cases. Sentence-BERT (SBERT) (Reimers and Gurevych 2019) resolves this by producing one vector per sentence, computed only once, so that the comparison becomes an arithmetic operation between vectors that already exist.

The contribution of SBERT that matters to this work does not lie in the architecture but in the training objective: the vectors are learned so that the distance between two of them reflects similarity of meaning. The authors report that an encoder of the BERT family used directly to compare sentences produces weak representations for that purpose, at times falling behind the simple average of word vectors (Reimers and Gurevych 2019). It follows that an encoder tuned to another task does not inherit this property by virtue of having been trained in the appropriate domain.

For financial text there are models trained specifically for the domain (A. H. Huang, Wang, and Y. Yang 2023; Zhuang Liu et al. 2020). The natural assumption is that they are superior

Table 2.2: The approaches considered for representing financial news, with the representation each one produces and the implications of adopting it.

Approach	Representation	Strengths and limitations
Term counts (Robertson and Zaragoza 2009; Salton, Wong, and C. S. Yang 1975)	Weighted word frequency	Transparent and fast; blind to synonyms
Financial lexicons (Loughran and McDonald 2011)	Word lists with assigned polarity	Fully transparent; limited to the compiled vocabulary
Word vectors (Mikolov et al. 2013; Pennington, Socher, and Manning 2014)	One fixed vector per word	Brings synonyms together; a single sense per word
Sentence vectors (Devlin et al. 2019; Reimers and Gurevych 2019)	One vector per sentence, sensitive to context	Compares sentences directly; larger model, less legible internally
Domain models (A. H. Huang, Wang, and Y. Yang 2023; Zhuang Liu et al. 2020)	As above, tuned to financial text	Domain vocabulary; superiority on this task is not automatic
Large models (Li et al. 2023)	Text generation over broad knowledge	High capability; expensive, opaque and outside the free-of-charge constraint

on this task for that reason. The assumption is not automatically true, and Section 5.3 measures it rather than assuming it.

The work belongs to the wider literature of natural language processing applied to finance. The review of Kearney and S. Liu (2014) covers the state of that area up to 2014, dominated by domain lexicons and by supervised classifiers over word counts; the contextual sentence representations used in this work are later and do not appear in that review. There is also a substantial line of research that uses text to anticipate price movements (Ding et al. 2015; Xing, Cambria, and Welsch 2018). The present work stands outside that line by design choice: text is used here to locate comparable earlier cases, whose outcome has already occurred and has already been measured, and not to generate a forecast.

One recent work in that line deserves individual examination, being the closest to what is done here. Du et al. (2024) train a triplet network over social media messages and price series, and present to the user, alongside each prediction, the most similar historical case with the same outcome and the most similar one with the opposite outcome. This is retrieval of analogues used as explanation, over the same materials this work uses, and it is therefore the closest published neighbour.

The differences are nonetheless fundamental, and worth stating because they delimit the contribution. The label that organises the triplets is the direction of the next day's movement, and the task evaluated is to predict it; magnitude enters only to define a dead band around zero. The sentence encoder is used frozen, as the input layer to a convolutional network and a recurrent network with attention, which are the ones actually trained, that is, the space in which news items are compared is not changed. And the explanation is demonstrated through three case studies with attention maps, with no measurement of its quality and no participants.

There is a result of their own in that work that deserves recording, because it concerns the difficulty of the task and not the method. In the ablation study, applying contrastive learning to the textual component alone degrades accuracy on all three benchmark datasets, falling below the variant that uses no contrastive learning at all; the gains come from the quantitative component, and the authors themselves conclude that textual information plays a complementary role. Anyone seeking to use text to anticipate the direction of a movement has here, published, an indication that the signal is weak; Section 5.3 reaches a conclusion of the same sign by an entirely independent route.

It is on those two differences that the fourth research question is built. If direction is the weak signal that their ablation study shows, and if the comparison space there remains unchanged, then two distinct questions are left open: whether the *magnitude* of the movement, which is not a direction and does not constitute a forecast, is learnable from text, and whether the gain appears when the encoder itself is adjusted rather than a network placed on top of it. Section 3.6.1 describes how both are put to the test, with an arm trained on magnitude and a replication arm trained on direction, and Section 5.4 reports what was obtained.

Representing each news item by a vector turns the identification of comparable cases into an information retrieval problem. The proximity measure adopted is cosine similarity, whose mechanics are described in Section 3.6. The operation is exact, since the whole case base is scanned and the result ordered, so it does not depend on any approximation. At substantially larger scales, exhaustive search can be replaced by an approximate index (Johnson, Douze, and Jégou 2021) without changing the representation, which is a path for growth and not a present necessity.

Since retrieval returns an ordered list, its evaluation resorts to order-sensitive measures (Manning, Raghavan, and Schütze 2008). The one adopted in this work is precision at the first k results, that is, the fraction of those k results that are relevant. The definition of relevance, which is where an evaluation of this nature is decided, is established in Section 3.6 together with the baselines against which the result is compared.

2.4.1 Retrieval with generation, and why the system stops before it

The architecture in current use for answering a question from a body of documents is retrieval-augmented generation, in which a retriever selects passages and a language model writes the answer from them (Lewis et al. 2020). The literature on the family is extensive and has been systematised (Y. Huang and J. X. Huang 2026).

The system described here executes the first half of that architecture and stops before the second, and the reason is the one that runs through the work. Retrieval returns verifiable objects: a headline, a date, a measured similarity and an observed price movement. Generation converts those objects into a statement in prose, whose faithfulness to what was retrieved is itself a property to be evaluated, and which the recipient of this system has no means to confront. The option does not follow from a technical limitation, and the distinction should be drawn clearly: adding generation is possible and cheap; what is not cheap is the evaluation that it does not displace the meaning of the evidence. No comparison with a complete architecture was carried out in this work, so what is asserted here is a design reason and not a result. Section 4.7 describes the grounded generation layer that was in fact built and evaluated, and the reason why it is not exposed.

2.4.2 Financial news corpora and sector labels

Evaluating retrieval over financial news requires a body of text aligned with prices, and that alignment is what distinguishes a usable corpus from a collection of headlines. The dataset used in this work is FNSPID (Dong, Fan, and Peng 2024), which brings together news and the corresponding price series and whose construction is documented.

Relevance between two news items is approximated, in this work, by belonging to the same sector. The option replaces a non-existent human annotation with an objective criterion, and carries the corresponding limitation. The validity of a grouping is measurable, by the separation between groups (Rousseeuw 1987) or by the agreement between alternative groupings corrected for chance (Vinh, Epps, and Bailey 2010). Neither of those measures was applied to the sector map adopted. What the label guarantees is objectivity and reproducibility, and not that the sector is the most informative partition of the corpus.

2.5 Event studies and return decomposition

This section presents the instrument with which the effect of an event on the price is measured, as well as the statistical correction on which the separation between the company component and the market component depends.

The instrument adopted is the event study, whose formulation goes back to Fama et al. (1969). The behaviour of the method over daily returns was examined by Brown and Warner (1985), who compare alternative ways of estimating the expected return and conclude that the current procedures, based on estimating the market model by least squares, remain well specified in short windows around the event. The reach of this conclusion is delimited by the authors themselves: the simplest variant, which subtracts only the historical mean of the stock, loses statistical power and may become misspecified when the event dates coincide. It is this result that motivates the order of magnitude of the one, three and five day windows adopted in this work. The transposition is not direct, and the difference is declared in Section 3.6, according to which the value presented to the user is the raw return and not the abnormal return that those authors study.

The existence of a statistical relation between published text and market movements is a long standing observation (Tetlock 2007). The appropriate verb is that of evidence of a statistical relation, and not that of a demonstration of causality.

One methodological point conditions the second question of the work, that of the separation between the company and the market. That separation requires estimating the sensitivity of each stock to the market, and the estimate is noisy, especially in volatile companies or those with a short history. Blume (1971) establishes that the coefficients so estimated tend to regress towards the mean between successive periods, from which followed the practice of shrinking the estimate with a fixed weight. This work adopts instead the shrinkage weighted by the imprecision of the estimate itself (Vasicek 1973), whose formulation and effect are presented in Section 3.5.

The theoretical framing that justifies the refusal to forecast also belongs to this area. The efficient market hypothesis, in its semi-strong form, holds that prices already reflect publicly available information (Fama 1970), from which it follows that future movements should not be systematically predictable from that information. It is one of the reasons why the system measures what has already happened rather than projecting what is to come; the second, of a practical nature, is that only the past is verifiable by the reader.

2.6 Case-based reasoning

This section frames the third technique of the system within an established line of artificial intelligence and delimits the part of that line the work implements.

Case-based reasoning (Aamodt and Plaza 1994) approaches a new problem through the retrieval of similar earlier problems and the use of what is known about them, dispensing with exclusive recourse to general domain knowledge. The source organises the method into a cycle of four processes, namely retrieve, reuse, revise and retain, and considers all four necessary to the complete resolution of a problem.

Figure 2.2 shows the cycle and marks where the system stops.

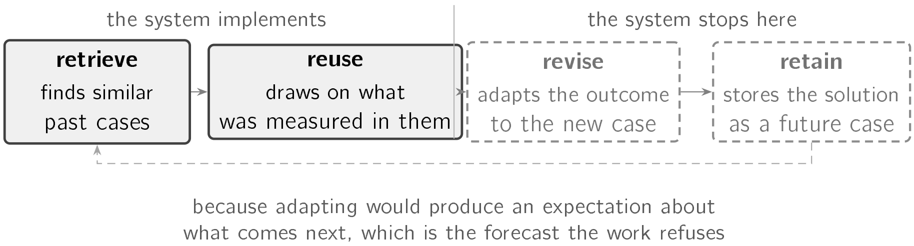

Figure 2.2: The four processes that Aamodt and Plaza (1994) consider necessary to the complete resolution of a problem. The system runs the first two in full and stops before the next two. The interruption is not a shortfall of implementation: *revise* adapts the outcome of a past case to the present one, and that adaptation is an expectation about the future, that is, exactly the forecast the founding constraint refuses. What the system presents is the measured outcome of each case, as it occurred. The first two processes, run in full and with the decision left to the person, correspond to what Kolodner (1991) calls case-based decision aiding.

The technique described in Section 3.6 implements the first two and stops deliberately before the rest: it retrieves similar cases and presents what was measured on them, without adapting that outcome to the present case. Adaptation would produce an expectation about what is going to happen, which is precisely the forecast the work refuses. The partial implementation of an established cycle is, in this case, a consequence of the founding constraint and not a shortcoming.

This stopping point has, moreover, a name of its own in the literature, and recognising it changes its status. Kolodner (1991) describes *case-based decision aiding* as a methodology in which the system augments the person's memory by supplying analogous cases, and in which it is the person, not the machine, who decides, using those cases as guidance.

The starting point is an observation from psychology: people reason well by analogy but remember the right cases poorly, whereas computers remember well. Put that way, what this system does ceases to be the execution of half a cycle and becomes the full execution of a distinct methodology, proposed in 1991 by the person who defined the area. The difference is not one of pride: a partial implementation explains what it lacks, whereas a methodology declares what it delivers and to whom the decision belongs.

It is worth recording, in the opposite direction, what case-based reasoning has actually done in this domain, so that the originality claimed is no larger than it is. Oh and T. Y. Kim (2007) build a daily indicator of the condition of the Korean financial market that classifies each day as a stable, unstable or crisis period, retrieving cases from a base built on the 1997 crisis. This is case-based reasoning applied to finance, and it is instructive for two reasons. The first is the object: the unit classified is the whole market, in a judgement about systemic risk, and not the movement of one company following a news item. The second is the destination of the retrieved cases, which serve to produce a label automatically; the article does not describe their presentation to anyone. Showing the cases, which is what is done with them here, is precisely what is left undone there.

The option finds additional support on the side of the reader. Research on explanation in the social sciences establishes that people look for contrastive and selective reasons, and not for an exhaustive enumeration of everything that contributed to an outcome (Miller 2019). A small set of concrete cases, with dates and outcomes, corresponds better to that expectation than a coefficient. It is for that reason that each alert presents a few precedents and the exact quantities that support the decision, rather than the totality of the values computed.

There is a second requirement, on the same side, that conditions not the content of the explanation but the frequency with which it is delivered. A recipient exposed to excessive warnings stops processing them, an effect measured on retail investors (C.-W. Liu, Chen, and Wen 2023) and discussed in Section 2.1. The consequence is that the decision policy forms part of the design. An alert that is not read is worth the same as an alert that was not sent, and that is why Chapter 4 treats the suppression criteria with the care it devotes to the methods.

2.7 Explainability and trust in automated systems

This section situates the explainability requirement that runs through the work and presents the cautions that the literature associates with the presentation of explanations.

The demand that a system show how it reached a conclusion is an area in its own right, with consolidated reviews (Adadi and Berrada 2018; Barredo Arrieta et al. 2020). The area divides into two routes, shown in Figure 2.3: transparent models, interpretable in themselves, and *post-hoc* methods, which explain an opaque model already trained and which can be organised by the type of object they explain (Guidotti et al. 2018).

The two most widely used *post-hoc* methods operate in distinct ways. LIME fits a simple model in the neighbourhood of an individual prediction, locally approximating the behaviour of the opaque model (Ribeiro, Singh, and Guestrin 2016). SHAP attributes a prediction to the variables that produced it using Shapley values (Lundberg and S.-I. Lee 2017). Both are generic and both are useful.

Both share, however, a property that is decisive in this context. Faced with decisions of serious consequence, Rudin (2019) argues for abandoning opaque models in favour of models

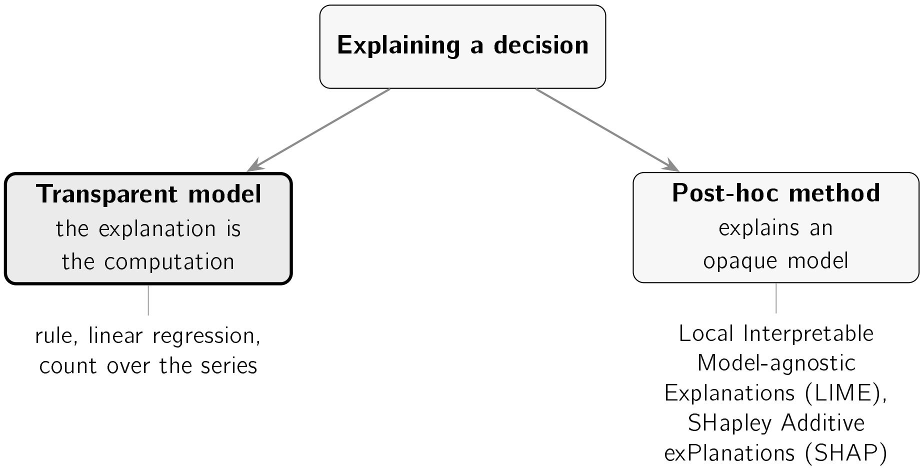

Figure 2.3: The two routes distinguished by the literature (Barredo Arrieta et al. 2020; Guidotti et al. 2018): building models that are transparent by design, or explaining an opaque model after training. This work follows the first.

interpretable by construction, on the grounds that every *post-hoc* explanation is itself an approximation, and an approximation can lead to mistaken readings. The user receives in that case two objects whose relation they are not in a position to assess, namely a decision and an explanation that may not correspond to it.

This work follows the conservative route. Instead of explaining an opaque model *post hoc*, it resorts to components in which the explanation and the computation coincide. A departure from a recent norm, an additive split whose sum reproduces the observed movement, and a set of earlier cases with the outcomes measured are objects presentable in full. The explanation is not a second object produced over the decision: it is the decision, stated at length.

Two cautions from the literature close this section, and both condition the organisation of Chapter 5.

The first is that interpretability does not designate a single property (Lipton 2018), so a system should declare the type of interpretability it offers rather than claiming the term without qualification. This work offers transparency of mechanism, that is, the user can follow the computation that led to the decision, and does not offer a causal explanation of the event in the world.

The second concerns the effect of the explanation itself on the person who receives it. Trust in automated systems fails in two symmetric directions, with the user ignoring a correct warning or accepting an incorrect one, so the objective is not to maximise trust but to make it appropriate to the real capability of the system (J. D. Lee and See 2004). Bansal et al. (2021) add that the presence of an explanation increases the probability of accepting the system's answer whether it is right or wrong, that is, that the observed gain does not come from the person becoming better at distinguishing the two cases. The design consequences that follow are described in Section 3.9.3.

These cautions are general. On the concrete choice made by this dissertation, which is to explain by cases, there exists empirical evidence gathered with people that points the other way, and relegating it to a footnote would be the cheapest way of neutralising it.

Waa et al. (2021) compared rule-based explanations, example-based explanations and no explanation, in two experiments with forty-five participants each, in a decision aid for insulin dosing. Rule-based explanations significantly improved identification of the decisive factor; example-based explanations were indistinguishable from no explanation at all, neither in factor identification ($p = 0.796$) nor in self-reported understanding ($p = 0.283$). The explanation the authors give for the result is the part that matters most here: both styles, they write, provide only the details relevant to a single decision, and not the underlying rationale or causality.

Within the financial domain itself, and with non-experts, there are two recent studies. Cau et al. (2023) compare three explanation styles in a stock trading task and conclude that the abductive and deductive styles improve performance when the model is confident, by comparison with the inductive style, which is the example-based style and therefore the one closest to what this system delivers. Bertrand, Eagan, and Maxwell (2023) test an automated life-insurance advisor with feature-based explanations and conclude that these improve neither understanding nor appropriate reliance, by comparison with giving no explanation at all, and that the more conversational format increases the user's trust, sometimes to the user's detriment.

Three consequences follow for this dissertation. The first is that the choice of case-based explanation, justified in Section 2.6 by the literature on explanation in the social sciences (Miller 2019), has no support in the measurements with people that exist, and in one case has evidence from the domain itself against it. The second is that the diagnosis of Waa et al. (2021) names exactly what this system adds to the cases: an additive decomposition whose sum reproduces the observed movement, which is the underlying computation whose absence the authors hold responsible for the result. The system does not deliver cases alone; it delivers cases and the arithmetic that accompanies them. Whether that is sufficient is an open empirical question, and the work does not answer it. The third, and the most consequential, is that these results make the study with users more necessary and not less: there is published literature pointing against the founding hypothesis, and it is in those terms that Section 6.4 states its absence.

The second study supports, by contrast, a decision already taken. If the conversational format is what produces excessive trust, then not exposing the conversational layer described in Section 4.8 ceases to be merely engineering prudence and acquires empirical support within the domain.

Furthermore, claims of interpretability demand rigorous evaluation and are not established by declaration (Doshi-Velez and B. Kim 2018). It is for that reason that Chapter 5 measures what is measurable by automatic procedure and explicitly identifies the part that remains unevaluated for lack of a controlled study with users.

2.8 Machine learning in a production environment

This section brings together the literature relating to the learned component of the system, organised according to the three questions its deployment raises.

The first question is that of determining against what stronger reference a simple model should be compared. Ensembles of trees trained by gradient boosting (Friedman 2001) capture interactions and non-linear effects that a logistic regression does not capture, and therefore play that role in Chapter 5. It is not claimed that they constitute the best existing method, since that would be a claim about an entire field.

The second question concerns the honesty of the value presented. A score is only interpretable as a probability if it is calibrated, and the raw scores of many classifiers are not; *post-hoc* calibration over a held-out validation block corrects that distortion (Niculescu-Mizil and Caruana 2005). The same source does not, however, favour the option taken in this work: it identifies logistic regression as already well calibrated to begin with and concludes that, above roughly a thousand calibration cases, isotonic regression matches or exceeds the sigmoid. The deployed model is a logistic regression with a validation block far above that threshold, so both statements apply to it. The option is justified by measurement and not by authority: the comparison of the two variants over the same validation set is reported in Table 5.7, on page 96.

An alternative route was considered and not adopted. Conformal prediction returns a set of labels with a distribution-free guarantee, valid in finite samples (Angelopoulos and Bates 2023; Vovk, Gammerman, and Shafer 2005). Two clarifications determine that it is not applicable under the conditions of this work. The coverage of the current variant is marginal: it holds on average over cases, and not case by case. And the guarantee requires exchangeability between the calibration block and the test block, an assumption that the chronological split with embargo of Section 3.7 refuses by construction. Section 6.5 returns to the method as a path to explore.

The third question is that of the persistence of performance over time. A deployed model faces change in the distribution of the data it observes, and the concept drift literature establishes the taxonomy of that change and the strategies for detecting and adapting to it (Gama et al. 2014). This work measures drift on its own data rather than assuming it absent.

There is, finally, a cost that is captured by no quality metric. D. Sculley et al. (2015) establish that machine learning components accumulate hidden maintenance, in particular through entanglement between parts and through unstable, undeclared data dependencies. Chapter 6 returns to this point with an example measured in the present system, in which part of the learning cycle stopped without any error being recorded.

2.9 Synthesis and positioning of the work

This section locates the work with respect to the review presented and states the scope decisions that follow from it.

The review allows the space in which the work is inscribed to be delimited precisely, and that space is not a technical void. Every component used exists, is published and is evaluated, as Table 2.3 makes explicit.

The gap is therefore not one of method, but of integration under constraint. The products that answer the three questions are professional terminals, unavailable free of charge or at a price accessible to the retail investor. Their price is not published in a citable form, and it is unavailability, and not a concrete value, that supports the argument. The free products that claim to answer the third deliver a statement without the evidence that supports it. The academic literature includes integrated systems and proposals for explaining past movements, as Section 2.2 documents. The distinction from MoFiN DSS includes the absence of prediction; with DeepTrust and Stock Pattern Assistant, that choice is shared. The

Table 2.3: Every component of the system exists in the literature in isolation. What does not exist is their combination under the zero-cost constraint, with the evidence delivered to the user at the moment the statement is presented.

Question	**What the literature offers**	**What the target audience lacks**
Is it unusual?	Anomaly detection, statistical and learned (Chandola, Banerjee, and Kumar 2009; F. T. Liu, Ting, and Zhou 2008)	A measure relative to the company's own norm, communicated in accessible language
Is it the company or the market?	Event study and sensitivity estimation (Brown and Warner 1985; Vasicek 1973)	Availability outside a professional terminal
Has it happened before?	Sentence representation and retrieval (Manning, Raghavan, and Schütze 2008; Reimers and Gurevych 2019)	A case base with measured outcomes, and not recalled ones
Why believe it?	Explainability and the trust literature (J. D. Lee and See 2004; Rudin 2019); measurements with people that do not confirm the gain (Bertrand, Eagan, and Maxwell 2023; Cau et al. 2023; Waa et al. 2021)	The evidence delivered alongside the statement and verifiable at the moment of reading; still to be demonstrated with users

contribution must therefore be assessed through the combination of rarity, decomposition and precedents, the operating constraints and the evaluation of the evidence delivered. The reviewed works delimit this contribution; they do not demonstrate a universal absence of comparable systems.

This observation determines the distinction that runs through the work and that Chapter 4 takes up when justifying the absence of a language model in the composition of the messages. A generated sentence without links to facts presents itself to the reader in a fluent register and gives no means of confronting it with what supports it. A sentence composed from computed objects is a statement accompanied by the evidence, in which every value presented points back to the measurement that produced it and the reader can confirm each element at the moment of reading. This is the property the system seeks to preserve, and it is what fixes the nature of the contribution. It is not the formulation of a new method. It is the selection, integration and critical evaluation of existing methods in a working system, under declared constraints, and with the results that do not favour the options taken included in the evaluation.

Four scope decisions follow from this positioning, and each one deliberately excludes a class of functionality.

Answer the three questions, and only those. No functionality is incorporated that does not serve one of them, which keeps the system at a size compatible with explaining it in full.

Produce no price forecasts. Neither direction, nor target, nor recommendation. Besides the verifiability reason stated in Chapter 1, the theoretical framing of Section 2.5 applies (Fama 1970).

Manage no portfolios and give no advice. The system does not know the user's positions and does not tell them what action to take, which avoids collecting personal data and keeps the work outside the boundary of regulated financial advice.

Use free sources exclusively. The constraint makes the system accessible to the audience it addresses and turns the resulting limitations into measured and reported quantities.

The last decision has a real cost that should be stated without mitigation. News arrives late and the coverage of events is incomplete. Chapter 4 quantifies both limitations and Chapter 6 assesses their consequences.

Chapter 3

Methods and materials

Chapter 2 established that the three questions of the retail investor have an answer in the literature but not in any free tool. This chapter describes the options taken to answer them. It first presents the research methodology and the datasets. It then describes the four methods selected, one for each decision the system has to take. It indicates the tools employed. And it finally states the technological and social challenges of the domain. Whoever reads it will know how each of the four decisions is taken, and what guarantee stops each one from using information that did not yet exist.

3.1 Research methodology

This dissertation is framed within design science research (Hevner et al. 2004), in which the object of study is an artefact built to solve a class of problem and the knowledge produced resides both in the artefact and in the process of its evaluation. It is the appropriate framing when the contribution is a system, and it differs from a purely empirical study in that the object evaluated does not exist before it is built.

The design science literature establishes that the evaluation of the artefact must be rigorous and constitute a central activity of the process, and not a final confirmation step (Hevner et al. 2004; Peffers et al. 2007). This requirement is met through three commitments made before the experiments were carried out.

The first consists in comparing detection, retrieval and triage with simpler alternatives selected before measurement, so that the result can be negative. Decomposition receives only a descriptive check of the split and fit. The second consists in fixing the success criterion of each comparison together with the metric, and not after observing the values obtained. The third consists in evaluating the artefact twice: over held-out historical data and, subsequently, in operation, over the decisions the system takes autonomously in production. This second evaluation produces the result described in Section 5.7.

The requirement is not met in full, and the gap is declared. Utility, in the sense in which the design science literature employs the term, is utility for whoever uses the artefact, and that property is determined with users. What this dissertation measured is the faithfulness of the explanations produced, a property that belongs to the system; utility, which depends on the relation between the system and whoever reads it, remains unmeasured.

The system takes four distinct decisions, shown in Figure 3.1. The first three correspond to the three investor questions stated in Chapter 1; the fourth is not a question of the investor, but of the system itself, and exists because answering the first three does not imply interrupting the user.

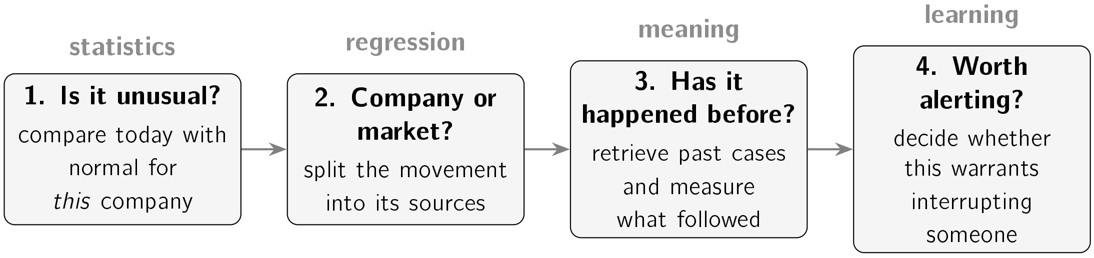

Figure 3.1: The four decisions of the system and the family of techniques associated with each. The first three correspond to the questions stated in Chapter 1. Each column corresponds to a section of this chapter, between Section 3.4 and Section 3.7.

3.2 Datasets

This section describes the datasets used in the training and evaluation of the system, as well as the records produced by the system in operation.

The historical layer uses the news component of FNSPID (Dong, Fan, and Peng 2024), which associates a date and a company with each headline. Although the published dataset also includes prices, in this work returns and labels are constructed by aligning the headlines with price series obtained separately. From the alignment comes the training set of the triage model, with 79,753 examples.

The source file is identified by revision, not by name. FNSPID is distributed in a repository whose default address points at the main branch, and a branch is a pointer that moves: a dataset referred to that way is not reproducible by construction. The corpus in this work starts from the file `Stock_news/nasdaq_exteral_data.csv` at a fixed revision, 23.2 GB with SHA-256 checksum prefix `1a7a3eb8`; the filtering is local and deterministic, and the corpus it produces has prefix `af61708c`. The full checksums, the per-company counts and the audit of the procedure are recorded in a versioned manifest in the repository, and an automated test refuses any corpus that is not this one.

Two construction decisions are declared here because they bear on how the results should be read. The first is that headlines are kept exactly as FNSPID delivers them: the corpus contains 1,704 exactly repeated headlines, 2.1% of the total, which were not removed. The second is that the filtering scans the whole file rather than stopping when it believes it has passed the symbols being sought; the file is not globally sorted by symbol, and stopping early would truncate the corpus in silence.

The set covers fourteen of the fifteen companies in the canonical map. The one missing is Meta, and the reason lies in the dataset rather than the procedure: FNSPID indexes the company under its legacy symbol `FB`, so selecting by the current symbol returns no records at all. Records under the legacy symbol do exist, but they amount to 432 headlines spread over 87 days between February and June 2020, four months in a six-year span. The company therefore appears in the detection evaluation, which rests on prices, and not in the news dataset.

The evaluation of the anomaly detector uses a distinct dataset, made up of three years of daily quotes. The choice of a separate set is deliberate: the evaluation requires variety of volatility across companies, whereas the list monitored in production is a product decision.

Evaluating the detector on exactly the set it was built to serve would produce a measurement with no informative value.

All prices used are closes adjusted for splits and dividend distributions. The adjustment is not a detail in the period considered, which contains a four-for-one split of Apple and two of Tesla: on unadjusted series, the detector would flag falls of thirty to eighty per cent on days with no event at all.

Table 3.1 brings together the datasets used, and Table 3.2 the company sets, whose counts differ for reasons described in the respective column.

Table 3.1: Datasets used in the work. The span of the triage training set refers to trading days, after alignment: the headlines run from 2018-01-01 to 2023-12-16, and the two extreme dates, a holiday and a Saturday, are carried forward to the following session. The last three rows correspond to the system in operation.

Dataset	**Span**	**Size**
Historical		
Triage training set	2018-01-02 to 2023-12-18	79,753 examples, 14 companies
Prices for the detection evaluation	2023-06-01 to 2026-06-01	15 companies, ≈ 750 days each
Operation		
Reconstructed precedent base	one year of own collection	38,214 cases with measured impact
Alerts delivered	2026-07-09 to 2026-08-13	367 messages
Triage decision log	2026-07-22 to 2026-08-20	36,925 decisions

Table 3.2: Company sets used. The counts differ because each set serves a distinct purpose. The sets are not nested: the list monitored in production contains two companies absent from every historical set, and the canonical map contains a third, Meta, that does not reach the triage dataset, for the reason given above.

Set	**Companies**	**Purpose**
Extended sector map	17	Universe covered by the movement decomposition
Canonical historical map	15	Relevance label in the retrieval evaluation
Prices for detection	15	Variety of volatility in the detector evaluation
Triage dataset	14	Result of the alignment between news and prices
covered by the training block	13	Companies actually present in training
covered by the test block	9	Companies present in the evaluation
List monitored in production	12	Product decision

3.3 Where artificial intelligence enters this system

The four decisions this chapter goes on to present draw on distinct families of technique, and the question of where machine learning sits in this work has a different answer for each of them. Table 3.3 gathers that answer, which would otherwise remain scattered across the sections that follow, and adds the decision that precedes all of them: the filter that determines which headlines come to be assessed at all.

Table 3.3: The system's five decisions, the origin of their parameters, and what learning each of them from data would require. The middle column separates three situations that the generic label of artificial intelligence conflates: parameters estimated in this work, parameters of a model trained by others and used without adjustment, and the absence of parameters. The values in the first row are measured and discussed in the subsection that follows.

Decision	Origin of the parameters	What learning it would require
Headline relevance	rule; no estimated parameters	labels independent of the rule itself. The only one available is its own output, which makes the exercise circular
Rarity of the move	none; mean and deviation of the preceding window	rarity labels that were not themselves a function of volatility
Attribution of the move	estimated by regression, with shrinkage	they are already estimated; what is missing is an observable reference value for the attribution
Comparability of news items	model pre-trained by others, used without adjustment	it was attempted, and is Section 3.6.1; the result is in Section 5.4
Materiality of the news item	estimated in this work, on the historical set	they are already estimated; the result is in Section 5.5

The decision that comes before the others, and is a rule

Before any of the four, the system decides whether a headline concerns the company at hand. The rule requires the company to be mentioned, by name or by one of its alternative names, and rejects a closed set of market-summary patterns. Measured over the ten companies the watched list covers, it retains 4 843 of 16 840 headlines, that is 28.8%, and discards 71.2%. Breaking the discards down locates the work: the mention rule accounts for 68.3% of the total and the pattern list for only 3.0%.

In proportion, it is the most aggressive filter in the system, and it is the only one that remains a rule. The reason is not convenience. Learning this decision would require headlines annotated for pertinence, and the only label available in quantity is the output of the rule itself, which would make the exercise circular. The honest alternative is manual annotation, which Section 6.5 lists among the lines requiring human judgement. Until it exists, the retention rate depends on the list of alternative names, and a company without an entry in that list falls back on requiring the ticker symbol, which almost no headline writes.

What this means for reading the results

The five decisions correspond to five distinct situations, and the table separates them one by one. Relevance is a rule with no parameters. Rarity has no parameters: it compares the day with the mean and deviation of the preceding window. Attribution has estimated coefficients, but estimated without labels, from that same series of returns. Comparability rests on a model trained by others and used without adjustment. And only materiality has parameters trained on labelled examples, which makes it the only decision learned in the ordinary sense of the word. The only place where a representation was trained for this problem is the contrastive adjustment of the encoder, whose answer is given in Chapter 5.

This distribution is not an accident of execution, and it is what gives meaning to the comparison running through the evaluation. The two decisions where this work's own learning was applied, materiality and the attempt to adjust the encoder, are also the two whose answers are negative; the affirmative answers come from a statistical rule and from a model used as it stands. The comparison between each adopted technique and the corresponding learned alternative is presented in each case study, and it is this work's most transferable finding. The generic label of artificial intelligence would poorly describe a system in which learning occupies one of five decisions; naming the place of each family of technique is how to describe it without promising what it does not do.

3.4 Detection of unusual movements: the rolling z-score

The first decision of the system consists in determining whether the movement observed on a day is unusual for the company in question. A fixed threshold rule, of the kind that flags changes above three per cent, is inadequate for this purpose, because the same percentage corresponds to events of very different rarity depending on the volatility of the company.

The method adopted belongs to the statistical family of the anomaly detection taxonomy (Chandola, Banerjee, and Kumar 2009), which defines normality from the recent distribution of the data themselves. The movement of the day is compared with the recent behaviour of the same stock and expressed in standard deviations:

$$z_t = \frac{r_t - \mu}{\sigma} \tag{3.1}$$

where r_t is the logarithmic return of the day, $r_t = \ln(P_t/P_{t-1})$, and μ and σ are the mean and the standard deviation of the returns of the twenty preceding days. The use of logarithmic returns follows the current convention in financial series, since it makes returns additive over time and symmetric between rises and falls. The values presented therefore differ from those displayed by a stock application: a logarithmic return of $+19.82\%$ corresponds to a price change of $+21.92\%$.

The window slides each day, and the day being assessed is excluded from the window that defines its own norm, as shown in Figure 3.2. Without this exclusion, the day would contribute to the mean and to the standard deviation against which it is compared, attenuating its own anomaly. The same discipline prevents the use of any information posterior to the day being assessed.

The separation is not merely a convention of writing: it is a slice. Listing 3.1 reproduces the calculation as it is executed. The line that guarantees it is the one that establishes the

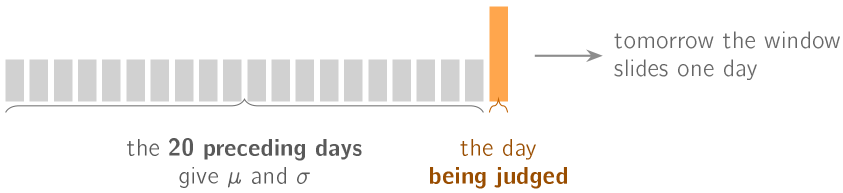

Figure 3.2: The window that establishes the usual behaviour of the stock. The day being assessed is **left out** of the computation of the norm against which it is compared, a separation that prevents the use of information unavailable at the moment of the decision.

`recent` window, whose interval ends at the penultimate element and excludes, by construction, the day being assessed.

```
r = pd.Series(list(returns), dtype="float64").dropna().reset_index(drop=True)
if len(r) < window + 1:
    raise ValueError(f"São precisos pelo menos {window + 1} retornos (recebidos {len(r)}).")

recent = r.iloc[-window - 1 : -1]  # janela ANTES do último (sem lookahead)
last = float(r.iloc[-1])
mu = float(recent.mean())
sigma = float(recent.std(ddof=1))
```

Listing 3.1: The window that excludes the day assessed. The interval `[-window-1:-1]` ends before the last element, so the return of the day under assessment does not enter the mean or the standard deviation against which it is compared. The body of the detection function is reproduced, without the indentation it carries in the source file. The comment is in the original.

When the standard deviation of the window is zero, Equation 3.1 is undefined. The system distinguishes two cases: holding the value constant does not constitute an anomaly, whereas a departure from that flat norm is flagged without a z value being assigned to it, stating that the preceding variance was zero. This distinction prevents the guard against division by zero from silencing precisely the first movement after a constant period.

Applying the method to the largest movement of the period studied, recorded by Tesla on 24 October 2024, produces a daily return of $+19.82\%$ against a mean of -0.92% and a standard deviation of 2.725% in the twenty preceding days, which corresponds to $z = +7.61$. The value does not follow from the absolute size of the movement, but from the fact that it exceeds by seven and a half times the usual oscillation of the stock in those weeks.

3.5 Decomposition of the observed movement: two-factor regression with shrinkage

The previous technique quantifies the size of the movement, but does not identify its origin. The second decision of the system consists in splitting the observed return into its market, sector and company-specific components:

$$r = \underbrace{\beta_m r_m}_{\text{market}} + \underbrace{\beta_s \tilde{r}_s}_{\text{sector}} + \underbrace{\varepsilon}_{\text{company}} \tag{3.2}$$

The variables r_m and $\tilde{r}_s$ denote, respectively, the daily return of the market index and that of the sector, whereas β_m and β_s quantify the historical sensitivity of the stock to each of those sources. The company-specific component is defined as the residual, so that the sum of the three parts always reproduces the observed movement. The market is represented by the exchange-traded fund that tracks the broad United States index, and the sector by the corresponding sector fund, chosen for being the most liquid of their class and for having a long history in free sources.

Three aspects determine the validity of the split.

The first is the orthogonalisation of the sector factor. A sector fund and the market index move in strongly correlated ways, and introducing them jointly in the regression produces unstable coefficients through collinearity. The factor used is therefore not the sector return, but the residual of a first regression of the sector on the market.

The second is the temporal restriction of the estimation. The window ends on the day before the day being explained, by the same discipline applied to detection.

The third is the treatment of noise. A sensitivity estimated over twenty observations is imprecise, and extreme estimates in agitated windows produce meaningless splits. The finance literature documents that estimated betas regress towards the mean between successive periods (Blume 1971), from which followed the practice of shrinking the estimate with a fixed weight. That practice applies the same correction to precise and imprecise estimates. The alternative adopted weights the shrinkage by the imprecision of the estimate itself (Vasicek 1973):

$$\beta = w\hat{\beta} + (1 - w)\,\beta_{\text{ref}}, \qquad w = \frac{\sigma^2_{\text{ref}}}{\sigma^2_{\text{ref}} + \mathrm{SE}^2(\hat{\beta})} \tag{3.3}$$

When the fit is precise, the standard error is small, w approaches unity and the estimate is practically preserved; when the window is noisy, the standard error grows and the estimate falls back on the reference value. The adjustment is gradual and not dichotomous. The original formulation estimates β_{ref} and σ^2_{ref} from the cross-sectional distribution of betas; in this work they are fixed, and each has its own justification. For the market, $\beta_{\text{ref}} = 1.0$ is adopted, which corresponds to the assumption that the stock follows the index in the absence of reliable local information, and is the value towards which the cross-sectional mean of the betas tends by construction of the index itself. For the sector factor, already orthogonalised with respect to the market, $\beta_{\text{ref}} = 0.0$ is adopted as a regularisation choice for no additional sector exposure when the estimate is imprecise. A zero-mean orthogonalised factor does not imply a zero coefficient. The common standard deviation of 0.5 determines how quickly the transition between the sample estimate and the reference value takes place: it is fixed so that a window with a standard error of that same order receives approximately equal weight from the two origins, which places the point of balance within the range of standard errors actually observed in twenty-day windows. Estimating these values from the sample was not done, and the reason is declared: twelve to seventeen companies do not constitute a representative cross-section, so a prior estimated over them would have an empirical appearance without being one.

Three safeguards complete the estimation. Below ten usable observations the regression is not attempted, unit sensitivity to the market and zero to the sector being assumed, a situation that is declared to the user. An estimate whose absolute value exceeds ten is treated as a numerical failure of the window and not as an economic sensitivity, falling back in the same way; the guard is meant for degenerate windows and does not replace the shrinkage, which is the mechanism that handles noise. Finally, the constant term is recomputed over the already shrunk coefficients, since the value returned by the regression is centred on the raw coefficients and reusing it would decentre the model actually reported.

Figure 3.3 presents the split of a real movement, in which the market and sector components act in the direction opposite to the observed movement.

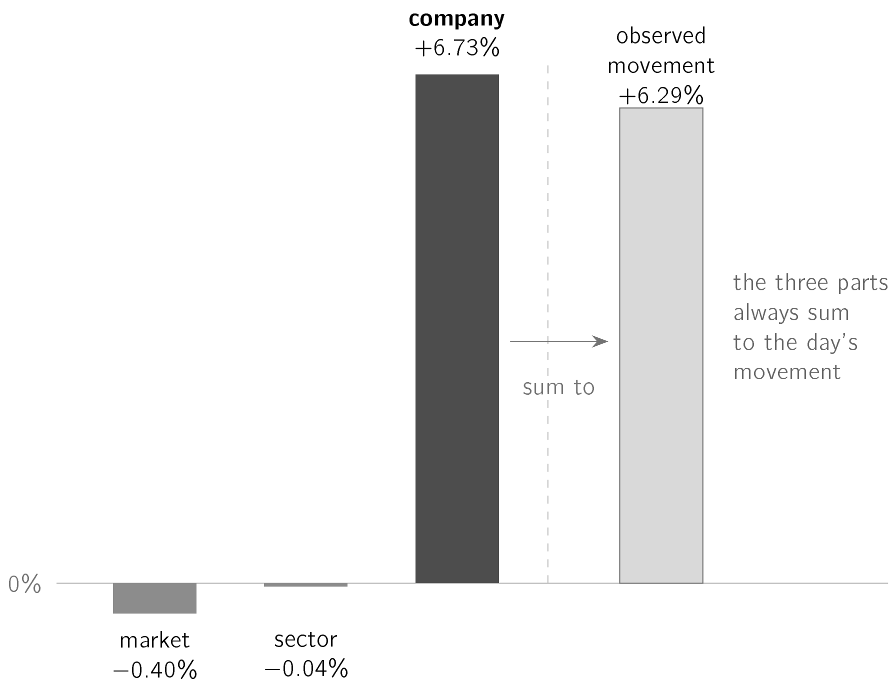

Figure 3.3: Split of an AMD movement into logarithmic returns. The bar dates were not preserved in this evaluation record. The market and sector components acted downwards, so the rise is attributed entirely to the company. The equality between the sum of the parts and the observed movement holds by construction and, for that reason, establishes nothing about the quality of the fit, a point taken up in Section 5.6.

In this AMD case, the logarithmic return of +6.2944%, corresponding to a simple price change of +6.50%, splits into −0.3992% of market, −0.0362% of sector and +6.7298% specific to the company, with $\beta_m = 2.0143$ and $\beta_s = 1.5888$ estimated over the twenty preceding days. The information conveyed exceeds the percentage alone: the stock rose despite the context acting in the opposite direction, and the market beta indicates a stock that historically amplifies the movements of the index.

The exact sum of the three parts does not constitute validation of the method. The equality holds by construction, since the specific component is defined as the residual, and a meaningless split would sum just as well. The informative measure of the quality of the fit is the coefficient of determination in the estimation window, whose value is reported in Chapter 5.

3.6 Precedent retrieval: sentence embeddings and cosine similarity

The third decision consists in identifying past news items comparable with the news item under analysis and measuring the subsequent behaviour of the price. Search by word matching is inadequate for two symmetric reasons: news items with the same meaning may share no vocabulary, and news items with common vocabulary may deal with distinct subjects.

The representation adopted is a sentence embedding model (Reimers and Gurevych 2019), which converts each headline into a 384-dimensional vector whose position in the space reflects the meaning of the sentence. The relevant contribution of this model is not the architecture, but the training objective: the vectors are learned so that the distance between two of them corresponds to similarity of meaning. An encoder of the same family trained for another task does not inherit this property, an observation that Chapter 5 confirms experimentally.

The proximity between two sentences is measured by cosine similarity, that is, by the angle between the vectors and not by the distance between their endpoints, which makes the measure independent of the length of the text. The vectors are returned with unit norm, so the cosine reduces to the inner product, a property that allows the complete case base to be scanned without approximation. On two real pairs from the case base, two headlines about falls in technology and semiconductor companies obtain $\cos = +0.956$, and a headline about Peloton compared with another about the impact of the pandemic on the automotive industry obtains $\cos = -0.086$. The similarity threshold used in production is 0.45.

The ordering of the candidates is not, however, the cosine alone. Cases close in topic but distant in time describe a market regime that is no longer the prevailing one, so the ordering applies an exponential decay over the age of the case, in the form $\cos \times 2^{-\mathrm{age}/h}$, with a half-life h of one hundred and twenty days. A case one year old thus enters with an approximate weight of 0.12 and prevails only in the absence of more recent material on the same topic.

The decay orders and is not displayed: the value shown next to each precedent is always the cosine, since presenting a composite score under the name of similarity would produce a number the user cannot verify. The two decisions compose in series, so the 0.45 threshold applies to a set already selected by the ordering, and not to the neighbours with the highest cosine. Figure 3.4 represents the property this measure exploits: headlines with the same meaning occupy nearby positions even when they share no vocabulary, whereas headlines that share vocabulary and deal with distinct subjects end up far apart.

The measurement of the outcome adapts the event study (Brown and Warner 1985): with the day of the news fixed, the price change is measured at one, three and five days. Two detailed decisions condition the result. News published outside trading days is aligned with the first following trading day. The measurement begins at the close of the day of the news, and not at the open, a conservative option that avoids attributing to the news the part of the movement prior to its publication. Figure 3.5 shows the two decisions.

The value presented to the user is the raw return and not the abnormal return the source normalises. The option is deliberate and has a declared cost. The raw return is verifiable by the user in their own application, but it incorporates the movement of the market over the period. In periods of generalised decline, the precedents appear to have worse outcomes than those attributable to the news. The market is discounted where it is indispensable, in the construction of the label described in the following section.

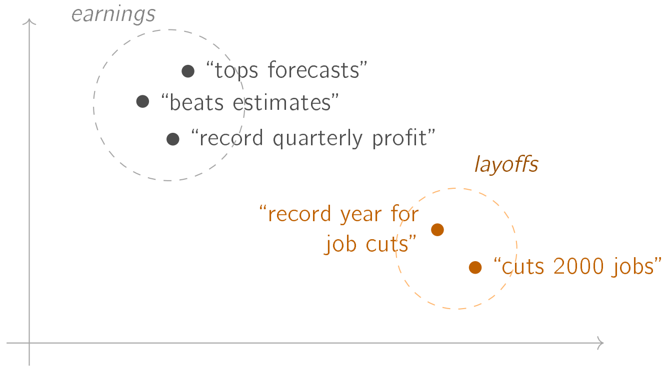

Figure 3.4: Each sentence corresponds to a point in the space. Sentences of close meaning occupy neighbouring positions even when they share no vocabulary, whereas sentences with common vocabulary ("record") and distinct subject end up far apart. The representation is two-dimensional to allow it to be read; the model uses 384 dimensions.

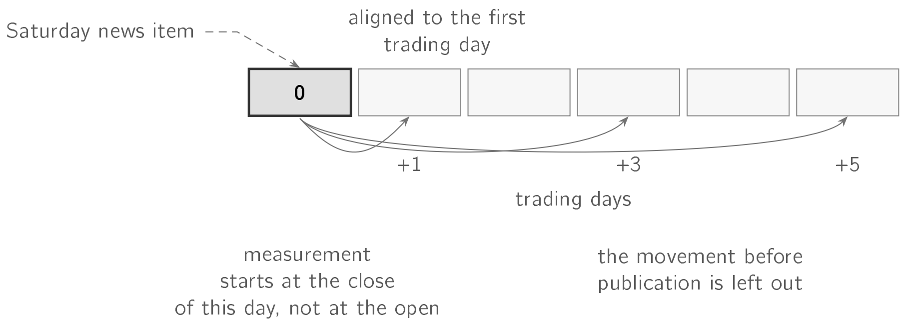

Figure 3.5: The two detail decisions that condition every outcome measured in this work. A news item published outside a trading day is aligned with the first following day on which the market opens, and measurement starts at the close of that day, marked solid. The second decision is conservative by construction: starting at the open would attribute to the news the part of the movement that preceded it.

Listing 3.2 reproduces the comparison as it is executed over the whole case base. The line that decides the behaviour of the system is the last: a record without a usable vector receives a similarity of zero and falls to the bottom of the ordering, instead of interrupting the cycle.

3.6.1 Adjusting the encoder for comparable materiality

The retrieval described above organises the case base by *theme*. A precedent on the same theme may nevertheless have moved by a magnitude very different from the one at hand, and the alert presents it alongside as though the two were comparable. The fourth research question asks whether the encoder can be adjusted so that proximity also reflects comparable materiality, and Section 1.3 justifies why that question does not contradict the no-prediction constraint.

```python
def cosine_similarities(query, matrix) -> np.ndarray:
    """Similaridade do cosseno entre um vetor `query` (1D) e cada linha de `matrix` (2D).

    Vetorizado: devolve um array (n_amostras,) com a similaridade de cada linha ao query.
    Linhas nulas (ou query nulo) recebem 0.0.
    """
    query = np.asarray(query, dtype="float64")
    matrix = np.asarray(matrix, dtype="float64")
    if matrix.ndim != 2:
        raise ValueError("`matrix` tem de ser 2D (n_amostras, dim).")
    if query.ndim != 1 or query.shape[0] != matrix.shape[1]:
        raise ValueError("dimensões incompatíveis entre `query` e `matrix`.")
    qn = float(np.linalg.norm(query))
    row_norms = np.linalg.norm(matrix, axis=1)
    denom = row_norms * qn
    dots = matrix @ query
    out = np.zeros_like(dots, dtype="float64")
    nz = denom > 0
    out[nz] = dots[nz] / denom[nz]
    return out
```

Listing 3.2: The comparison at the core of retrieval, vectorised over the whole case base. The decision that matters is the last one: a null vector receives a similarity of zero rather than raising, because a record without an embedding must fall to the bottom of the ordering and not interrupt the cycle. Reproduced from the source file; the comment is in the original.

The training pairs

The target of a pair of headlines is $1 - |p(a) - p(b)|$, where p is the percentile of the subsequent movement in the distribution of movements of the training block. Two reasons support the percentile rather than the raw difference. The first is that returns are heavy-tailed: half a percentage point does not mean the same thing at the centre of the distribution and in the tail, and a target built on the raw difference would spend almost its whole domain distinguishing extremes from one another. The percentile is invariant to monotone transformations and treats the two regions by the same standard. The second is that the target lies in $[0, 1]$ by construction, which is the domain expected by the loss functions used, with no arbitrary scale to justify.

The percentile map is fitted *only* on the training block and reused on the others. Fitting it on the complete set would constitute information leakage, and leakage of a kind that no boundary check would catch: the distribution of the test block would enter training without a single pair crossing the boundary. The embargo block is always left out, since it exists precisely to separate training from test.

The sampling, and why it cannot be at random

The first construction drew two indices at random. The difference of percentiles between two uniform points is *triangular*, with mean 1/3 and almost no density at the extremes, so the target ends up concentrated around a single value: measured on the data, mean 0,665 and standard deviation 0,237. With such a target, the minimum of the loss is obtained by predicting the mean for every pair, and predicting the mean for everything is, in practice, collapsing the representation. Section 5.4 reports that this is exactly what happened.

The construction adopted draws the desired distance d first and only then the starting point within the margin that d admits, $p_a \sim U(0, 1-d)$, so that $|p(a)-p(b)| = d$ by construction. On the real data the target then has mean 0,499 and standard deviation 0,289, that is, the intended uniform distribution.

The two arms, and what the comparison between them establishes

The main arm trains on the *magnitude* of the movement, taken in absolute value. The replication arm trains on the *signed* return, reproducing in this same harness the direction objective of Du et al. (2024). The comparison between the two is what makes the distinction between magnitude and direction verifiable: were direction as learnable from text as magnitude, the two arms would be indistinguishable.

The extension relative to that work lies in what is adjusted. Du et al. (2024) train a triplet network over text and price series, taking as positive the pair that shares the *direction* label, and keep the sentence encoder *frozen*. Here the encoder is adjusted, and the objective of the main arm is magnitude and not direction.

Configuration and reproducibility

Both arms start from the same `all-MiniLM-L6-v2` used in Section 3.6, so that the comparison with unadjusted retrieval is like-for-like. The loss is *CoSENT*, which orders pairs by graded similarity without imposing a scale on the output of the cosine. Training runs over forty thousand pairs, one epoch, batches of thirty-two examples and a learning rate of 5×10^{-6}, with the seed fixed at forty-two for the generators of `random`, `numpy` and `torch`. The complete configuration is kept alongside the weights.

The two measures, and the diagnostic that precedes them

The measure of the objective is **materiality comparability**: the mean absolute difference, in percentage points, between the magnitude of the query's movement and that of the precedents returned. Lower values are better. The second measure is the **precision@5 by sector** of Section 5.3, and it is present to show what the adjustment costs, or does not cost, in thematic relevance. Measuring by sector alone would be evaluating the main arm by the metric of the arm it is not.

Before either is interpreted, a degeneration diagnostic runs, measuring the mean cosine between *different* headlines, the norm of the mean vector and the standard deviation per dimension, and further how far the space has moved relative to the starting model and whether the geometry of the similarities has been preserved. The reason is that a collapsed encoder produces a plausible and entirely meaningless table of metrics: every query returns the same neighbours, and nothing in the table gives it away.

3.7 News triage: a model learned from labelled examples

The fourth decision determines which news items justify interrupting the user. Unlike the previous ones, it has no obvious analytical formulation, and constitutes the natural point for learning from the data.

The label formalises the question as the occurrence of an abnormally large movement after the news. The return of the stock over the following three days is measured, the movement of the market over the same period is subtracted, and the example is classified as positive when the absolute value of the residual exceeds two per cent. The use of the absolute value is deliberate: the model learns about materiality and never about direction, which preserves the founding constraint of the work.

Subtracting the movement of the market assumes unit sensitivity, which is an inconsistency with the technique described in Section 3.5, which refuses precisely that assumption. In a high-beta stock, the residual retains part of the movement of the market, and the label may classify as material a day on which only the market moved. All the families of model evaluated in Chapter 5 are trained and evaluated against the same label, so the comparison between them is internally consistent under this definition of target. That does not guarantee, however, that the ordering survives an alternative definition, since the error introduced may be correlated with the company and with volatility.

The model receives nine inputs: the volatility of the twenty preceding days, the momentum of the five preceding days, the return of the day itself, the length of the headline and five sector indicators. An additional variant incorporates the 384-dimensional vector of the headline. Figure 3.6 groups the inputs by the level of information each one describes, a distinction that Chapter 5 shows to be decisive.

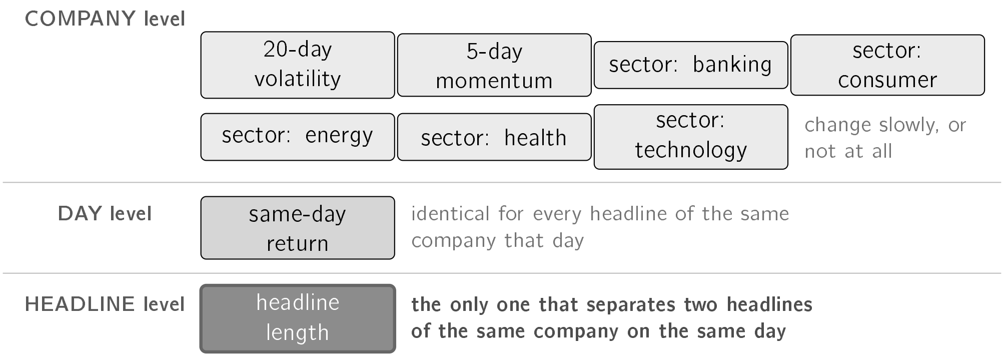

Figure 3.6: The nine inputs of the deployed model, grouped according to what each one allows to be distinguished. Seven characterise the **company** and vary slowly or stay constant; one characterises the **day** and takes the same value for every news item of that day; **only one** varies between consecutive headlines. The ablation in Section 5.5.5 measures the dominant dependence on company; the shared label cannot evaluate distinctions between headlines from the same day.

Seven inputs describe the company and vary slowly or are constant; one describes the day and is identical for every news item of that day; and only one, the length of the headline, distinguishes two headlines of the same company on the same day.

The split of the data is chronological and not random, as shown in Figure 3.7. A random split would allow training on news posterior to that used in the evaluation, inverting the temporal order the system faces in operation. Between consecutive blocks there is a five-day embargo, necessary because the label of a news item depends on the following three days: without the embargo, the label of the news items at the end of a block would be determined by days belonging to the following block. The embargo discards 820 of the 79,753 rows, that is, 1.03%, leaving 28,574 examples for training, 17,710 for validation and 32,649 for testing.

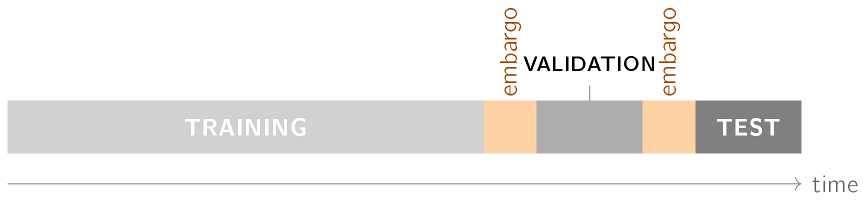

Figure 3.7: The split of the data follows **chronological order** and not a draw. Between consecutive blocks a five-day *embargo* is inserted, since the label of each news item depends on the following three days: without it, the end of the training block would be determined by days belonging to the following block.

The chronological discipline is a claim about what the dataset contains, and a claim of that nature is not established by declaration. It is therefore enforced by an executable procedure, reproduced in Listing 3.3, which builds two series identical up to the day of the news item and divergent after it, and requires the two properties that separate a legitimate input from a leak: the inputs of the model have to remain equal, and the label has to change. An input that came to depend on the future would break the first property; a label that did not depend on it would break the second.

```
def test_anti_lookahead_mutar_o_futuro_muda_o_rotulo_mas_nao_as_features():
    base = [100.0] * 25
    sobe = _serie(base + [100, 110, 120, 130])   # movimento anormal enorme
    plano = _serie(base + [100, 100, 100, 100])  # nada acontece
    mercado = _serie([100.0] * 29)
    d = 25
    assert event_features(sobe, d) == event_features(plano, d)  # features iguais
    assert abnormal_label(sobe, mercado, d, 0.02, 3) == 1  # rótulos diferentes
    assert abnormal_label(plano, mercado, d, 0.02, 3) == 0
```

Listing 3.3: The procedure that enforces the temporal separation. The two series coincide up to the day of the event and diverge afterwards. The inputs have to come out identical, because none of them may observe later sessions; the labels have to come out different, because it is precisely the future that they measure. The test runs on every execution of the automatic verification. Identifiers and comments are in the original.

The test block contains more examples than the training block. The cut is made on trading days, according to a proportion of seventy, fifteen and fifteen per cent, and the asymmetry results from the density of news, which grows markedly over the period covered by the corpus. Splitting by days, and not by number of examples, is necessary so that news items of the same day are not distributed across distinct blocks.

The score produced by the model is not, in itself, a probability. Since the value is presented to the user, it is converted through *post-hoc* calibration, fitting a two-parameter sigmoid over the held-out validation block (Niculescu-Mizil and Caruana 2005):

$$p_{\text{calibrated}} = \frac{1}{1 + e^{-(a\, p_{\text{raw}} + b)}} \tag{3.4}$$

The learned values are $a = 3.700$ and $b = -2.313$. The negative sign of b indicates that the model, without calibration, produces values systematically above the observed frequency. Part of this behaviour follows from the reweighting of the classes during training, which shifts the scores upwards by construction.

The two parameters are not chosen by hand. Listing 3.4 reproduces the fitting as it is executed: a logistic regression of one variable over the held-out validation block, which is what makes the two values above reproducible rather than asserted.

```
class PlattCalibrator:
    """p_calibrada = sigmóide(a·score + b), com (a,b) ajustados na validação."""

    a: float
    b: float

    def __call__(self, scores: np.ndarray) -> np.ndarray:
        z = self.a * np.asarray(scores, dtype="float64") + self.b
        return 1.0 / (1.0 + np.exp(-z))


def fit_platt(scores_val: np.ndarray, y_val: np.ndarray, seed: int = 42) -> PlattCalibrator:
    lr = LogisticRegression(max_iter=1000, random_state=seed)
    lr.fit(np.asarray(scores_val, dtype="float64").reshape(-1, 1), y_val)
    return PlattCalibrator(a=float(lr.coef_[0][0]), b=float(lr.intercept_[0]))
```

Listing 3.4: Where a score becomes a probability. The two parameters are not chosen: they are fitted by a logistic regression over the held-out validation block, which is what makes the value in Equation 3.4 reproducible. The class and the fitting function are reproduced from the source file; the comment and the identifiers are in the original.

The calibration is fitted on a block whose prevalence of positive cases differs from that of the test block, a consequence of the chronological split. The effect is measurable and is reported in Chapter 5. The transformation is monotonic and preserves the ordering between news items, which is what the decision policy uses; it is the absolute value presented that is affected.

3.8 Tools and infrastructure

The system is written in Python 3.12. The dependencies that produce the results of this dissertation are pinned to exact versions, and not to ranges, so that the environment can be recreated on another machine: `numpy` 2.1.3, `pandas` 2.2.3, `scikit-learn` 1.9.0, `matplotlib` 3.11.0, `sentence-transformers` 5.6.0 and `torch` 2.12.1 in the CPU variant. No result requires a graphics processing unit, a consequence of the options described in this chapter.

Triage training, calibration and resampling by company and day use seed 42. The five retrieval repetitions, including the evaluation with temporal precedence, use seeds 42 to 46. The random baseline in the budget analysis uses 40 seeds, from 0 to 39. These choices fix the random operations of each protocol; they do not make datasets collected at different times identical.

All data sources are free. Prices come from a chain of alternative providers consulted in a fixed order, since no free source is sufficiently reliable on its own; the system always records which one answered. News comes from three complementary providers, whose comparison is presented in Chapter 4. Delivery uses the Telegram bot interface.

3.9 Technological and social challenges

This section states the risks that a system with these characteristics raises and the design decisions that respond to each of them.

3.9.1 Privacy and data protection

An investment portfolio reveals income, risk aversion and life plans, information that the General Data Protection Regulation treats as deserving reinforced protection when it allows a person to be profiled.

The design decision that responds to this risk is minimisation: the system does not know the users' positions and does not ask for them. There is no portfolio, there are no positions and no amounts, and the public channel does not identify recipients. An alert about a company is identical for every reader. The cost of this option is the impossibility of personalisation, since the system alerts on a fixed list of companies and not on the positions of whoever reads it.

There is one exception worth naming. Users who choose to receive personalised alerts through the Telegram bot are associated with two elements: the conversation identifier assigned by the platform and the list of companies selected. No name, contact or amounts are stored. The processing rests on the express request of the user, who begins it by subscribing and ends it through a command that deletes the record.

3.9.2 Security

An autonomous system that consumes external services holds credentials. The keys reside in environment variables and in the platform's secret store, never in the repository, and the text sent to the logs passes through a function that masks parameters with a credential name. This second safeguard responds to the fact that the error messages of communication libraries reproduce the address of the request, and that some interfaces carry the key in that address.

The sentence encoder is fetched on demand and checked against a checksum fixed in the code. Without that verification, a corrupted or substituted file would silently modify the similarity criterion used by the system.

3.9.3 Ethical and social questions

The founding constraint of the work is also its main ethical safeguard. A system that announced the future behaviour of the price would issue a forecast not verifiable at the moment of reading; a system that indicated instruments to buy would enter a regulated activity. This boundary is one of design, drawn out of prudence and not from an analysis of the applicable law: there was no legal opinion, and operating the system as a service would require obtaining one first.

The guarantee applies to what the system writes and not to what it reproduces. Third-party headlines are quoted in full, since they constitute the fact that triggers the alert and allow verification at the source, and some are themselves speculative. From this follows a limitation without a solution within the scope assumed: the relevance filter determines whether a headline is about the company, and not whether it is factual. The distinction between news and opinion would require a classifier the system does not have.

Repeated interruption is the second risk. The effect is documented in clinical decision support systems, where Ancker et al. (2017) measure that the probability of accepting an alert decreases as the volume of alerts received increases, and that the effect is accentuated by repetition of the same alert. The domain is distinct and what transfers is the mechanism: whoever is interrupted frequently stops distinguishing the interruptions, from which point a correct warning and an incorrect warning have the same effect. The system responds with a daily alert budget, a rising threshold for repeated alerts of the same company, and suppression of news that reproduces a story already communicated.

The third risk follows from the explanation itself. Trust in automated systems fails in both directions, with the user ignoring a correct warning or accepting an incorrect one (J. D. Lee and See 2004), and the presence of an explanation increases the acceptance of wrong answers (Bansal et al. 2021). The system therefore presents past cases individually, with date and outcome. The average that accompanies the cases in the alert is descriptive and cannot anticipate the outcome of the current news item. That this option produces appropriate trust is a hypothesis and not a result, since the controlled study needed to verify it was not carried out.

3.9.4 Use of artificial intelligence tools

This subsection declares, as the Statement of Integrity refers, how generative artificial intelligence tools were used in this work and for what purpose.

The conception is the author's: the problem, the research questions, the constraints that define the system (free sources only, explainability first, and the refusal to forecast prices), the design of each experiment, the success criterion fixed before running it, and every decision about what to build, keep or discard. No conclusion in this document was delegated.

The execution was assisted, and to a substantial extent. Those tools wrote and restructured a considerable part of the system's code and of its tests, implemented the evaluation procedures, drafted and edited prose in this document, and conducted reviews that found defects in the software and in the text. Where a measurement contradicted a claim of the author, the claim was withdrawn rather than defended, and Appendix A records those that were withdrawn.

The content of this document was reviewed by the author. It, the errors that remain in it, and its defence are his responsibility.

Chapter 4

Implementation: InvestiGator

This chapter describes the realisation of the system: its architecture, the way each method of Chapter 3 is put into effect, the policy that decides what is communicated to the user, the construction of the message delivered, and the life cycle of the learned component. Whoever reads it will know the path of a news item from collection to message, and why nine in every ten candidates are never communicated.

Chapter 3 presented four methods in isolation. Composing them into a working system introduces problems that none of them presents on its own: the unavailability of the sources, the volume of candidates per cycle, the coherence between the values computed and the text presented, and the silent degradation of the components over time. The description that follows is organised around those problems.

The system developed is called InvestiGator and is in continuous operation, delivering alerts to a messaging channel and making the record of its decisions available in a web application. The values presented in this chapter were read from the log of the system in operation, and the dates on which each measurement was taken are indicated alongside each one.

4.1 System architecture

The architecture of InvestiGator was defined according to the following requirements, concerning cost, modularity, verifiability and resilience:

- to operate exclusively on free data sources, with no subscriptions and no per-request costs;
- to isolate each method in a component with a single responsibility, replaceable without changing the others;
- to guarantee that any value presented to the user originates in the component that computed it, and not in a second independent derivation;
- to tolerate the unavailability of any external source without interrupting the service;
- to record every decision taken, including the decisions not to communicate.

The system is organised into five components, shown in Figure 4.1: detection, retrieval, triage, explanation and delivery. The decomposition of the movement is a sixth component, activated only on the path started by a price movement, since the split applies to the return of the day and there is no object to split when the event that triggers the assessment is a news item.

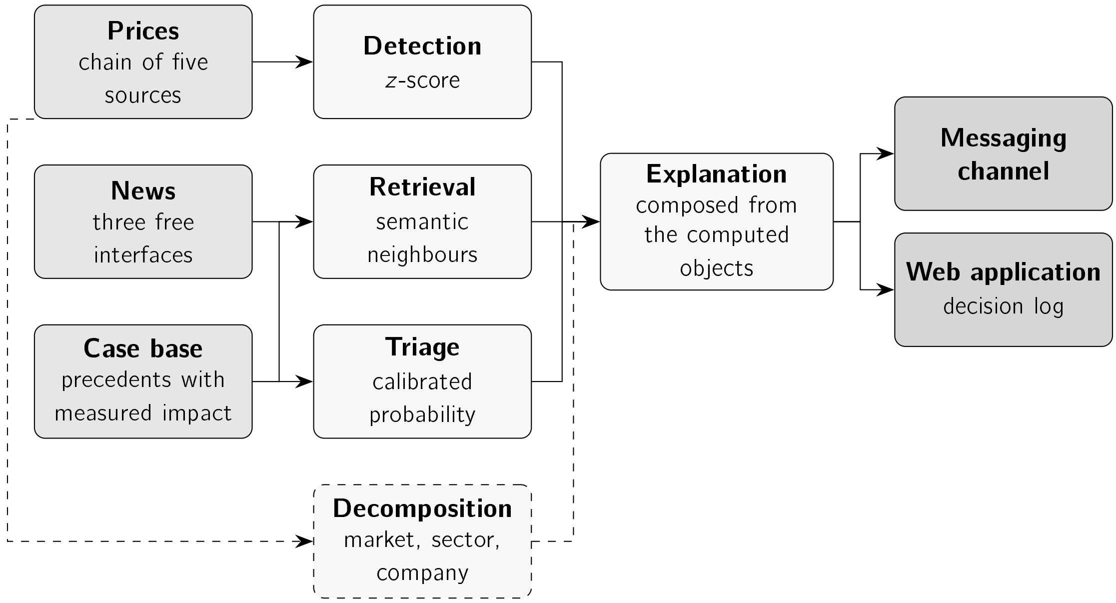

Figure 4.1: Components of the system and their connections. On the left, the data sources; in the centre, the methods of Chapter 3 and the component that composes the text; on the right, the two delivery channels. The decomposition is dashed because it is the only component that does not take part in every assessment.

One construction rule runs through the whole implementation: the text delivered to the user is composed from the objects the components produced, and not written independently. When the detector computes a z value equal to $+7.61$, it is that value the message presents. The alternative, which would consist in keeping one place where the text is written and another where the quantity is computed, admits divergence between the two without that divergence being observable. The constraint eliminates that possibility by construction.

4.1.1 Communication between components

The components communicate by direct passing of objects within a single process, and not by network protocol. The decision follows from the usage profile: the system serves a small list of companies, with a sixty-second assessment cycle, and the volume does not justify the coordination cost of a distributed architecture. The separation between components is ensured by explicit interfaces, which preserves replaceability without introducing communication latency.

Two exceptions operate across process boundaries. Delivery uses the bot interface of the messaging channel, by HyperText Transfer Protocol (HTTP) request. The web application that exposes the decision log runs as a separate process and reads the same data store, without writing to it. This second separation guarantees that consulting the log does not interfere with the assessment cycle.

4.2 Data acquisition

The zero-cost constraint determines the acquisition decisions. A retail investor who considers brokerage costs high does not subscribe to a financial information terminal, and the usefulness of the system for the recipient defined in Chapter 1 depends on the system operating without that cost. Free sources present, however, three systematic limitations: they block access for excess of requests, they answer with an error without warning, and they publish late. The decisions described in this section respond to those limitations.

4.2.1 Quotes

Prices are obtained through a chain of five sources, consulted in a fixed order, the first that answers with valid data being adopted. The identifier of the source that served each value is recorded together with the value, since the provenance of a quantity is part of its explanation.

The order of the chain is neither arbitrary nor definitive. One of the sources was demoted after it began answering with an anti-robot check, which makes it unusable in a reliable way in automated operation; it remains in the chain, in a lower position, as an opportunistic attempt. The decision resulted from observing what the source returned, and not from declared preference.

The redundancy of the chain has a consequence that goes beyond availability. Without it, a day on which the single source blocks access is indistinguishable from a day on which the market did not open, and the system has no way of separating the two situations.

4.2.2 News

News is obtained per company, from three free interfaces. The material received includes a high volume of unusable content: automatic lists of the day's largest movers, texts that mention the company only in passing, and press releases from law firms. A relevance filter, applied on entry, requires the headline to name the company explicitly and rejects the known patterns of automatically generated text.

The use of three sources rather than one was assessed by measurement, over twelve companies and three days. The criterion adopted is not the number of headlines returned by each source, but the number of headlines that survives the relevance filter, since material rejected on entry never comes to exist for the system. Table 4.1 presents the result.

Table 4.1: Material delivered by each free source after the relevance filter is applied. The *exclusive* column indicates the headlines that one source brings and none of the others brings. That value is an upper bound, since the comparison between headlines is made by textual matching after normalisation, and two sources that describe the same event in different words produce two headlines counted as distinct.

Source	**Relevant**	**Precision**	**Freshness**	**Coverage**	**Exclusive**
Finnhub	432	**35%**	15.8 h	**12/12**	401
Alpha Vantage	141	24%	**9.3 h**	**12/12**	119
Polygon	429	27%	52.6 h	8/12	**418**

The three sources have complementary profiles, and it is that complementarity, and not the aggregate volume, that justifies using them together. Finnhub has the highest labelling precision and covers all the companies. Alpha Vantage has the lowest median age of the news, at 9.3 hours against 15.8, which makes it the relevant source for reducing the delay of the alerts. Polygon contributes the largest number of exclusive headlines, with a median age of 52.6 hours, which makes it suitable for feeding the case base and unsuitable for alerting in useful time. That is the use assigned to it.

The three sources together produce 970 distinct relevant headlines, against the 432 obtained with Finnhub alone. The gain concentrates on the companies with the lowest individual coverage, which are precisely the cases in which the system previously had no material to comment on.

Aggregation does not, however, guarantee a reduction in latency in general. The three sources publish with delays of their own, and the gain in a concrete case depends on which of them observed the event first. The median freshness improves, and that improvement is measured; the claim that alerts now arrive earlier would require a new end-to-end latency measurement over an extended period, which was not carried out.

4.2.3 Historical base and deployed base

Precedent retrieval requires a body of past cases with a known outcome. The system uses two distinct sets, with distinct purposes, and their separation matters for reading the results of Chapter 5.

The first is FNSPID (Dong, Fan, and Peng 2024), a public set of financial news aligned with the corresponding prices. From the alignment described in Section 3.2 result 79,753 examples between 2018 and 2023, each with the subsequent price movement. It is on this set that the methods are evaluated.

The second is the case base the deployed system consults, and it results from merging two origins that should not be confused. The first is a reconstruction of one year of history, obtained by running the collection and maturation components over the past, with 38,214 cases; the separation from the log of the running system is deliberate, since a replayed case is not an observed case. The second is the live base, fed by the scan, with 11,445 cases as of 20 August 2026, and growing with operation. Retrieval scans both together. The vectors are stored as a single-precision matrix mapped on disk. The format is not indifferent: the same representations in the language's native structures exceeded the memory available in the execution container, and the change of format was the condition that made their use possible.

4.3 The path of an event

The system admits two distinct paths, depending on whether the event that triggers the assessment is a news item or a price movement. The two converge in the explanation component and use the same decision policy, described in Section 4.4.

4.3.1 Path started by a news item

The path started by a news item comprises nine stages, shown in Figure 4.2. Five of those stages are decision points at which the candidate can be discarded, in which case the system does not communicate.

The operation of the path is illustrated by an alert delivered on 12 July 2026, concerning Meta, with the headline *"Mark Zuckerberg Said Meta's AI Bets 'Haven't Come to Fruition*

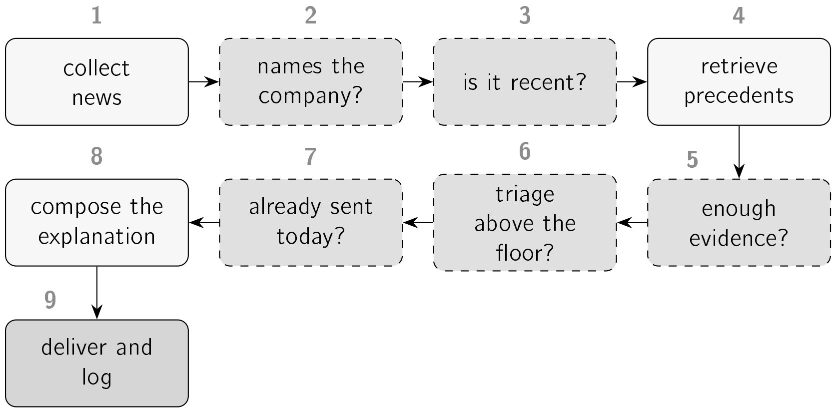

Figure 4.2: The nine stages of the path. The dashed boxes correspond to decision points at which the candidate can be discarded. Section 4.4 quantifies the proportion of candidates eliminated at each one.

Yet' as Shares Fell 5%". The headline names the company and matches none of the automatic text patterns. The date falls within the two-day interval required. The three closest precedents are Microsoft of 2023-11-07, with similarity 0.46, Nvidia of 2023-08-02, with 0.51, and Nvidia of 2023-09-21, with 0.46; the best of them is above the 0.45 threshold. The triage model assigns a probability of 54%. And the alert is the first concerning the company on that day. The message is then composed and logged.

The precedents retrieved in this case belong to companies distinct from the one the alert concerns, and are presented nonetheless. The option is deliberate and follows from the formulation of the second research question: comparability is established by the topic of the news and not by the identity of the company. News items concerning investment in artificial intelligence at large technology companies are cases comparable with one another.

The outcomes of the three precedents diverge, with movements of +2.70%, −3.87% and +5.05%. The system presents the three individually, alongside their range and average. The average summarises only the displayed cases; on its own, it would hide the absence of a common outcome for similar cases and would not support a forecast for the current news item.

4.3.2 Path started by a price movement

The second path is substantially shorter. The z-score is computed by two routes. After the close of each session, the system assesses the return of the closed day of each company on the watched list against the norm of the twenty preceding days, as described in Section 3.4. During the session, it assesses the return in progress, obtained from the current quote against the previous close, against the same daily norm, formed in that case by the twenty most recent complete days, since the day in progress does not yet belong to the series. The second route brings the detection forward without changing the definition of normality, and is what allows a movement to be communicated while the session is under way; it is also the one that does not depend on the historical series source, whose bar for the day may arrive late. On either route, crossing the threshold is a sufficient condition for assessment.

It is on this path, and only on this one, that the decomposition of the movement intervenes. A line is added to the alert with the split computed according to Section 3.5, in the form

"NFLX -2.11% *today* $= -0.08\%$ *market* $\cdot$ -0.32% *sector* $\cdot$ -1.71% *company itself"*, followed by a sentence that names in plain language the dominant part. When estimating the sensitivity to the market is not possible, the line is presented all the same, declaring that unit sensitivity was assumed and that the split is indicative.

Two capabilities were added to the path after the system entered operation, both from observing its behaviour in production. The first consists in assigning severity levels: a z value of 1.6 and a value of 3.2 came to be described by distinct terms, instead of producing the same wording. The second consists in comparing with the remaining companies of the same sector in the same session, since an alert that reports a fall without indicating that the whole sector fell conveys incomplete information.

The second correction resulted from reading the first thirty alerts delivered. In nine of them the text was internally contradictory: one line indicated that the movement appeared to be common to the sector, and the split presented two lines below attributed the largest part to the company. The two lines were correct in isolation, and the contradiction resulted from the comparison with peers considering only the direction of the movement of the neighbouring companies, and not its magnitude. The comparison came to consider both.

4.4 Decision policy

Communicating every candidate assessed is equivalent to communicating nothing. A channel that delivers fifteen messages a day stops being read, and the system loses the usefulness that justifies its existence. The decision points of Figure 4.2 respond to this constraint.

4.4.1 Observed selectivity

The joint effect of the decision points is considerable. Over the news dated between 1 and 3 September 2026, the system assessed 743 distinct headlines from twelve companies and delivered 15 alerts, which corresponds to a ratio of roughly one in fifty. The window is short, and the decision log that supports it keeps three days, which fixes its limit. Figure 4.3 presents the distribution by company.

The budget is, in the configuration in operation, the constraint that actually binds. Over the production decision log, and on each of the six days measured, between 77.7% and 99.8% of the assessments end there, against a maximum of 4.6% at the similarity threshold and of 14.0% at the same-day repetition check. The reading is not that the remaining gates are useless: it is that the reduction in volume that matters has already occurred before them, in the relevance filter and in the selection of one headline per company per cycle, and that from there on the governed quantity is the daily total. It is also the direct consequence of the policy change described in Section 5.7: the model ceased to exercise a veto and came to rank within the budget, so the budget being the active constraint is the policy behaving as it was designed, and not a departure from it.

Two clarifications condition the reading of these numbers, and the second is the more important. The first is that the unit is the distinct headline: the system re-assesses the same headlines at every cycle, so the log contains 1,321 rows for the 743 headlines, and counting assessments would inflate the funnel. The second is that the total of fifteen alerts does not measure the raw material available, but the policy: the daily budget of five was used in full on all three days, so fifteen is the ceiling the policy imposes and not an observed quantity. What the distribution by company measures is the ordering within that ceiling, and that is not imposed: no company has a quota of its own.

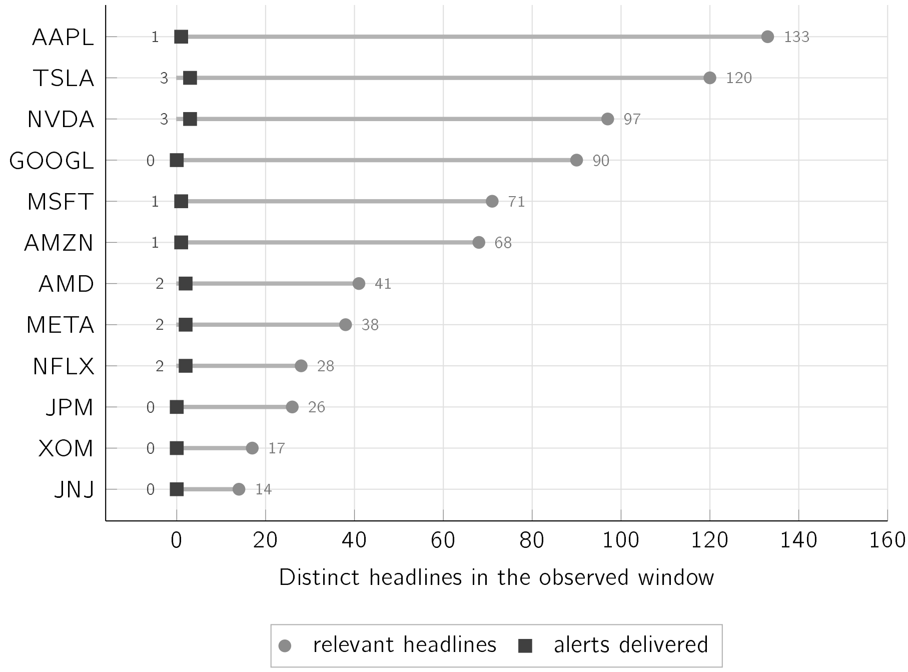

Figure 4.3: Distinct headlines assessed and alerts actually delivered, by company, between 1 and 3 September 2026. The total is 743 headlines and 15 alerts, and the two quantities are counted over the same window. The total of alerts is the ceiling the daily budget imposes, and not a measurement; what the figure shows is the distribution of that ceiling, which the policy does not fix. Eight of the twelve companies receive at least one alert, and the company with the most headlines assessed is not the most alerted. Section 5.7 examines the effect of this policy over a much larger window.

The behaviour on a single day makes it possible to locate where candidates are eliminated. Figure 4.4 distributes the 5,060 assessments recorded on 15 August 2026 by the decision point at which each one ended. The log read corresponds to part of the day and not to the complete day, so the values should be read as proportions.

Reading these values exposes an instrumentation defect. The 333 assessments that crossed every decision point do not correspond to 333 distinct candidates approved for delivery. The system re-assesses the same headlines at every sixty-second cycle, and a headline already delivered that day crossed every decision point again at every cycle. Delivery was prevented at the end, by a duplication check that, unlike all the others, was not recorded as a stage.

The consequence matters for the purpose of the log. The view built to make the system's decisions inspectable reported 333 approvals on a day on which five messages were delivered. The duplication check became a stage with an identifier of its own, like the rest, and there is an automatic test that fails if it is omitted again. The values of Figure 4.4 predate that correction.

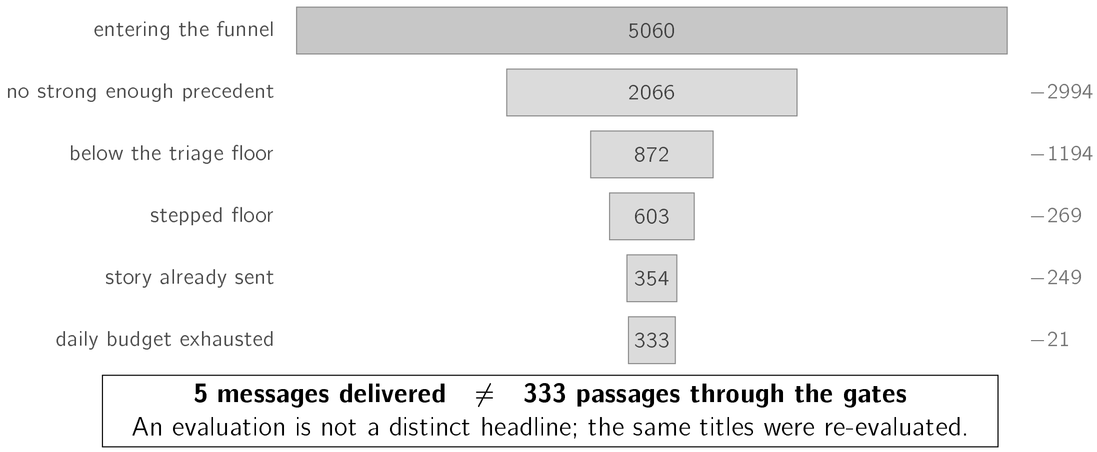

Figure 4.4: The 5,060 assessments recorded during part of 15 August 2026. Each band represents what remains after the gate indicated, and the value on the right what that gate eliminated; the eliminations sum to 4,727 and leave 333 of the 5,060 assessments. The five messages delivered are a separate count, not a category of assessments. The daily budget was used in full. The date is also that of the day on which the triage floor ceased to exercise a veto and came to rank, so the band of 1,194 assessments eliminated at that point belongs to the configuration prior to the one described in Table 4.2; on the following days that point stops eliminating candidates. The same applies to the stepped-floor band, which Figure 4.5 records at zero over the six later days.

4.4.2 Thresholds and their provenance

Each decision point uses a constant, and the origin of those constants is heterogeneous. Table 4.2 distinguishes the values derived from an analysis from the values chosen, a distinction that matters because the two cases admit defences of a different nature.

The similarity threshold requires separate treatment, since it is the least grounded of the seven. In a scan measured in July 2026, when the watched list had ten companies and the daily budget did not yet exist, seven candidates were eliminated at this point, four of them by differences of 0.04 or less, and it was then the point that eliminated the most. It ceased to be: under the policy in operation, the order of magnitude of what each point eliminates is that of Figure 4.5. The hypothesis that higher similarity values correspond to more informative precedents was tested, and the hypothesis is not confirmed: the agreement in direction between the past case and the case under assessment is at the level of chance on both sides of the threshold. The threshold controls the volume of alerts and ensures thematic coherence, and does not select cases with greater predictive value.

4.5 Construction of the explanation

The message delivered is not a single sentence, but a structure with four parts, always present and always in the same order:

1. the observed fact, expressed in values, with the source indicated;
2. the past cases retrieved, presented individually, with date, similarity and subsequent movement;

Table 4.2: Constant associated with each decision point and the origin of its value. The point that checks whether the headline names the company uses no constant, since it corresponds to a textual matching rule. The daily budget acts after the decision points shown in Figure 4.2 and constitutes, in the configuration in operation, the mechanism that actually limits the volume.

Decision point	**Value**	**Provenance**
Unusual movement	$\lvert z \rvert \geq 1.5$	Chosen, and distinct from the value of 3.0 under which the detector is evaluated in Chapter 5. The evaluation fixes a demanding value in order to characterise the method; the deployment adopts a more permissive value as a product decision, offset by the severity levels that distinguish 1.5, 2.0 and 3.0 in the wording presented. The two values answer different questions.
Names the company	—	Textual rule, defined after inspecting the material that was crossing the initial filter.
Is it recent	2 days	Chosen, so as to cover a weekend.
Enough evidence	0.45	Chosen. It eliminated most candidates under the previous configuration; the daily budget now dominates.
Triage above the floor	50%	Derived from a cost analysis, which produces 0.49 under the condition of symmetric cost between an uncommunicated movement and a false alarm; the deployed value is the rounding of that result. In the configuration in operation it exercises no veto: it ranks the candidates of the same cycle.
Daily budget	5 per day	Chosen. The five slots are assigned in order of arrival and not to the five highest-scoring candidates of the day, since the decision is taken cycle by cycle, without knowledge of later candidates. It is additional to a limit of two messages per company per day, which remains in force and is checked after this one. The two answer distinct problems: the per-company limit prevents saturation by a single name, and the global budget prevents twelve companies at two messages each from producing twenty-four interruptions in a day.
Stepped floor	second alert: 0.64	The configuration keeps the derived values 0.49 and 0.64, and the first is explicitly ignored when the global budget is active: the first daily alert of each company has no triage floor. The mechanism thus acts only as a limiter of repetition within the same company.

DECISION POINT	VALUE	ORIGIN	OF WHAT IT ELIMINATES
Names the company	`rule`	chosen	before the log
No news at all	`--`	chosen	0 to 3.9%
Is it recent	`2 days`	chosen	0 to 3.5%
Enough evidence	`0.45`	chosen	0 to 4.6%
Triage above the floor	`50%`	derived	no longer vetoes
Same story	`--`	chosen	0 to 1.8%
Already sent today	`--`	chosen	0 to 14.0%
Stepped floor	`0.64`	derived	0% over the six days
Daily budget	`5 per day`	chosen	77.7 to 99.8%

The point that eliminates almost everything is one of the chosen ones, and it is so on every day measured. Of the two whose value was derived from a cost analysis, the triage floor ceased to veto on 15 August 2026 and the stepped floor remains armed and eliminated nothing on any of the six days measured, because the budget runs out before a company reaches a second alert. The relevance filter acts before a stage log exists, so the fraction it eliminates is not observable here.

Figure 4.5: Each decision point, the origin of the value it uses and the fraction of assessments that ends there, measured over the production decision log across six days, between 19 and 21 August and between 3 and 5 September 2026. The column presents the range across those days and not a single value, since the log keeps only three days at a time and the fractions vary with the day. The unit is the assessment and not the news item: the system re-scores the same headlines at every sixty-second cycle, so these values measure where the system's time is spent and not how many distinct stories each point eliminated. The distinction between a derived value and a chosen value is the same as in Table 4.2. The nine include the five shown in Figure 4.2 plus four that act outside it: the absence of news, the suppression of repeated stories, the stepped floor and the daily budget. The reading the figure allows, and that the table alone does not give, is that the reduction in volume is not made by the points whose value was derived: it is made by the budget, whose value was chosen, and so it is on each of the six days. The values come from the decision log published with the system.

3. the caveat that the cases presented are observed history and not a forecast;

4. the estimated probability of a market reaction, with no indication of direction.

The order responds to two constraints. The fact precedes the statistic because an alert that opens with a z value loses the reader on the first line, and the recipient defined in Chapter 1 does not have the training that would make that value immediately interpretable. The estimate occupies the last position because it is the only quantity in the message that refers to the future, which ensures that a reading interrupted before the end retains facts and history.

On the path started by a price movement, the first part also carries the statistical characterisation in plain language, since in that case the observed fact is the movement itself. On the path started by a news item, that position is occupied by the headline and the link to the article.

The number of precedents presented is likewise a design decision. The system displays three of the cases retrieved rather than all of them, and the reason is not available space. Research on explanation establishes that people look for selective and contrastive reasons, and not exhaustive enumerations of the factors that contributed to an outcome (Miller 2019). A list of thirty cases would be more complete and less usable. The same principle determines that the message displays the quantities that support the decision and not the full computation that produced them.

Figure 4.6 reproduces an alert exactly as it was delivered, copied from the channel log without editing. The headline it quotes is reproduced verbatim, since translating or paraphrasing quoted material would break the verifiability that the system exists to preserve.

```
[1] News alert for AMZN (Amazon) (2026-08-13)
  "Prediction:  Amazon Will Join Apple in the $4 Trillion Club Before 2030"
[2] 3 similar past headlines.  Their 5-day move ranged -2.73% to -2.73%
  (average -2.73%):
  -2.73% in 5d · AAPL 2026-08-05 (8d ago) · "If You Invested $100 In Apple
    Stock 10 Years Ago, You Would Have This Much Today" (sim 0.56)
  -2.73% in 5d · AAPL 2026-08-05 (8d ago) · "Apple's Selloff Is Overdone"
    (sim 0.51)
  -2.73% in 5d · AAPL 2026-08-05 (8d ago) · "Apple's Buyback Is Retiring
    Less Stock Just As The Memory Bill Grows" (sim 0.47)
[3] 3 of 3 shown cases moved down - topic-similar past cases, not a
  prediction for this news (an observed pattern, not a forecast).
  Observed past outcomes after similar news, not a price prediction and
  not advice.
[4] Materiality estimate (learned triage):  57% chance of an unusually
  large move over the next few days, in EITHER direction - raised by
  recent volatility (20d) and sector; lowered by today's own move and
  recent momentum (5d).  This estimates whether the market reacts, never
  which way:  it is not a price forecast and not advice.
```

Figure 4.6: Alert delivered on 13 August 2026, reproduced from the channel log, with typographic markers suppressed and the numbering of the four parts added; no number, word or punctuation mark was altered. The numbers in square brackets identify the four parts of the structure. Every value comes from the component that computed it: the similarity, from the retrieval engine; the five-day movement, from the impact measurement; and the probability of 57%, from the calibrated model. Part [1] contains a headline that is itself a forecast, with a target and a horizon; it is third-party content reproduced literally, and Section 3.9.3 discusses the reach of the no-forecast guarantee in that context. Part [2] displays the similarity as a raw value, faithful to what the engine computed and not interpretable by whoever reads it.

The probability presented on the last line is not the raw score of the model. It results from its conversion into a probability, by fitting a sigmoid over a held-out validation block (Niculescu-Mizil and Caruana 2005). The distinction is operationally relevant: a raw score orders the candidates without admitting interpretation in isolation, whereas a calibrated probability establishes that, among the headlines to which the system assigns 57%, approximately that fraction does show an unusual movement. It is the difference between a quantity usable only internally and a quantity that can be presented.

The guarantee has declared limits. Section 5.5 reports the mean error of this probability and records that the calibration is fitted on a block whose prevalence does not coincide with that of the block to which it is applied, which shifts the scale by about five percentage points in the optimistic direction. The ordering between candidates is not affected, since the transformation preserves the order; the absolute value announced is.

The last line of the alert does not merely present the probability: it names the inputs that raise it and those that lower it. That enumeration is not drafted, it is read off the calculation itself. Listing 4.1 shows the operation that produces it. In a logistic regression with standardisation, the logit is the sum of the constant term with the product of each coefficient by the standardised value of the corresponding input, so each term of that sum is the exact contribution of one input, and not an approximation obtained over the model after training. The hundreds of dimensions of the headline vector are summed into a single term, so that the message remains legible; the context inputs appear one by one. It is from this ordering that the words *"raised by recent volatility (20d) and sector; lowered by today's own move"* of Figure 4.6 come.

```
scaler = pipeline.named_steps["scale"]
lr = pipeline.named_steps["lr"]
x_scaled = scaler.transform(np.asarray(x_row, dtype="float64").reshape
    ↪(1, -1))[0]
contribs = lr.coef_[0] * x_scaled

grouped: dict[str, float] = {}
for name, c in zip(feature_names, contribs, strict=True):
    if name.startswith("emb_"):
        key = "headline content"
    elif name.startswith("sector_"):
        key = "sector"
    else:
        key = _FRIENDLY.get(name, name)
    grouped[key] = grouped.get(key, 0.0) + float(c)
return sorted(grouped.items(), key=lambda kv: -abs(kv[1]))
```

Listing 4.1: The decomposition that the last line of the alert states. The product of the coefficient vector by the standardised vector gives the contribution of each input to the logit; the terms are grouped and ordered by decreasing magnitude. The explanation presented to the user is this sum, and not a second model fitted over the first. The internal documentation of the function is omitted.

4.5.1 Disagreement between the claim and the evidence

The alert of Figure 4.6 presents three precedents with an identical impact of −2.73%, and an identical date. The coincidence is not accidental: the impact is measured per pair formed by company and day, so three headlines of the same company on the same day share the same value by construction. The wording *"3 of 3 shown cases moved down"* presents as agreement between three cases what is one day observed three times.

The extent of the problem was measured over the 247 news alerts with precedents delivered between 11 July and 13 August 2026. In 36.8% of them the cases displayed rest on a number of distinct days lower than the number of cases presented, and in 11.3% the three cases correspond to the same day. Of the 120 alerts that claim unanimity of direction, 23.3% rest on a single day.

The effect is circumscribed. No metric reported in Chapter 5 is affected, since precision at k counts headlines and direction agreement is measured pair by pair over the historical corpus. What is affected is the strength the alert claims before the user, and Chapter 6 records the fact as a limitation.

4.5.2 Web application

The same information is made available in a web application, which allows what was delivered and, above all, what was not, to be consulted. It is in that application that the decision log described in Section 4.4 resides. The design of the interface is not a contribution of this dissertation and is therefore not described; presenting it is justified by its being the only place in which the system's absence of communication is observable. Figures 4.7, 4.8, 4.9 and 4.10 present, in that order, the state of the day, the evidence for one company, the arithmetic behind that evidence, and the record of the decisions taken about one company's news.

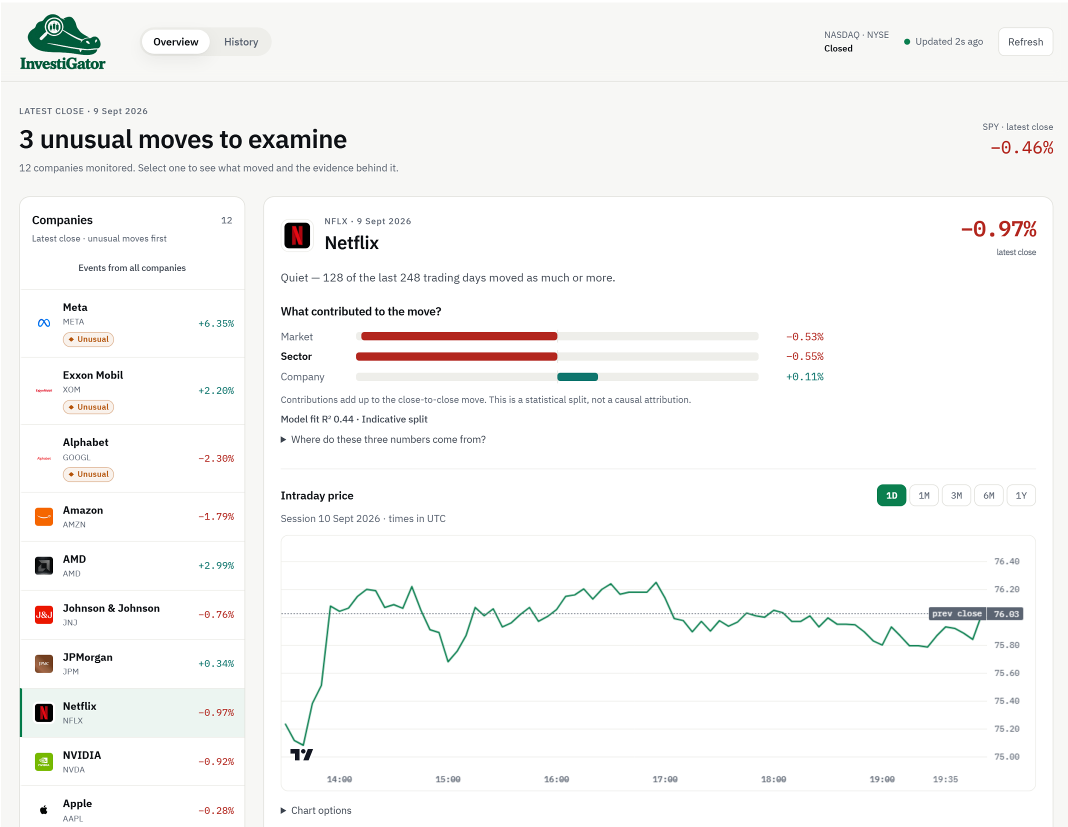

Figure 4.7: Web application on 9 September 2026, with the United States session closed. The heading counts the movements the detector marked as unusual, and the list beside it holds the twelve watched companies ordered by that same criterion, each carrying its state on the surface. Choosing a company governs the whole page: the analysis, the chart and the list of events below all follow it, and widening the list back to every company is a separate control, since that choice does not change the company under analysis. The captures are generated from a snapshot of the application itself, and not collected by hand, with the capture width fixed in the generator of each one, since it is on that width that the interface text stays legible after being reduced to the text box of this document.

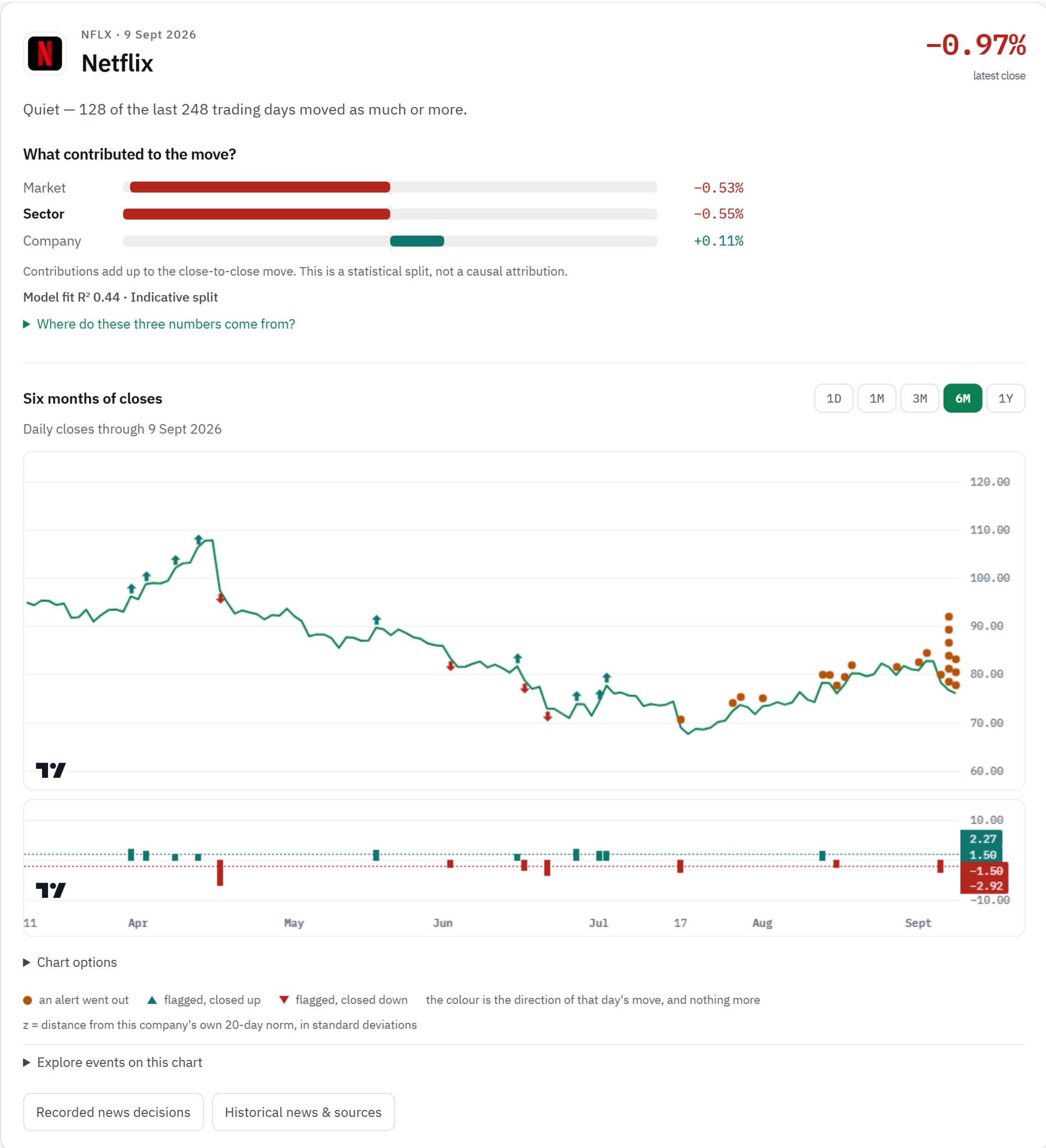

Figure 4.8: View of one company: the verdict in words before any number, the split of the movement into the three parts, and the price series. The marks separate the two things this work distinguishes: the dot marks a day on which an alert was delivered, and the arrow a day the detector flagged that no message accompanied. The lower band shows the z value of those days against the ± 1.5 threshold, on an axis of its own since it shares no unit with the price. The direction of the arrow is that of the close and nothing more; the date and the z value of each day are on the cards below, and not on the line, so that they do not overlap. The interface opens on the current day; the figure shows the six-month range, since that is the one in which the distinction between flagging and communicating becomes observable. The line beneath the split states the quality of the fit that produced it, compared against the median across the seventeen companies of the sector map, and is what lets the reader know when the split is to be trusted.

▼ Where do these three numbers come from?

1. The equation. The day's move is split into what the whole market did, what Netflix's sector did beyond the market, and the rest.

```
move = β_market × market return + β_sector × sector return + company
```

2. This day's numbers. The two β values are Netflix's measured sensitivities, estimated over the 20 trading days before this one. The β and the factor return are shown rounded to two places, while the parts are computed on the unrounded values, so redoing a multiplication by hand can differ in the last digit.

market	1.14 × −0.47%	−0.53%
sector	−0.91 × +0.60%	−0.55%
company	what is left over	+0.11%
total		**−0.97%**

3. Why the company figure has no formula. Market and sector are products: a sensitivity multiplied by what that factor did today. The company figure is not calculated that way at all. It is the remainder — whatever the market and the sector do not account for is attributed to the company by subtraction. That is why the three always add up to the move exactly, and why anything the model failed to capture lands in this last line.

4. How much to trust it: R^2. R^2 answers one question — over the 20 trading days before this one, how much of Netflix's day-to-day movement did this same market-and-sector model manage to track?

```
R² = 1 − (variation the model missed) ÷ (total variation)
```

A model that tracked every day perfectly would leave nothing missed and score 1. A model that tracked nothing would score 0. Here it is **0.44**, so the model followed about **44%** of Netflix's daily movement and missed the other 56%. That is below the 46% median measured across the seventeen companies of the sector map. Treat the split as indicative: the weaker the fit, the more of the move ends up in the company line simply because the model could not place it.

Figure 4.9: The panel that reconstructs the three figures of the previous illustration. It is the only place in the product where a reader can redo a number instead of accepting it, which is why it is shown: the equation, the equation with the day's values substituted, and the coefficient of determination stated in the terms of the question it answers. The third step is the one that matters most, and it is the answer to what a reader asks on seeing three parts that add up exactly: market and sector are products, a sensitivity multiplied by what that factor did on the day, whereas the company part is not computed at all. It is the remainder, so the sum closes by construction and anything the model failed to capture is attributed to the company. That is the same property the artefact of Section 3.5 declares, stated where the reader meets the number.

The company of Figure 4.8 is an illustrative case of the investor's second question, through the disagreement between the percentage change and the split. The stock fell 0.97%, and the contribution of the company itself was *positive*, at +0.11%: what pulled it down was the market, at −0.53%, and the sector, at −0.55%. An investor reading only the percentage would conclude that something had happened at the company, and the split states the opposite. The three parts add up to the observed movement by construction, as Figure 4.9 shows, so the closing of the sum is not evidence that the split is well estimated; and this statistical split does not rule out unmodelled external causes. The quality of the fit reported for the day, 0.44, falls below the median of 0.460 across the seventeen companies of the sector map, and the interface marks the split as indicative for that reason. The verdict classifies the day as ordinary: 128 of the last 248 trading days recorded an equal or larger change.

The same figure shows the caveat that accompanies it. The coefficient of determination for this company is 0.44, below the median of 0.460 across the seventeen companies of the sector map, and the interface states this beneath the split, with the instruction to read it

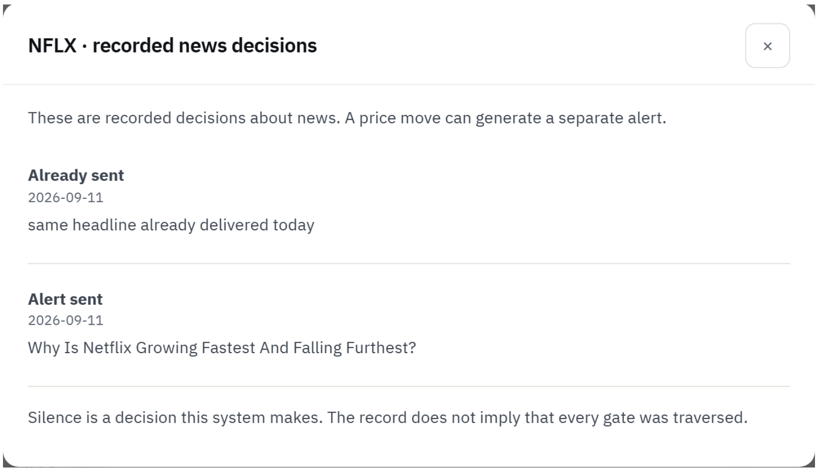

Figure 4.10: The decisions recorded for one company on one day, each naming the headline it concerns and, where the message was withheld, the concrete reason. Silence is here a recorded decision and not an absence of information. The interface declares what the record does not establish, in the line that closes it: these are the decisions that were logged, and the list does not imply that a headline traversed every gate of the policy of Section 4.4, since a headline stopped at the first gate never reaches the others.

as indicative. The two statements coexist by design: the split is the best estimate available and, for this company, the market and the sector track under half of the daily variation. Omitting the second would make the first more persuasive than the fit that supports it, which is the defect that Section 6.4 names.

Figure 4.10 shows the same requirement on the side of silence. Each decision names the headline it concerns and, where the message was withheld, the criterion that withheld it, so a reader can tell a headline the system never judged material from one it judged material and suppressed for a reason declared in advance. None of the decisions is presented as a failure, which avoids the inverse reading that an earlier version of the interface allowed. The record is also explicit about what it does not establish: it lists the decisions that were logged, and a headline stopped at the first gate never reaches the others, so the list is not a traversal of the policy.

4.6 Deployment and operation

The system runs as a permanent process, with a sixty-second scan cycle, on hosting credits included in an academic plan. The first version used a free best-effort scheduler, whose effective period was approximately one and a half hours. The change to a permanent process was motivated by reducing the delay of the alerts.

Latency was measured over the 278 news alerts delivered between 29 July and 30 August 2026 that record a publication time and a send time. The median of the complete path is 353 minutes, with a ninetieth percentile of 964 minutes. The breakdown locates that time without ambiguity: the median between publication and detection is 353 minutes, and the median between detection and the arrival of the message is 5 seconds, with a ninetieth percentile of 16 seconds. The system delivers in seconds what the source gives it.

The separation by configuration does not confirm the expectation that motivated the change. Over the 28 alerts delivered by the scheduler the median is 196 minutes; over the 250 delivered by the permanent process it is 402 minutes. The shorter cycle did not reduce the observed latency. The comparison is not interpretable as an effect of the cycle, since the two configurations did not operate in parallel, the news sources and the calendar period changed between them, and the samples are of very unequal size. What the measurement establishes is that shortening the cycle would only pay off if the time were in the cadence with which the system consults the source, and it is not.

Three causes contribute to the delay in discovery, and this history does not separate them. The first is the source listing the news hours after the time it itself declares. The second is that the most recent item in the feed is not the most recent *relevant* one: in a sample collected on 7 August 2026, the feed of one of the companies carried 250 headlines with the most recent published at 11:39, but the most recent among the 30 relevant ones was from 08:14. The alert goes out on the correct headline, and it is that headline that is old. The third is that the daily budget is already spent, which does not appear in this measurement because an alert that is not delivered records no send time. Only the first would be mitigated by a paid news service, and its contribution is not measured.

All these values are a lower bound on the latency felt by the user, since the publication time is the one declared by the source and not the instant of the event.

4.6.1 Growth of the case base

Every relevant headline observed by the scan is kept as a candidate precedent. Eight days later, when the impact on the price becomes observable, the candidate is measured by the same alignment rule of Section 3.6 and joins the case base. The waiting interval is not inefficiency: it is the condition that prevents a case from knowing its own outcome. Figure 4.11 represents the cycle.

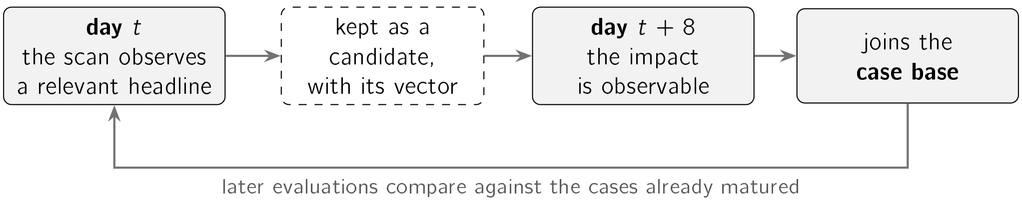

Figure 4.11: Path of a headline between its observation and its availability as a precedent. The eight-day interval follows from the impact of a case being observable only after the price movement. The cycle keeps up to date the evidence the system presents, and not the parameters of the model: no component is retrained.

The mechanism updates the evidence displayed, without touching the parameters of the model. In the concept drift taxonomy of Gama et al. (2014), the system has detection of the change and not adaptation to it.

It was found, with the system already in operation, that this cycle had been interrupted for nineteen days without any error being recorded. The two processes run on separate machines whose file system is reinitialised at every start, and the intermediate maturation files were not being published to the shared data store. It corresponds to the class of cost that D. Sculley et al. (2015) call an undeclared data dependency: each component fulfils its contract, and the assumption that links them is expressed nowhere. Chapter 6 returns to the case as a limitation.

4.7 Life cycle of the learned component

The system trains a single model, the nine-input logistic regression described in Section 3.7. Measured against a one-variable baseline, that model performs worse, a result reported in Section 5.5.

The engineering contribution does not lie in the size of the model, but in the infrastructure that makes that result, and the negative conclusion it supports, defensible. That infrastructure has seven components, each of which exists because its absence would produce a favourable and unverifiable result. Figure 4.12 represents the two halves of the cycle.

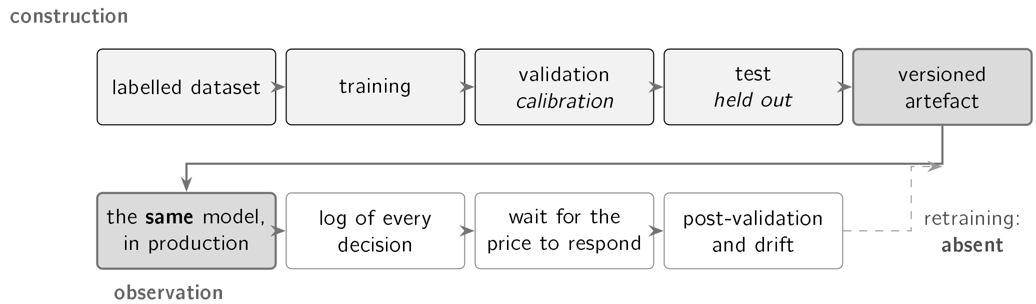

Figure 4.12: The two halves of the cycle. In the upper half, construction: the data are labelled, the split is chronological, the calibration is fitted on the validation block and comparisons are carried out on the held-out test block. Reusing that block limits the independence of the comparisons, as discussed in Section 5.5. In the lower half, observation: the model in production is the model evaluated, every decision is logged, and days later it is confronted with the movement actually observed. The dashed link would close the cycle and does not exist: drift is measured and is not corrected.

Construction of the dataset. FNSPID supplies dated headlines, and the prices come from a distinct source. The alignment of the two series, the assignment of a Saturday news item to the corresponding trading day, and the derivation of the label from the future movement constitute the work that produces the 79,753 examples of Section 3.2.

Declaration of the label as a decision. The existence of an abnormal movement in the following three days is not a fact present in the data: it is a definition, with a chosen threshold and horizon. Section 5.5 measures nine alternative definitions instead of assuming the one adopted.

Verification of the temporal separation. To the chronological blocks and the embargo is added an executable test, reproduced in Listing 3.3: if the future of an example is perturbed, its inputs must stay unchanged and its label must change. The absence of leakage thus goes from a claim to a property verified at every run.

Calibration with measured bias. The output of the classifier is a value in the unit interval, which serves both to rank candidates and to exclude them. The Platt fit runs on a block that never took part in training, and the prevalence discrepancy between that block and the test block is declared in Chapter 5 instead of being left implicit.

Versioning of the artefacts. The trained model, the calibrator and a descriptor of the run, containing the size of each block, the prevalence, the seed and the metrics, are kept with the dissertation. A metric unaccompanied by the artefact that produced it is not reproducible.

Identity between the model evaluated and the model deployed. The sentence encoder used in the evaluation depends on a library whose size exceeds the production container. Instead of substituting another model, which would make the dissertation describe a system distinct from the one in operation, the same model was exported to a lightweight execution format and the fidelity of the conversion was measured.

Observation of the deployed system. Every triage decision is logged. Once the necessary days have passed, a second procedure labels those decisions by the same rule used in training and measures the contribution of the model. It was by this route that the absence of contribution reported in Section 5.5 was established. The drift of the inputs is measured by the same mechanism.

4.7.1 Components developed and components consumed

The only model consumed from outside is a pre-trained sentence encoder, used without any tuning. The work claims no contribution over that component beyond its selection by measurement, carried out against four alternatives, among them a larger model, two more recent encoders and a model specific to the financial domain, which had the worst performance of the set. The dependency is essential, since there is no semantic retrieval without sentence representation, and it is at the same time replaceable: the interface that isolates it has an alternative implementation that degrades to lexical overlap when the model is not available.

Everything else was developed within this work. The labelled dataset and the alignment procedure that protects it from temporal leakage. The detector, with the estimation window that excludes the day being explained. The split of the movement with shrinkage of the coefficients. The precedent base with measured impact. The triage classifier and its calibration. The decision policy and the constants of Table 4.2. The layer that composes the text from the computed objects. And the mechanism that observes the system after deployment.

The system does not integrate a language model in the production of the text delivered to the user. The absence is deliberate and follows from the argument developed in Section 2.7: a generated sentence is a statement, whereas the system described delivers statements accompanied by the evidence that supports them.

A grounded generation layer is, even so, implemented and evaluated, and is not exposed to the user. Section 4.8 describes it, with the guarantee it offers and the reason it remains closed.

4.7.2 The missing link: retraining architecture

The cycle does not include retraining. The parameters correspond to the period from 2018 to 2023 and were not readjusted, even though drift is measured and the population stability index places the main input in the band of significant displacement. The operation would be technically possible, since new labels derive from observed prices; the impediment is that retraining with the production log would change the values reported in this document, and assessing that change requires observation time which, at the date this chapter was written, did not exist.

The instrumentation that would make it possible was built and deployed on 4 September 2026, and the log came to store, per decision, the value of each input, the date of the last price bar used and the identity of the model. Two restrictions limit what can be drawn from it, and both were fixed before any candidate existed. The first is that only decisions whose price bar is prior to or equal to the date of the news are usable, since a later bar would describe a market that has already observed the outcome the label measures. The second is that the unit of analysis is the pair formed by company and day, and not the headline, which substantially reduces the number of independent observations available in a short window. Nothing is asserted in this document about the result of that comparison.

The absence of execution does not, however, dispense with defining the architecture. Figure 4.13 presents the continuous training cycle designed for this system, with the mechanisms that would make it safe. Three decisions characterise it.

The first is the **trigger**, conditional and not temporal. Retraining is set off when the Population Stability Index (PSI) of any input exceeds 0.25, the conventional threshold of significant displacement, or after three months without review. The second is the **rolling window**: training covers the twenty-four months preceding the trigger, and validation and test the following periods, with the five-day embargo of Section 3.7. The split stays chronological, because the problem does not cease to be temporal just because the model is new.

The third, and the one that distinguishes a continuous training cycle from dangerous automation, is the **promotion gate**. A newly trained model does not replace the model in force by virtue of being more recent: it replaces it only if it beats it over the same held-out test block, and if the confidence interval of the difference excludes zero. A model that does not pass that gate is logged and discarded, and the model in force remains with the drift declared. It is the difference between adapting to change and chasing noise.

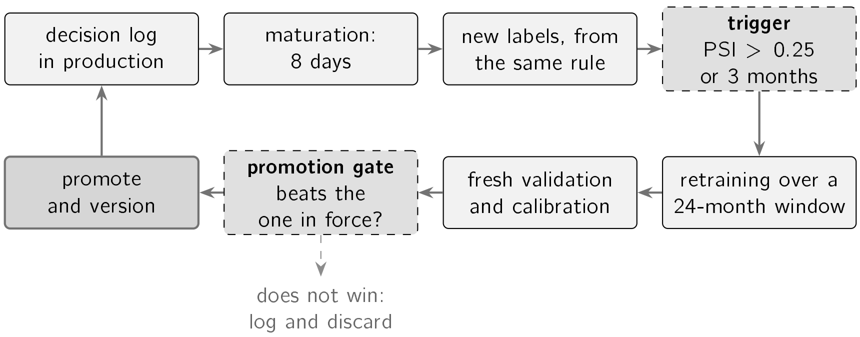

Figure 4.13: Continuous training cycle defined for this system and not executed within this work. The trigger is conditional and not temporal; the window is rolling and the split stays chronological with embargo; and the promotion gate prevents a new model from replacing the model in force without beating it over a held-out block. The dashed link is the rejection path, and its existence is what distinguishes the cycle from automation that chases noise.

The cycle likewise includes no form of human ground truth. The three labels used in this work, corresponding to abnormal movement, membership of the same sector and percentile of movement, are verifiable approximations that were never confronted with the judgement of a human assessor. It is the limitation on which the others most depend, and Chapter 6 returns to it.

4.8 The intelligence layer

The previous sections described a system that composes text from computed objects, with a closed vocabulary. This section describes the layer that does the same with a language model, under a different guarantee, and the reason it is not exposed to the user.

The current architecture for answering a question over a body of documents is retrieval-augmented generation (Lewis et al. 2020), discussed in Section 2.4.1. The system in production runs the first half and stops before the second. The layer described here runs both, and exists to determine what it would cost to do so under the same discipline of evidence.

4.8.1 The contract: language does not produce facts

The object that crosses this layer is an *evidence bundle*. Every fact in it carries a citable identifier and a declared origin, one of three: measured from prices, computed by one of the engines of Chapter 3, or produced by the triage model. No fact has a *generated* origin. That separation is the rule that structures the layer: the language model writes the language and never the numbers.

Figure 4.14 shows the traversal. Two calls to the model, and the division between them is deliberate. The first writes no text: it translates the question into a plan the system knows how to run. The second decides nothing: it drafts over facts that already exist. Between them, the components described in this chapter produce the evidence, and neither call has access to the market.

A deterministic router covers the first call when no interface key is available. It is not a fallback of convenience: the constraint of using only free sources implies that the layer has to keep answering when the usage limit runs out, and a conversational interface that falls silent at that point would not serve the audience Chapter 1 defines.

4.8.2 Two levels of guarantee, and this is the weaker one

The grounding check refuses any sentence whose number does not belong to the fact it cites. The formulation matters: the link is per sentence and not over the whole bundle, because otherwise any number in the bundle could be attached to any claim.

The guarantee is not, however, of the same nature as the one the alert offers, and Table 4.3 separates the two.

The asymmetry is justified by risk and not by performance. The alert interrupts the user without being asked, and the text of this layer is consulted by someone already looking at the evidence that accompanies it.

4.8.3 What was measured, and what is not claimed

The check was confronted with a set of attempts to circumvent it. Over twenty-three attacks, all were refused; over eight faithful texts, none was; and in the sections actually composed not a single violation was delivered.

Table 4.3: The two ways of guaranteeing that the text does not claim more than the components computed. The difference is not one of measured effectiveness, but of nature: an enumeration of what is permitted is closed, and an enumeration of what is forbidden is a list that may not be complete.

	Delivered alert	**Text of this layer**
How it reaches the reader	pushed, unrequested	pulled, with the evidence beside it
How it is composed	closed vocabulary	language model
What the check enumerates	what is permitted	what is forbidden
Strength of the guarantee	closed by construction	limited by the list

These numbers delimit themselves. The first two are deterministic and reproduce over the same body of attacks; the third is a count over one concrete run. The measured strength is a lower bound, since the body of attacks was built by the author and nothing guarantees it exhausts the ways of circumventing the check.

The layer is not exposed, and the reason is not the performance of the check. It is the nature of the guarantee that Table 4.3 describes: exposing to the user a text under the weaker guarantee, when the rest of the system operates under the stronger one, would make the product's promise uneven depending on the screen. No comparison with a complete retrieval-augmented generation architecture was carried out, so nothing is claimed about what it would add.

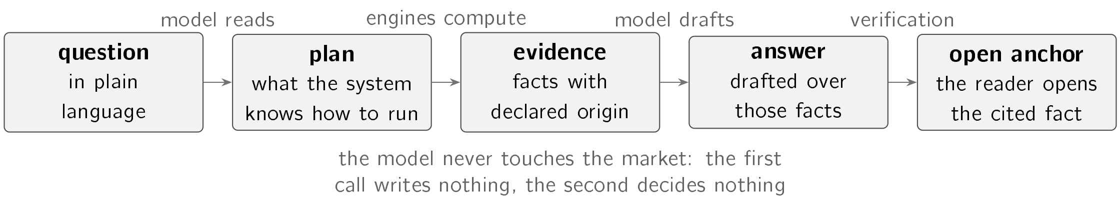

Figure 4.14: The two calls to the language model are separated on purpose. The first translates a human sentence into a plan the system knows how to run and writes no text. The second drafts over facts that already exist and decides nothing. Between them, the engines described in this chapter produce the evidence, and each fact arrives with its origin declared. The final verification refuses any sentence whose number does not belong to the fact it cites, and that is what lets the reader open the anchor and confront the claim with what supports it.

Chapter 5

Case studies

The two previous chapters described the methods chosen and the system that puts them to work. This one determines what each of them is worth. The experimental design and the metrics used are described first. Four case studies follow: three correspond to the research questions, and the fourth to the behaviour of the system in operation, where the choices of the previous chapters meet data that were not available when they were made. Whoever reads it will know what each technique is worth against a simpler alternative, and where the measurement did not confirm the deployed choice.

5.1 Experimental design and metrics

This section fixes the form common to all the case studies and defines the measures employed, with particular attention to the value each one takes in the absence of information.

Each case study follows four steps, shown in Figure 5.1: the question, the measurement procedure and the alternative against which the comparison is made, the result obtained, and what that result does not allow to be concluded. The fourth step is what distinguishes an evaluation from a demonstration. The values measured over frozen historical sets are produced by automatic procedures with a fixed seed and reproduce exactly. Two classes of value do not have that property, and are identified where they appear. The values read from the system in operation are dated snapshots, which grow with the channel. Those that depend on quotes readjusted by the source may vary in the third decimal place.

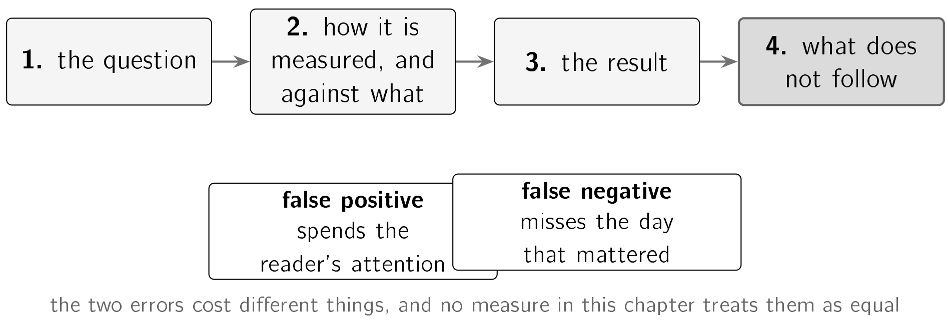

Figure 5.1: The form followed by all the case studies. The fourth step delimits the reach of each result. Below, the two types of error that the measures of this section balance, and the distinct cost of each one in the application domain.

All the measures rest on the four combinations between what the system claims and what is observed. Precision is the fraction of correct calls among the occasions on which the system spoke, TP/(TP + FP); recall is the fraction of the relevant cases that were flagged, TP/(TP + FN). The two vary in opposite directions, and a rule that flags almost everything reaches high recall with low precision.

F_1 is the harmonic mean of the two, $2PC/(P + C)$, and stays close to the worse of the two values, which prevents an extreme from being rewarded. For the z-score, with precision 0.381 and recall 0.800, it gives 0.516; for the fixed threshold rule, with precision 0.122 and recall 0.992, it gives 0.218. The two terms are rounded to the third decimal place, so the calculation redone over the printed values may differ in the last.

Precision@k applies to retrieval, which returns an ordered list of which the user reads only the beginning. It measures the fraction of the first k results that are relevant (Manning, Raghavan, and Schütze 2008), with $k = 5$, the value that corresponds to what fits in an alert.

Precision-Recall Area Under the Curve (PR-AUC) applies to triage, which returns a probability and not a decision. Each possible threshold produces a precision-recall pair, and the set of those pairs draws the precision-recall curve; the area under that curve summarises the model across all thresholds simultaneously, instead of judging it at a manually chosen threshold. It is preferable to the percentage of correct answers because, with unbalanced classes, invariably answering that the news item is not material is right 62.2% of the time without any learning.

The Brier score looks at the value of the probability and not at the ordering, being the mean squared error between the probability announced and the observed outcome, with lower values corresponding to better performance. The square penalises confident mistakes more than proportionally, which is the appropriate property in a system that displays percentages to a recipient without the means to question them.

A measure read without the value that corresponds to it in the absence of information leads to the wrong conclusion, and this chapter documents three occasions on which that happened. Table 5.1 fixes that value for each measure.

The first row of Table 5.1 is the only measure in the chapter that uses no labels. The firing-rate spread is the difference between the rate at which the detector flags in the company where it speaks most and the rate in the company where it speaks least. It rests on a single principle: a universal detector should behave in a similar way in distinct companies. Its least obvious property, and one that Section 5.2 develops, is that the optimum is attainable without any information.

5.2 Detection consistent across companies

The first case study answers RQ1: a rolling z-score detects unusual movements more consistently across companies than a fixed threshold rule. The answer is affirmative, with a margin of more than twenty times on the principal measure.

5.2.1 Procedure

The evaluation uses three years of daily quotes from fifteen large-capitalisation companies, between June 2023 and June 2026, which corresponds to approximately seven hundred and fifty days per company. The set is distinct from the list of twelve companies monitored in

Table 5.1: Each measure, the question it answers, the value a method without information obtains, and the case study in which it is used. The central column is the one that prevents the optimistic reading of an isolated percentage.

Measure	**Answers**	**Value without information**	**Used in**
Firing-rate spread	Does the detector behave the same way in distinct companies?	zero is the optimum, and is attainable without information	RQ1
F_1	Is it right when it speaks, and does it speak when it should?	depends on the data	RQ1
Precision@5	Of the five presented, how many are relevant?	0.117, measured by random choice	RQ2
PR-AUC	Does it rank well, whatever the threshold?	0.378, the prevalence of the test block	RQ3
Brier score	Does the percentage announced match the observed frequency?	approximately 0.235, announcing the training prevalence	RQ3

production, for the reason stated in Section 3.2: the evaluation requires variety of volatility, and the deployed list is a product decision.

The principal argument uses no labels. A detector meant to flag rare days should fire at a similar rate in distinct companies; if it fires frequently in one and rarely in another, what it is measuring is the volatility of the company and not the rarity of the day. As a supporting argument an approximate label is used, which classifies as extreme a day in the ninety-ninth percentile of the company's own returns. The label is imperfect, being itself relative to volatility, and therefore occupies the secondary position.

5.2.2 Result

Figure 5.2 summarises the two measures. It is the first that supports the conclusion: what distinguishes the methods is not the level at which they fire, but the fact that the fixed threshold follows the volatility of each stock while the z-score stays almost constant across them.

The firing-rate spread is 0.015 for the z-score and 0.344 for the fixed threshold, a difference of more than twenty times. Against the approximate label, the z-score obtains $F_1 = 0.516$ and the fixed threshold 0.218.

The comparison involves an asymmetry of treatment. The window of the z-score is varied later in this section, whereas the three per cent threshold of the fixed rule is the conventional value and was not optimised. What the comparison establishes with confidence is not that no fixed threshold would obtain a better result, but that a fixed threshold measures a quantity different from the one intended, because the same percentage corresponds to distinct rarities in distinct companies. This argument does not depend on the value chosen.

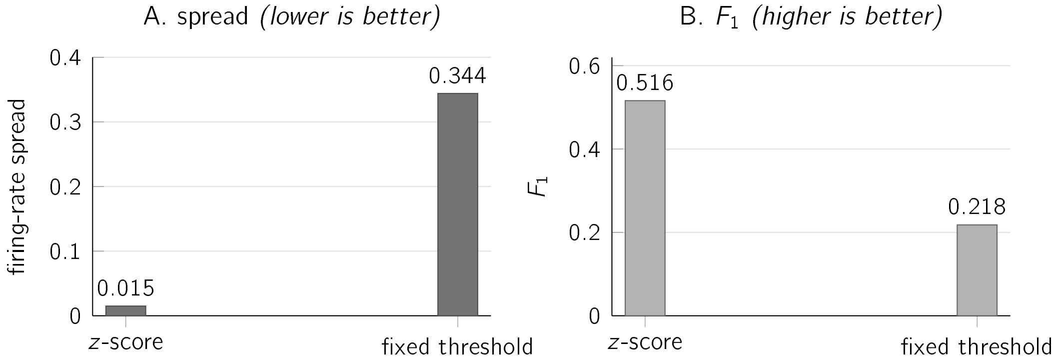

Figure 5.2: The two measures, side by side. In A, the difference between the maximum and the minimum rate across the fifteen companies; the fixed threshold flags 0.9% of the days in the calmest company and 35.3% in the most volatile, while the z-score stays between 1.6% and 3.1% in all of them. In B, the performance against the approximate label.

5.2.3 What the principal measure does not exclude

The firing-rate spread has zero as its optimum, and that optimum is attainable by a method with no information at all. A detector that flagged in each company a fixed fraction of days chosen at random would show a practically null spread, beating the z-score precisely on the measure this section elects as principal. The claim follows from the construction of the method and requires no execution.

The measure is not invalidated; what it requires is that it be read together with the second, and that is why there are two. The spread excludes the fixed threshold, which fires as a function of the volatility of the company and not of the rarity of the day. F_1 excludes random firing, which sits at the level of chance against any label. Neither measure is sufficient on its own, and only the rolling rule satisfies both.

An asymmetry in the standard of evidence applied throughout the chapter should also be recorded. The values of this section and of the next are points, without confidence intervals, whereas those of Section 5.5, where the result that contradicts the initial hypothesis resides, come with resampling by groups. The partial justification lies in the order of magnitude of the differences observed here, which no plausible resampling would reverse. The justification is not complete, since days of the same company are also not independent observations in this case, for the same reason of volatility clustering invoked later on.

5.2.4 Comparison with learned detectors

The most demanding comparison sets the statistical rule against machine learning methods designed to detect anomalies. Two were evaluated, over the same information and under the same discipline of not using information posterior to the day assessed: *Isolation Forest* (F. T. Liu, Ting, and Zhou 2008) and the *Local Outlier Factor* (Breunig et al. 2000). Both ran in a single configuration, without a hyperparameter search, which limits the reach of their defeat for the same reason invoked later for the tree model. What is established is that these detectors, parameterised in this way and over these inputs, do not beat the rule; not that no parameterisation would. Figure 5.3 presents the result.

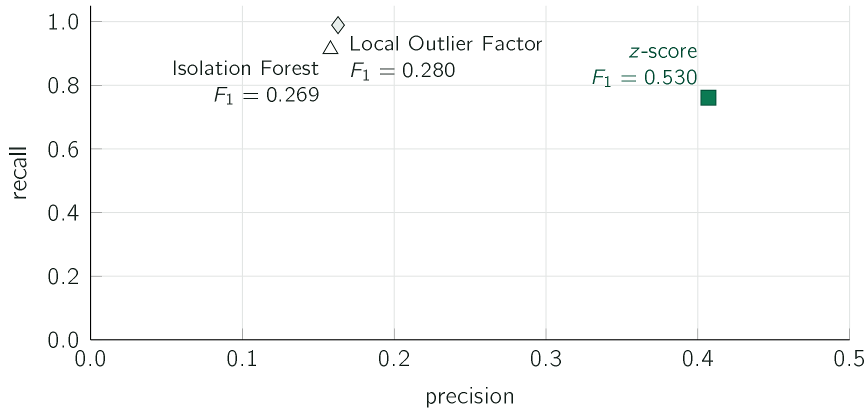

Figure 5.3: The three detectors over the same data and under the same protocol, restricted to the region all three can score. The two learned methods occupy the region of near-total recall and low precision; the rule trades some recall for precision. The F_1 beside each point summarises that trade-off. The firing-rate spread follows the same pattern, at 0.017 for the rule against 0.140 and 0.183 for the learned detectors.

The transparent rule beats both methods, with almost twice the F_1 and with substantially better consistency across companies. This is the first of three cases, throughout the chapter, in which a more sophisticated technique is compared under equivalent conditions with a simpler one and does not obtain a better result. The choice of the simple method thus becomes justified by measurement and not by preference.

The explanation does not lie in the quality of the methods. Both determine what is unusual from the geometry of the data, by opposite routes. *Isolation Forest* identifies the rare point by how easily successive random cuts separate it from the rest (F. T. Liu, Ting, and Zhou 2008). The *Local Outlier Factor* compares the density in the neighbourhood of each point with that of the points surrounding it (Breunig et al. 2000).

These are non-parametric methods, outside the statistical family that Chandola, Banerjee, and Kumar (2009) characterises by the modelling of the distribution of normal data, and they were designed for sets with many interacting variables. The problem treated here has another form: one series of returns per company, whose relevant structure is temporal. Volatility clustering, formalised by Autoregressive Conditional Heteroscedasticity (ARCH) and Generalized Autoregressive Conditional Heteroscedasticity (GARCH) models (Bollerslev 1986; Engle 1982), moves over time the threshold of what constitutes an unusual day; a rolling window follows that movement by construction, whereas a detector fitted to the complete set has to infer it.

A note on the compatibility between the values: the z-score appears with $F_1 = 0.530$ in this comparison and with 0.516 in the previous one. The two values measure the same detector under distinct protocols. The second covers all the days assessed, while the first is restricted to the region all three detectors can score, the only one in which the comparison between them is fair. The spread follows the same rule, being 0.017 under this protocol and 0.015 under the previous one.

5.2.5 Two comparisons that do not confirm the deployed choices

Two additional experiments were designed to justify options taken, and neither of them confirmed them. Both are reported as they came out.

The first concerns the volatility estimator. The alternative to giving the same weight to the twenty days of the window consists in giving greater weight to the recent days, through an exponentially weighted average. Figure 5.4 presents the result.

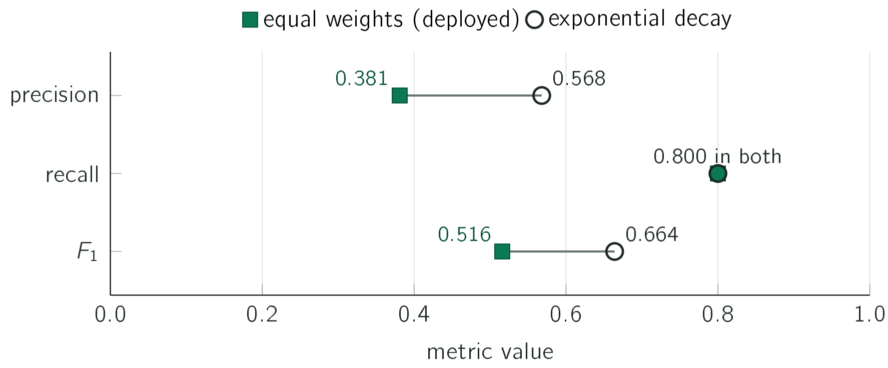

Figure 5.4: The two estimators, with the same recall; each segment links the deployed result to the alternative on the same measure. The exponentially weighted alternative removes almost half the false alarms, raising precision from 0.381 to 0.568, and is even more consistent across companies, with a spread of 0.012 against 0.015. The system keeps the estimator that obtained the inferior result, for the reason set out in the text.

With identical recall, the alternative obtains $F_1 = 0.664$ against 0.516 for the deployed estimator. The mechanism is understandable. Large movements occur in clusters. A twenty-day average with equal weights dilutes a shock over twenty observations and keeps flagging the following days for weeks; an average that favours the recent past absorbs it immediately.

The system keeps the estimator that obtained the inferior result. The reason is the one that runs through the work: the mean and the standard deviation of the last twenty days can be explained in two sentences to any reader, and exponentially decaying weights cannot. In a tool whose central argument is explainability, a gain in precision obtained at the cost of the explanation is not unequivocally a gain. It is recorded, however, as a choice and not as a result: the alternative is measured, the gain is quantified, and adopting it would be a small change should explainability cease to be the dominant constraint.

The second experiment concerns the size of the window, and Figure 5.5 presents it.

A sixty-day window obtains $F_1 = 0.678$ against the 0.516 of the twenty-day window. Two reasons justify keeping the shorter window. The first is responsiveness: a sixty-day window takes three months to react to a change of regime, and a company moving from calm to volatile would continue to be judged by the previous norm.

The second concerns the validity of this particular measurement. The label is built from the percentile of movement of the company itself, and is therefore relative to volatility; a longer window produces a less adaptive norm, which flags the extreme tail more often, which is precisely what this label rewards. Circularity exists, and it favours the long window by construction. This is why the principal argument of this section is the consistency of

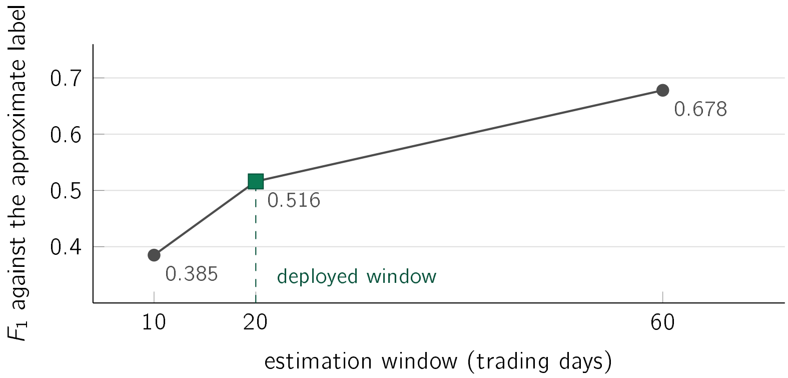

Figure 5.5: Performance against the approximate label for three window sizes. The value grows monotonically with the size of the window, and the deployed window is not the one that obtains the best result. The text sets out the two reasons why the longer window was not adopted, the second being a reservation about the validity of this measurement itself.

the firing rate, which uses no labels. The formulation that survives is that the twenty-day window was chosen for responsiveness, and that the only measurement available on this point does not support it.

5.2.6 Limits of the conclusion

The label is approximate and relative to volatility, which is why the argument that carries weight is the one about consistency, which does not depend on it. The fifteen companies are all large-capitalisation and were selected in 2026, with the evaluation running over the past: they survived by construction, and none of them was delisted, absorbed or dropped from the list. What is measured is the behaviour of the method in companies that persisted, and not in the population from which they were drawn. The comparison covers this set and this period only, and nothing in it allows the result in another market or another decade to be anticipated.

5.3 Precedent retrieval

The second case study answers RQ2: retrieval by meaning finds news from another company in the same sector better than the text and ranking alternatives evaluated. The answer is affirmative against those comparators; the advantage over chance holds in all five sectors. The date restriction is evaluated separately, in aggregate over the historical corpus.

5.3.1 Procedure

Evaluating retrieval systems faces the absence of a list of correct answers, since no set of pairs of news items is annotated as related. A verifiable approximation is therefore used: a retrieved news item is considered relevant if it belongs to the same sector as the query news item. The approximation is imperfect, because two news items from the same sector may deal with distinct subjects, but it is objective and was fixed before the results were observed.

A further constraint is imposed that makes the task harder: the retrieved news item must belong to a different company. Without it, the system would perform well by returning news from the company itself, which almost always belongs to the same sector by construction. The corpus is collected from the per-company news service, over a window of twenty-eight days closing on 25 June 2026, and is pinned by checksum with a versioned manifest. The collection calls for a precaution that is only discovered by measuring: the service returns at most around two hundred and fifty items per request and, on reaching that limit, ignores the start date of the request. Asking for five, thirteen or twenty-seven days of the same company returns exactly the same headlines, all from the last days of the window, with no error and no warning. The window is therefore requested in one-day slices, which ensures that no request reaches the limit and that the declared window is the window collected. Five company-day pairs remain above the limit, all from the same company, corresponding to 1.2% of the pairs: the day is the service's finest granularity, so that fraction is irreducible and is declared.

The corpus so obtained contains 18,599 news items, and its sector composition is not uniform: 84% belong to technology, because seven of the fifteen companies monitored are technological and are also those of greatest news volume. Under such a composition, aggregate precision stops measuring the ability to retrieve by topic and starts measuring the composition, as Section 5.3.4 demonstrates. The main measurement therefore uses a subset **balanced by sector**, with 569 news items from each of the five, 2,845 in total, selected with a fixed seed and likewise pinned by manifest. The quota is the size of the smallest sector, so that no news item is counted twice.

The measurement uses five hundred queries per repetition across five repetitions with distinct seeds. The span of the corpus delimits what the measurement establishes: twenty-eight days do not support claims about temporal generalisation. The protocol does not require the candidate to be prior to the query, and only 31.1% of the neighbours are. The check requiring earlier candidates was made in aggregate over the historical corpus, where the margin over chance holds, and not per sector.

5.3.2 Result

Figure 5.6 presents the method used, a larger variant and the trivial alternatives.

Random choice is right in 11.7% of cases, and that is the reference value, and not one in five, because the protocol excludes candidates from the query's own company and in the sectors served by two companies that exclusion removes half the sector. Search by words in common raises the result to 19.6%, and comparison by meaning to 39.5%.

The lexical baseline is the simplest form of that family: word counts projected by hashing and compared by cosine, without weighting by the rarity of terms, without frequency saturation and without normalisation by the length of the text. It belongs to the tradition that runs from the vector space model (Salton, Wong, and C. S. Yang 1975) to probabilistic ranking functions such as BM25 (Robertson and Zaragoza 2009), sitting on the first step of that family. The twenty percentage points that the sentence representation (Reimers and Gurevych 2019) adds therefore measure the distance to an elementary starting point and not the distance to a tuned lexical system. The comparison with a well-parameterised ranking function is work still to be done, and it is the one that would make this conclusion stronger.

The encoder specific to the financial domain obtains 0.275, the lowest value of all the models assessed. The model assessed is the public checkpoint `ProsusAI/finbert`, identified in this way because it is the exact artefact the measurement bears on; it belongs to the family of

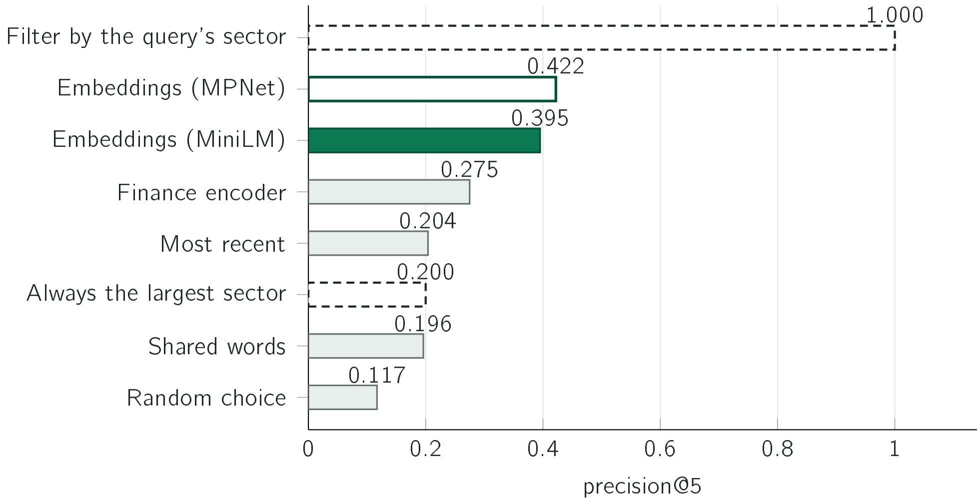

Figure 5.6: Precision@5 of the deployed MiniLM, in solid green, against seven alternatives. The green outline identifies the larger MPNet variant; the grey bars bring together two baselines, a lexical one and an encoder specific to the financial domain. The two dashed bars dispense with any model: always returning news from the largest sector obtains 0.200, which in this subset is the one-in-five value itself by construction, and filtering the candidates by the query's known sector satisfies the label in every position of the list whenever five eligible candidates exist. On the complete corpus, whose composition is not uniform, the first of those alternatives obtains 0.840 and overtakes the method, which Section 5.3.4 examines. The 1.000 is not a measurement: it is an arithmetic consequence of the definition of relevance, and it appears here as the highest reference value this task admits. It is not a product alternative, since it would return cases from the same sector without looking at the topic of the news item.

financial encoders of which Zhuang Liu et al. (2020) and A. H. Huang, Wang, and Y. Yang (2023) are the versions documented in peer-reviewed publications. It was used with the mean of its vectors as the sentence representation, which is the way to apply it without further training. It is an encoder tuned for sentiment classification and not for positioning sentences in a space where distance means similarity, and it is precisely that training objective that Sentence-BERT adds (Reimers and Gurevych 2019). The measurement establishes that this domain model, used in this way, obtains a result inferior to a generic model trained for similarity; it does not establish that domain knowledge is irrelevant to the task.

5.3.3 The trivial alternative the aggregate hides

The complete corpus is not balanced: 84% of the 18,599 news items belong to the technology sector. There is therefore a strategy that dispenses with any model and does not even look at the query, consisting in invariably returning five technology news items. It is right on every technology query and on none of the others, so its precision equals exactly the fraction of queries that are about technology, 0.840 in that corpus. Measured against this alternative, and not against chance, the margin of the method turns from positive to negative: 0.779 against 0.840.

The misleading element is, however, the aggregate value. The trivial strategy obtains 1.000 on technology queries and 0.000 on all the others, returning semiconductor news to someone querying about an oil company. What it exploits is not a property of the problem, but the composition of the corpus, and that is the reason the main measurement of this section uses the balanced subset, where the same strategy obtains 0.200.

The comparison within each sector checks whether the advantage over chance also holds outside technology. Panel A of Figure 5.7 presents it.

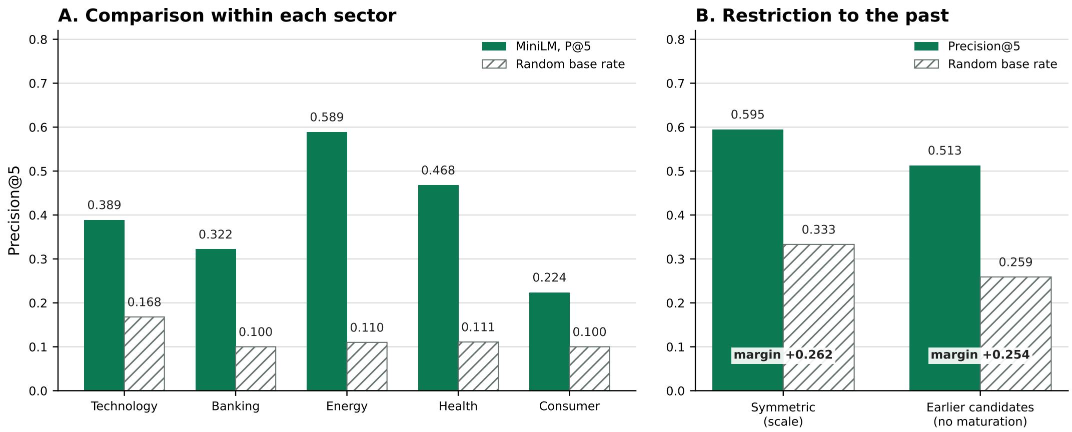

Figure 5.7: In A, precision@5 and the corresponding chance value within each sector: the method sits above its reference value in all five. In B, the symmetric protocol of the scale test and the protocol requiring earlier candidates, without reproducing the eight-day maturation in production, each beside its reference value. Precision falls from 0.595 to 0.513 and the chance value from 0.333 to 0.259, so the margin varies only from +0.262 to +0.254.

The chance value is 0.168 in technology and stands around 0.100 in each of the four remaining sectors, the difference following from the number of companies serving each sector. The method obtains 0.589 in energy, 0.468 in health, 0.389 in technology, 0.322 in banking and 0.224 in consumer. The gains over the respective reference value are +0.479, +0.357, +0.220, +0.221 and +0.123, that is, between a little over twice and more than five times chance. The claim the work supports is therefore the narrowest one: semantic retrieval beats the base rate within each of the five sectors, against the comparators evaluated.

This comparison does not measure the advantage over a rule that knows the query's sector. That sector is available in the company map: filtering candidates from other companies by the same sector would fill every slot correctly, provided five eligible candidates existed. This follows from the label, not from an additional experiment. The evaluation measures retrieval of this proxy from the text; it demonstrates neither superiority over a sector filter nor the additional semantic relevance the product aims to provide.

The larger variant obtains 0.422 ± 0.010 against 0.395 ± 0.011 for the one used, over five seeds; the difference is small relative to the dispersion and no significance test was carried out. Two more recent encoders obtain 0.375 and 0.395, the second indistinguishable from the deployed one. The adoption of the smaller model in production followed from computational cost and not from an advantage in quality, and is an explicit trade whose price in precision is quantified. The apparently most reasonable alternative, which consists in presenting the most recent news, obtains 0.204, indistinguishable from the lexical baseline of 0.196 given deviations of 0.009 and 0.004, and both far below any encoder. The two alternatives without a semantic model are thus tied at the bottom of the table, which is consistent with there being no reason for the most recent news item about another company to be related to the news item under analysis.

5.3.4 Sensitivity to the composition of the corpus

The previous subsection uses the balanced subset, and this one measures what happens without it. The same measurement over the complete corpus, 84% technology, produces Table 5.2.

Table 5.2: The same measurement, the same protocol and the same period, over two compositions of the same corpus. On the left the balanced subset, with 569 news items per sector; on the right the complete corpus, in which 84% of the news items belong to technology. The right-hand column does not correct the left-hand one: it shows what aggregate precision measures when composition dominates.

Precision@5	balanced	complete
Embeddings (MPNet)	0.422	0.778
Embeddings (MiniLM, deployed)	0.395	0.779
BGE-small	0.395	0.778
E5-small	0.375	0.775
Financial encoder	0.275	0.763
Words in common	0.196	0.746
Random choice	0.117	0.685
Always the largest sector	0.200	**0.840**

The complete-corpus column has two readings, and neither is a correction of the other column. The first is that **all the encoders become indistinguishable**: the five sit between 0.763 and 0.779, and the domain encoder, which in the balanced subset obtains the lowest value of all by a margin of more than twelve hundredths, rises to within two hundredths of the rest. The second is that the trivial strategy, which does not look at the query, obtains 0.840 and overtakes any method.

The explanation is the same in both cases and involves none of the methods. In a corpus where 84% of the news items belong to one sector, returning news from that sector is almost always the right answer, regardless of how the candidates were chosen. Aggregate precision stops measuring the ability to retrieve by topic and starts measuring the composition of the corpus.

This result is the reason the main measurement uses the balanced subset, and it is also a caveat that applies to any work reporting aggregate precision by membership of a class without declaring the distribution of those classes. The comparison by sector, which the previous subsection presents, is unaffected: by construction it compares each query with the reference value of its own sector.

5.3.5 Behaviour at scale and with earlier candidates

The previous evaluation bears on a recent corpus of limited size. To verify that the result does not depend on that corpus, the same measurement was repeated over the complete historical set, with approximately eighty thousand news items from six years. Under this symmetric protocol, which still admits candidates posterior to the query, precision@5 rises to 0.595. The rise should not, however, be read in full as a gain: the chance value rises likewise, from 0.117 to 0.333, because the composition by sector is different. Measured against the respective reference value, the margin is +0.274 in the first case and +0.262 in the second. The result establishes that the method does not degrade when the corpus goes from recent and short to six years, and not that it improves with more data.

Scale also allows the comparison the preliminary corpus could not support, and which is the first one a retrieval result invites. The lexical line in Figure 5.6 is word overlap, which neither weights rare terms nor corrects for headline length. Over the historical set, and under the same paired protocol, BM25 (Robertson and Zaragoza 2009) was measured in its place, as the reference method of lexical retrieval: it reaches 0.521, against 0.595 for the semantic model, with chance at 0.333. The paired difference is +0.073, positive across all five repetitions, and no query was left with zero-scoring candidates.

The figure forces a more moderate reading. Measured against chance, the margin of BM25 is +0.188 and that of the semantic model +0.261: lexical matching, done well, captures about three quarters of the distance between the method and chance. The gain from semantic representation is real, consistent and measurable, but it is the smaller of the two effects. This result was absent from the preliminary evaluation because the lexical comparator used there is weaker, and its absence left the semantic advantage looking larger than it is.

The previous protocol grants a freedom the system in production does not have. The only constraint imposed is that the candidate belong to another company, which allows it to be posterior to the query, and that case is common: in the recent corpus, 38.7% of the neighbours returned are dated after the query and another 30.2% are from the same day. The deployed system does not have that freedom, since the precedent base only receives a case eight days after it is observed, as Section 4.6.1 describes.

The criterion for a hit is sector membership, which does not use the future price outcome. Nevertheless, admitting later candidates changes the retrieval task. What is at stake is the correspondence between the value reported and the task the research question describes, which speaks of past news. The measurement repeats the protocol adding to the mask the condition that the candidate be strictly prior to the query, and reproduces the symmetric value in the same run so that the comparison is paired. None of the 2,500 queries was left without sufficient past. The result appears in panel B of Figure 5.7: precision falls by 0.082, the reference value falls by almost as much, and the margin varies by 0.008. The method keeps practically all of its advantage when earlier candidates are required. The mask neither imposes eight-day maturation nor checks the availability of outcomes at three and five sessions. The result therefore does not evaluate the base actually available at the instant of a production decision.

This is the third occasion in the chapter on which reading a precision without its corresponding reference value would lead to the wrong conclusion. It happened in the move from the recent corpus to the complete one, it happens with the precision within the budget treated in Section 5.5, and it happens here. Precision is a quantity with no meaning of its own, which it acquires only alongside the alternative that has no information.

5.3.6 Limits of the conclusion

Membership of the same sector is not equivalent to a relation between news items; it is an approximation built because it is verifiable and not because it corresponds exactly to the property intended.

The second observation has consequences for the product: retrieval captures topic and not direction. Two news items can deal with the same subject and the price move in opposite directions, which is what happened in the example presented in Chapter 4. Figure 5.8 quantifies it.

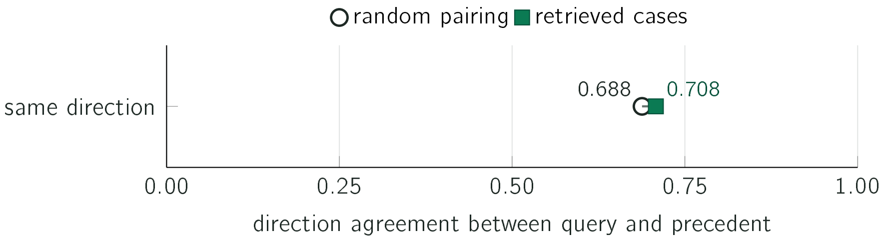

Figure 5.8: Fraction of retrieved cases whose price moved in the same direction as that of the query news item, against the value obtained by random pairing under the same constraints, on a full scale between zero and one. The difference is 0.020. The system therefore presents the cases individually, and the text of the alert states that the similarity is one of topic.

The direction agreement between the query news item and the retrieved cases stands at 0.708 against a chance value of 0.688. It is for this reason that the system presents the cases one by one and accompanies the range and mean with these individual cases. The mean describes the retrieved cases, not the next return; the alert states that the similarity concerns topic.

5.4 Adjusting the encoder for comparable materiality

5.4.1 The measurement that motivates the question

The retrieval evaluated in the previous section is measured by sector membership, and nothing in that metric concerns the magnitude of the movement. The first measurement of this section is therefore the one that justifies asking the question: over the test block, the unadjusted encoder obtains a materiality comparability of 2.173 percentage points, against 2.259 for random pairing under the same restrictions. The standard deviations between repetitions are 0.062 and 0.107, so the two values are not distinguishable. **On the quantity that Section 3.6.1 sets out to improve, the representation used by the system is indistinguishable from chance.** There is, then, room to improve, and it is that room the experiment attempts to occupy.

5.4.2 The first attempt collapsed the encoder

The first run used pairs drawn at random, and the outcome was what Section 3.6.1 anticipates as the consequence of a concentrated target: the encoder stopped distinguishing headlines. Table 5.3 gathers the diagnostic.

The mean cosine between different headlines rose from 0.205 to 0.994 and the standard deviation per dimension fell from 0.045 to 0.004: the model came to map every headline to practically the same vector. **The retrieval metrics of that run were discarded without being read**, because a collapsed representation produces a plausible table whose meaning is nil.

Table 5.3: Degeneration diagnostic, run before any retrieval metric is interpreted. The first column is the starting model; the second is the first attempt, with pairs at random; the last two are the two arms with stratified sampling. A cosine close to unity between *different* headlines means that every query returns the same neighbours, and no table of precision would give it away.

	unadjusted	at random	magnitude	direction
Cosine between different headlines	0.205	**0.994**	0.271	0.277
Norm of the mean vector	0.454	**0.997**	0.521	0.527
Standard deviation per dimension	0.045	**0.004**	0.043	0.043
Displacement from the initial model	—	0.269	0.782	0.775
Correlation of the similarities	—	0.234	0.607	0.630

With stratified sampling both arms pass the diagnostic, and pass it similarly to one another. The last two rows of the table are the ones that separate *not having degenerated* from *having learnt*: both arms displaced the space substantially relative to the starting model, with a mean cosine of 0.782 and 0.775 between the initial and the adjusted representation, and preserved the geometry of the similarities, with correlations of 0.607 and 0.630. The training, therefore, worked.

5.4.3 Result

The measurement runs over the 32,649 headlines of the test block, with five hundred queries per repetition across five repetitions, five neighbours per query and a three-day horizon. The queries are drawn before any model is applied, so they are identical across the three arms and the comparisons are paired. Table 5.4 presents the results.

Table 5.4: Materiality comparability and precision@5 by sector, over the test block. Comparability is the mean absolute difference, in percentage points, between the magnitude of the query's movement and that of the precedents returned, and *lower values are better*. Precision@5 is the metric of the previous section and is here to show what the adjustment costs in thematic relevance.

Arm	Comparability (pp) ↓	Precision@5 ↑
Unadjusted	$\mathbf{2.173 \pm 0.062}$	$\mathbf{0.771 \pm 0.011}$
Adjusted by magnitude	2.185 ± 0.065	0.746 ± 0.009
Adjusted by direction	2.157 ± 0.071	0.744 ± 0.006
Random pairing	2.259 ± 0.107	0.629 ± 0.007

On the quantity the adjustment exists to improve, nothing happens. The magnitude arm sits 0.012 points *above* the unadjusted model, that is worse, and the direction arm 0.016 below. Both differences are about a fifth of the standard deviation between repetitions of the arm itself, so they are not effects.

And the adjustment has a cost that can be measured. Precision@5 by sector falls from 0.771 to 0.746 and 0.744, a loss of about 0.026 with the same sign in both arms and larger than the dispersion between repetitions. What the adjustment changed in the representation was enough to degrade thematic relevance and not enough to improve materiality comparability.

Since the diagnostic of the previous subsection establishes that both arms trained and moved the space, **the reading is that the hypothesis is not confirmed, and not that the setup failed**. The comparison between the two arms adds the second part: being practically indistinguishable from one another, the experiment does not find the asymmetry between magnitude and direction that motivated the question.

5.4.4 Under the anteriority restriction

The previous protocol does not require the precedent to be prior to the query. With the measurement repeated under that requirement, which is the one production meets, the conclusions hold: comparability becomes 2.287 unadjusted, 2.309 for the magnitude arm and 2.289 for the direction arm, against 2.336 for chance, and precision@5 falls from 0.747 to 0.722 and 0.720.

There is in this protocol an observation the previous one does not permit. The margin of the unadjusted model over chance *shrinks* from 0.086 to 0.049 percentage points when anteriority is required. **Under the protocol production actually meets, the representation is even closer to chance on this quantity than the first measurement of this section indicated**, which strengthens the motivation for the question while making the negative answer sharper: the room to improve was larger than supposed, and the adjustment did not occupy it.

Two of the two thousand five hundred queries were excluded, there being not a single candidate prior to them within the block. The exclusion is declared here and in the generated report, and the filter is applied before any model is, so it is identical across the three arms.

5.4.5 The control arm collapsed, and the reason is measurable

The third arm starts from a domain encoder, to answer the objection that the result above might follow from having used a general-purpose model. It trained with the same stratified sampling, the same learning rate and the same number of pairs as the other two, and cost 35,386 seconds against 3,351 for the main arm, that is 28.3 seconds per step against 2.7.

It collapsed. The mean cosine between *different* headlines stands at 0.970, the standard deviation per dimension at 0.006, and the correlation of the similarity geometry with the starting point at 0.281. **Its retrieval metrics were not read**, for the same reason those of the first attempt were not.

The explanation is measured in the diagnostic itself, and is not a conjecture. The domain encoder, *before* any adjustment, shows a mean cosine of 0.599 between different headlines, against 0.205 for the sentence encoder. It starts, therefore, from a far less dispersed space, and a learning rate the sentence encoder tolerates is enough to tip it over. The threshold beyond which the adjustment degenerates is not a property of the recipe: it is a property of the starting model, and has to be determined for each one.

This measurement corroborates, by an independent route, what Section 5.3 establishes about the same encoder: applied in this way, the space in which it places sentences is little dispersed before any training, which is precisely the property a similarity measure requires and which it does not have.

What this arm does *not* establish is that a domain encoder cannot be adjusted. It establishes that it cannot be adjusted at this rate. A lower rate was not attempted, and the reason is cost: each run of this arm occupies the machine for about ten hours.

5.4.6 Limits of the conclusion

The result is negative about one training objective, one target and one configuration, and not about the idea of adjusting encoders. Three limits delimit it. The first is the target: magnitude enters as the percentile of the subsequent movement, and a target built differently might behave differently. The second is the configuration: one epoch ran over forty thousand pairs with a low learning rate, chosen because it is the one that avoids collapse, and no hyperparameter search was carried out. The third is the loss function: the classical alternative, which minimises the squared error between the cosine and the target, was declared in Section 3.6.1 and was not run. To these is added the learning rate of the control arm, which the preceding subsection shows must be determined per starting model and which was not determined for the domain encoder.

What the result does establish, and which does not depend on those limits, is the value of the gate: a collapsed representation produces metrics of normal appearance, and without the diagnostic of Table 5.3 the first run would have produced a complete and meaningless table.

5.5 News triage

The third case study answers RQ3: a trained model prioritises news better than a baseline built only on volatility. The answer is negative, and it is the result with the greatest informative value in the work.

5.5.1 Procedure

The dataset contains 79,753 examples between 2018 and 2023, each corresponding to a news item with the market context of the day and the label defined in Section 3.7. The split is chronological with an embargo, and the prevalence of positive cases in the test block is 0.378. The principal metric is PR-AUC, appropriate when the two classes are unbalanced.

For practical purposes, a PR-AUC difference below 0.02 is treated as indistinguishable, even when the paired interval excludes zero. The limit is a choice of the author, declared before the results, which requires it to be applied in both directions.

Two precautions in the setup ensure that the comparison does not favour the conclusion obtained. The first is the inclusion of an ensemble of trees trained by gradient boosting (Friedman 2001), the most expressive of the models compared, since it captures interactions and non-linear effects that logistic regression does not capture by construction. This model was run with the library's default parameters, without a hyperparameter search, which limits the reach of its defeat: it establishes that text produces no advantage in this setup, and not that it would produce none in a tuned non-linear model. The second precaution is the conversion of all scores into probabilities by *post-hoc* calibration over a held-out validation block (Niculescu-Mizil and Caruana 2005), without which the comparison by the Brier score would not be valid across models.

One input is not available in the same reference frame in production. The same-day return corresponds, in the historical set, to the complete movement of the day of the news, whereas in production the news is scored before that day closes. The result presented therefore evaluates the model with the complete daily variable and does not demonstrate parity with intraday scoring. The input belongs to the context block; the winning baseline uses only the volatility of the twenty preceding days, closed on the previous day. Correcting the reference frame can change the fit and ordering of the context models in either direction. Its effect remains to be measured through retraining and evaluation; the negative result applies to the protocol executed here.

5.5.2 Result

Figure 5.9 presents the five models and the no-information reference compared. The complete curves, of which these values are the area, appear in the artefacts identified in Appendix A. In them the volatility curve sits above the rest over almost the whole extent of the axis, which establishes that the difference is not an artefact of the aggregate value.

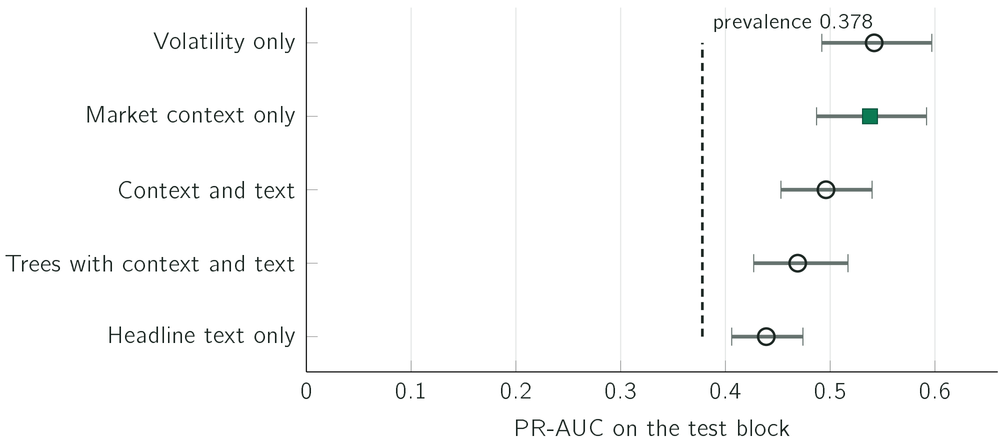

Figure 5.9: Area under the precision-recall curve of five models, with marginal ninety-five per cent intervals obtained by resampling the 1,951 groups. The dashed line marks the prevalence of the test block, which is the value obtained by a model that assigns the same probability to every example. The values are points, and the intervals are wide and overlapping: [0.492; 0.597] for volatility, [0.487; 0.592] for context and [0.453; 0.540] for context with text. The comparison the following subsection supports is the paired difference between models over the same resamples, and not the distance between two points.

The highest value belongs to the simplest model of all, the one that uses only recent volatility. Adding the rest of the context to it does not change the result distinguishably, at 0.538 against 0.542; adding the text of the headline degrades it distinguishably, at 0.496. The Brier score follows the same ordering, with 0.218 for the volatility model against 0.229 for the model with text. Always announcing one yields 0.622, but is a weak reference. The constant training prevalence, $q = 0.3854$, yields approximately 0.235 on the test: $\pi(1-q)^2 + (1-\pi)q^2$, with $\pi = 0.3781$. This is a descriptive calculation using the rounded prevalences in Figure 5.17, on page 92, and not a new run. The initial hypothesis, that the text of the news would bring information absent from the prices, is not confirmed over this corpus.

5.5.3 Three checks on the experimental setup

The negative result was submitted to three checks intended to determine whether it follows from the experimental setup.

The first concerns the fit of the text model. The comparison was repeated granting the text the best conditions available, with tuned regularisation, reduction of the text block to thirty-two dimensions and a language model specialised in finance. The best result of the text rises to 0.533 and stays below the 0.542 of volatility.

The second concerns sampling noise. The test block was resampled a thousand times by groups formed by company and day, since news items of the same day about the same company share the label and are not independent observations. The difference between volatility and the variant with text is +0.048, with an interval between +0.032 and +0.066, which excludes zero and exceeds the practical limit of 0.02. The value is the mean over the resamplings and does not coincide exactly with the difference between the two points, of +0.046. Of the comparisons in this section against the volatility baseline, it is the only one that exceeds it.

The reach of this interval is, however, limited: the resampling takes place within a single test block, with two hundred and twenty-one trading days, so it measures the sampling noise in that period and establishes nothing about stability across periods. The distinction is not theoretical in this work, since Section 5.7 reports a significant displacement in the distribution of the principal input. A rolling origin with three or four successive cuts would answer the wider question, and was not carried out.

The third concerns the definition of the label, which uses a threshold of two per cent and a horizon of three days, both chosen by the author. The comparison was repeated for nine combinations of threshold and horizon, and Figure 5.10 presents the result.

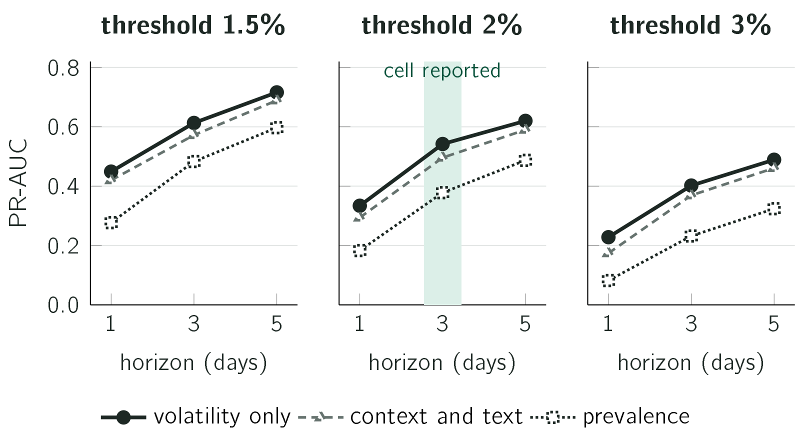

Figure 5.10: Performance of the two families under the three horizons, in three panels corresponding to the thresholds, with the prevalence of each cell as reference. The green band identifies the definition used in the rest of the chapter. Volatility matches or beats the model with text in all nine combinations, with prevalences between 0.082 and 0.597.

The negative result survives the three checks. Volatility matches or beats the model with text in all nine label definitions, with prevalences between 0.082 and 0.597, which removes from the concrete choice of threshold and horizon the status of a point of attack.

A fourth check bears on the composition of the blocks and leads to a conclusion that acts in both directions. Company composition changes across blocks, with substantial overlap. The training block runs between 2 January 2018 and 3 March 2022 and contains thirteen companies; the test block runs between 2 February and 18 December 2023 and contains nine. Five companies from training do not appear once in the test, and one company in the test, accounting for 17.1% of its rows, never appears in training. Thus eight of the nine test companies appear in training and account for 82.9% of test rows. For the companies present on both sides, the positive rate shifts markedly, and Apple goes from 0.448 in training to 0.183 in the test. These values are read directly from the identifier and block columns of the frozen set and require no additional execution.

This is not use of future information nor an error in the split: the cut is made by day, and the proportion respects the one defined in Section 3.7. It is the composition of the corpus that varies with time, for the same reason that makes the test block larger than the training block, since companies and their weights in the corpus vary between 2018 and 2023.

The signal of the model is essentially company identity, as Section 5.5.5 establishes. The overlap allows evaluation on known companies, but their positive rates change between periods. The comparison does not separate model quality from the transfer of these rates and the change in composition; this would require an analysis with constant composition. The difference between 47.0% positives in validation and 37.8% in the test can combine both effects, whose individual contributions were not measured.

5.5.4 Usefulness under the budget constraint

The product presents at most five alerts per day, so the question with operational meaning is what fraction of the five news items chosen each day are hits. Figure 5.11 isolates two references that look equivalent but are not. The model and the volatility-based alternative are compared in Figure 5.18, on page 93.

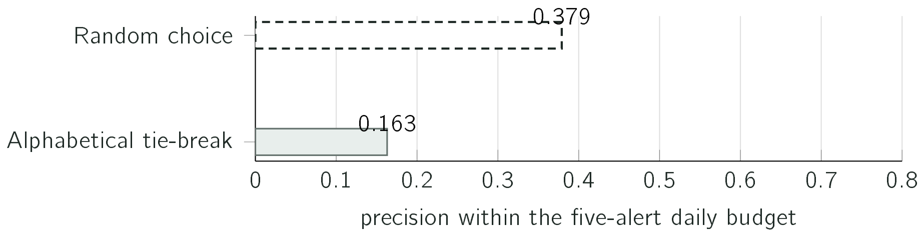

Figure 5.11: Two ways of choosing the five news items of the day, over the same test block. The alphabetical tie-break does not measure blind choice; random choice obtains 0.379 ± 0.017 over forty seeds. Both policies choose knowing the complete day; these values describe deferred selection and do not bound online precision.

The alphabetical reference requires an explanation: the original name, "always alert", is misleading. The model that always alerts assigns the same score to every news item, and when they all tie the metric selects by the order in which the rows appear in the file, which is sorted by date and by company name. That value of 0.163 measures choice in alphabetical order and not random choice, and every row it selects belongs to the company that comes first in the alphabet. Compared with the correct reference value, the gain of the model is 1.67 times and not the almost four times the first reading would suggest. The lesson is general and applies to the rest of the chapter: checking what the baseline measures is as necessary as checking what the model does.

The comparison with volatility, in Figure 5.18 on page 93, contains the uncomfortable result. A table of thirteen constants, corresponding to the mean volatility of each company, computed exclusively over the training block and without reading a single headline, obtains 0.662 and beats the trained model. The value is not that of the *volatility only* row of Figure 5.12, which is a regression on daily volatility and obtains 0.632 on this metric: here volatility enters as a constant per company, and Section 5.7 reconciles the two. The comparison has a reservation that makes it less clean than it looks: 17.1% of the rows of the test block belong to a company absent from training and receive the global fallback value instead of a constant of their own. It is, even so, the third case in the chapter in which the simpler method obtains the better result. What the evidence supports is that ranking by volatility pays off, and not that learning pays off.

5.5.5 Ablation of the model inputs

The previous result suggests that the deployed model recognises the company and not the news item. Checking that hypothesis consists in trying to reproduce it with a lookup table that assigns to each company a single number fixed on the training block, corresponding to the fraction of its news items that proved material. That table does not look at the headline, does not look at the day and invariably returns the same value for the same company. Figure 5.12 compares it with the deployed model and takes the model apart input by input, under the frozen protocol.

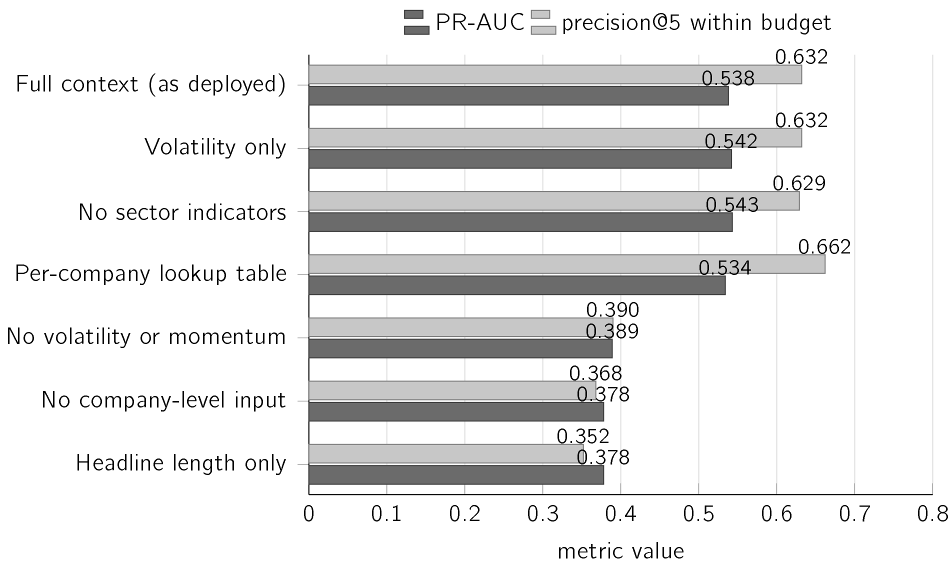

Figure 5.12: Each row corresponds to a variant with a subset of the inputs or to an alternative reference; the rows do not form a sequence of cumulative removals. The decisive row is that of the lookup table, which does not look at the news item and matches the deployed model on the first measure and beats it on the second. The two lowest rows, which have no company-level input at all, sit on the PR-AUC exactly at the prevalence of the test block.

Three readings follow from the figure.

The lookup table matches the model and beats it on one of the measures. On PR-AUC it obtains 0.534 against 0.538; on precision within the budget, which is the measure closest to real use, it obtains 0.662 against 0.632. This second difference was not accompanied by resampling by groups, unlike the PR-AUC differences of this section, and is therefore reported as an observation and not as established superiority. One number per company, fixed months earlier and never updated, produces the same effect as the trained model.

Without the company-level inputs no signal remains. Removing them, PR-AUC falls to 0.378, which is precisely the prevalence of the test block and the value defined in Section 5.1 as corresponding to the absence of information.

The only input that describes the news item adds nothing. The length of the headline, on its own, produces the same value, which makes the only per-headline information the deployed model receives indistinguishable from the absence of information.

These readings require two claims to be separated. The scientific conclusion of RQ3 stands: the variant with text has three hundred and eighty-four numbers per headline, and therefore has real information about the news item, and still obtained 0.496 against the 0.542 of volatility, that being the comparison that answers the question. The product decision, for its part, was wrong for a distinct reason: what was deployed as a decision gate was the variant without text, chosen by a metric that did not pose the question the product needed.

Keeping the nine-input variant does not follow from its having the best value among those that fitted in the execution container. By the criterion fixed before the results, four values are indistinguishable from one another: the 0.542 of volatility, the 0.543 of the variant without sector indicators, the 0.538 of the deployed variant and the 0.534 of the lookup table. None of the differences between them exceeds 0.009, far below the 0.02 limit. Applying that limit only when it disfavours a conclusion would amount to not having fixed it. The only difference the criterion separates in this comparison is the one that distinguishes these four variants from the block with text; measured against the volatility baseline, it is +0.048, with an interval between +0.032 and +0.066. The choice among the four is therefore entirely one of explainability and has no measurable price in PR-AUC: only the nine-input variant allows the alert to present the contributions that raise and lower the score, as Figure 4.6 shows.

The methodological lesson stands: PR-AUC over the complete set can take a high value when the variation between companies dominates, even in the absence of any ability to distinguish news items within the same company. The company-day label cannot evaluate differences between headlines in the same group; that ability requires a label per news item.

5.5.6 Text as an increment and not as a substitute

Comparing volatility, context and text in isolation answers which of the representations is better, and not whether the text adds information. Answering this second question requires adding it to a base that knows the company without knowing the news item.

The variant that combines context and text is not that base, since it adds the text to a block that contains information about the news item through the same-day return input. The appropriate base is the per-company lookup table, and the reason is methodological and not one of performance: it contains everything the model knows about the company and nothing about the news item, so adding the headline to it isolates the contribution of the text. The lookup table is not the best known predictor, since on PR-AUC volatility alone sits above it, at 0.542 against 0.534.

The criterion was fixed before the measurement: an increment only counts if the ninety-five per cent confidence interval of the difference, obtained by resampling by groups formed by company and day, excludes zero. The text representation was not tuned for this experiment, using the same reduction to thirty-two dimensions as the previous check, so that the result does not follow from searching for the convenient configuration. The resampling uses a thousand repetitions over 1,951 groups. Figure 5.13 presents the three intervals.

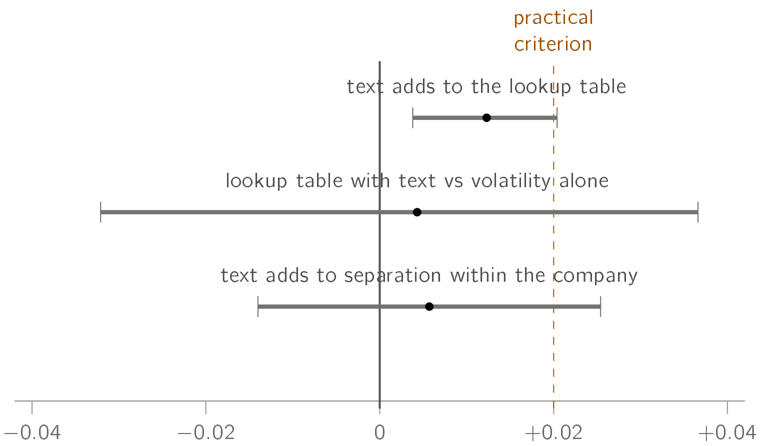

Figure 5.13: Differences in performance and their ninety-five per cent confidence intervals, obtained by resampling by groups. The first two rows are PR-AUC differences and the third is a difference in ranking ability within each company. Only the first excludes zero, and it does so below the practical criterion fixed before the measurement.

The headline adds to that base +0.012 of PR-AUC, with an interval between +0.004 and +0.020, which excludes zero. It is the first occasion on which a text representation shows a detectable increment on this task, and the value sits below the practical criterion of 0.02 fixed before the results. The increment and the base are rounded to three decimal places, so adding them may differ in the last place from the value the next reservation reports.

Three reservations delimit the reach of this increment, and all of them are measurements. The intervals of the first and the third are the ones Figure 5.13 draws, and neither of them excludes zero.

1. **The sum does not beat volatility alone.** The variant that adds the headline to the lookup table obtains 0.547 against the 0.542 of volatility, and the interval of that difference contains zero.

2. **The increment does not reach the product.** Precision within the budget of five alerts per day is 0.662 with and without the text, without a decimal place of difference. The increment exists in the ordering of the complete set and disappears when only five news items per day are selected.

3. **No ranking gain within the company is demonstrated.** The ranking ability within each company is 0.500 by construction for the lookup table and rises to 0.512 when the headline is added to it. For the deployed model it is 0.502, with an interval between

0.462 and 0.542, and rises to 0.508 with the text, an increment whose interval contains zero. Neither of the two is distinguishable from chance.

The answer to RQ3 remains negative and becomes more precise. No model with text beats volatility. The small aggregate increment does not yield a budget gain or demonstrate better ranking within the company. The shared label cannot identify information specific to each headline on the same day; that evaluation requires a label per news item.

A final reservation applies to the method itself. This is one more model evaluated over the same test block, which has already served several comparisons in the chapter, and the more often a test set is looked at, the more likely it becomes to find a small difference in it. The two defences used, which are not tuning the configuration and declaring the criterion before the measurement, reduce that risk without eliminating it. The formulation that survives is the weakest possible: it is not possible to reject that the headline adds information, and the increment, if it exists, is below the practical criterion fixed before the measurement. The observed value does not establish a guaranteed bound on the effect of the text.

5.5.7 Limits of the conclusion

The result does not establish that news text is useless. A single representation was evaluated, made up of headlines without the body of the article, over a single corpus. With full texts and a larger volume of data the answer could be different, which is future work.

There is also an alternative hypothesis that these measurements do not exclude. The label discounts the movement of the market assuming unit sensitivity, through the inconsistency declared in Section 3.7. In a stock of high sensitivity, the residual of the subtraction retains movement of the market, so the error of the label is not random: it is larger in the stocks most sensitive to the market, which are typically also the most volatile. And the winning baseline is precisely volatility. With what is measured it is not possible to separate the hypothesis that volatility better predicts materiality from the hypothesis that volatility better predicts the error introduced by the label. The experiment that would resolve the question consists in rebuilding the target with estimated and shrunk sensitivities, as Section 3.5 does, and repeating the six families. It was not carried out. The level and the ordering of the metrics remain conditional on the label used.

5.6 What remains measurable in the decomposition

The decomposition of the movement does not give rise to a research question. This section establishes the reason for that absence and what, in spite of it, remains falsifiable in the technique.

The difference from the three previous techniques is the existence of an external target. For detection there is a criterion of extreme movement, which allows hits and misses to be counted; for retrieval there is a verifiable notion of relevance, from which precision@k follows; for triage there is a label built from subsequent prices. In all these cases the technique can turn out to be wrong against a target external to it.

In the decomposition that target does not exist. No data source publishes what the true market share of the movement of a stock on a given day was, since that quantity is not observable and exists only relative to a model. With the sensitivities fixed, the split is an accounting identity and not a prediction that can turn out right or wrong. Constructing a measure of accuracy at this point would produce a value without a referent.

The technique is not left unevaluated, but unevaluated in that way, which requires determining what in it remains falsifiable. There are two properties, and both are measured over the seventeen companies of the sector map, which is the set the evaluation procedure covers, distinct from the list monitored in production. Figure 5.14 presents the two.

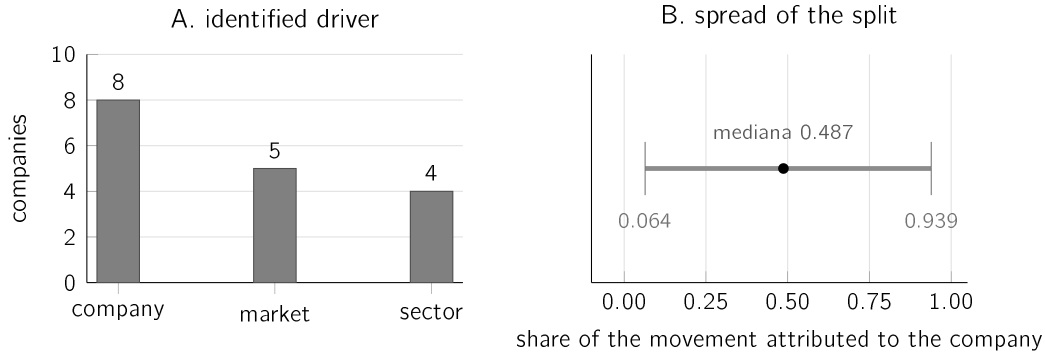

Figure 5.14: The seventeen companies of the sector map, decomposed on the same day. In A, the component identified as the largest part of the same sign as the observed movement. In B, the dispersion of the share attributed to the company. The split produces distinct answers for distinct companies on the same day, which is the minimum condition for it to be useful.

The first property is the ability to discriminate. A decomposition that attributed the whole of the movement to the company on every day would be useless even if arithmetically correct. On the day measured, the driver identified was the company itself in eight cases, the market in five and the sector in four, and the share of the movement attributed to the company had a median of 0.487, with extremes at 0.064 and 0.939.

The second property is the quality of the estimation of the sensitivities, and it is the one that can actually be wrong. The appropriate measure is the coefficient of determination in the estimation window, whose median was 0.460. This is not the coefficient of the least squares fit, which, computed within the sample and with a constant term, could never be negative, but the coefficient of the model actually used, with the sensitivities already shrunk and the constant term recentred. This is the appropriate measure, since it corresponds to the model whose split is presented to the user, and it is for that reason that it can take a negative value. In one of the seventeen companies it did, which means that, in that window, the market and the sector did not explain the variation of that stock better than its own mean, and that the split of that day rests on a fit that does not describe the data.

The result does not establish that the split is correct because the parts sum. They always sum, with a null difference to machine precision, because the company part is defined as the residual, and a split with no meaning at all would sum just as well. The exact sum is an implementation check and not a validation of method. The result also does not establish that the value presented to the user is reliable on every day, which in the case of a negative coefficient it is not, and it is for that reason that the limitation appears in Chapter 6.

5.7 The system in operation

The fourth case study bears on the behaviour of the system after deployment, on the decisions it takes autonomously. It is the less common of the two evaluations described in Section 3.1, and the one that produces the diagnosis that changed the system.

5.7.1 What the decision gate was separating

The deployed gate was a fixed threshold on the calibrated probability, which let through the candidates above 0.50. The question with meaning is whether that threshold separated headlines or separated companies, and the measurement was carried out over the 36,925 decisions actually taken in production between 22 July and 20 August 2026. Figure 5.15 presents the result.

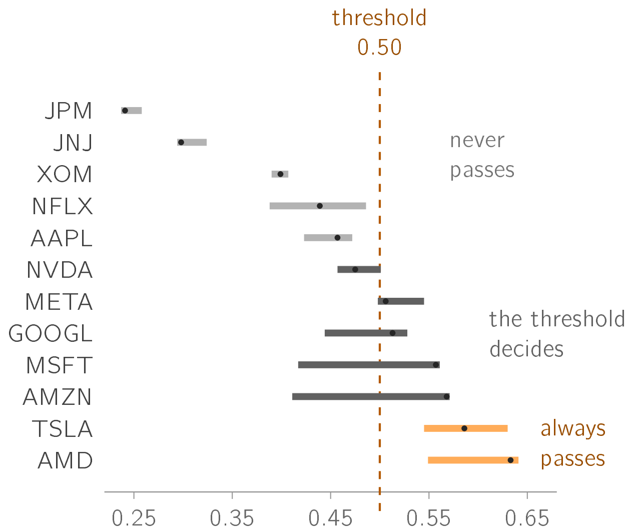

Figure 5.15: Range of all the scores the model assigned to each company in production, over the 36,925 decisions logged between 22 July and 20 August 2026, with the median marked. The light bars never reach the threshold, the dark ones cross it and the orange ones sit always above it. Seven of the twelve companies lie entirely on one side, which means that for those the decision was taken before the headline was read.

The values confirm the reading of the figure, and they reproduce across two distinct observation windows, as Table 5.5 shows.

What the score separates is companies, not headlines. In none of the assessments logged did Apple reach the threshold, with an observed maximum of 0.472, whereas AMD exceeded it in all of them. It is the same company whose movement Section 3.5 splits.

The fraction of cases in which the outcome was determined before the headline was read admits two counts, and both are reported. Counted once per distinct headline it is 48%; counted per logged decision it rises to 60%. The first is the one cited, and the reason is that the system re-scores the same headlines at every sixty-second cycle: that duplication is greater in the companies with fewer news items, which are precisely the ones that never exceed the threshold, so counting decisions shifts the fraction upwards, and in the direction that suits the conclusion.

Table 5.5: Selectivity indicators for the decision gate across two observation windows. In both, the spread between companies is several times the spread within each one: the conclusion that the score separates companies, and not headlines, does not depend on the window chosen. Both spreads are rounded to three decimal places and the ratio is computed on the unrounded values, so the division redone on the printed numbers may differ in the place shown.

Indicator	**36,925 decisions 22/07 to 20/08**	**4,366 decisions to 15/08**
Mean spread within each company	0.072	0.064
Spread between the company medians	0.392	0.385
Ratio between the two	5.4×	6.1×
Companies always above the threshold	2 of 12	3 of 12
Companies where the threshold decides	5 of 12	4 of 12

The defect has the same form as the one identified in Section 5.2, one layer up. In that case, a fixed threshold on the return measures the volatility of the company instead of the rarity of the day; in this one, a fixed threshold on a score that is almost constant per company ranks companies instead of news items.

None of the existing automatic tests covered it. The model was evaluated over the distribution of the training set and deployed behind two filters that alter it, and each component fulfilled its contract. The test that would have caught it is the one this work now runs: comparing the spread of the scores within each company with the spread between companies. What was wrong was an assumption about the link between them, stated nowhere. It corresponds to the class of cost that D. Sculley et al. (2015) place under data dependencies and entanglement between components, with the symptom the authors describe: the system goes on producing normal-looking values.

There is a second possible cause, which the logs do not allow to be separated from the first. One of the nine inputs is the same-day return. In training it corresponds to the complete movement of the day of the news. In production it is computed over the last daily bar available, which during the session is the previous close or an incomplete bar. The two causes produce the same symptom, and the log, at the date of this measurement, stored the probability and the outcome but not the value of the inputs at the moment of scoring, so separating them would require logging those inputs and observing again for weeks.

That instrumentation was built and deployed on 4 September 2026, and the log came to store, per decision, the value of each of the nine inputs, the date of the last price bar used and the identity of the model that scored. Two consequences follow from it and neither is a result. The first is that the decisions prior to that date remain non-reproducible from the log, since recomputing today the inputs of a past day would use a price series that already contains what came after; the measurements of this chapter rest on those decisions and are not affected by the change. The second is that the observation window needed to separate the two causes began on that date, and the acceptance criterion of any future comparison was fixed before a candidate existed, for the same reason that Chapter 3 invokes for the remaining experiments. Nothing is asserted here about its outcome.

The apparent correction would consist in replacing the fixed threshold with a threshold relative to each company. Simulating that alternative over the same decisions solves half the problem, since the twelve companies come to be represented, and introduces another. In the companies whose score barely varies, the percentile falls on the constant and almost every decision ties above it; one of them comes to generate 874 alerts. The relative threshold stops choosing by materiality and starts choosing by tie-break, which is precisely the artefact documented in Section 5.5. Within a company the score of the model barely varies with the headline, and no decision rule applied to that score can make it sensitive to the news, because the information that would distinguish one headline from the next is not present.

As a consequence, the model ceased to be a gate and became a ranking criterion, which is the use the measurement supports, and volume control moved to a daily budget of five alerts. This change brings together two policies that diverged: the dissertation evaluated a budget policy and the system deployed a threshold policy, so the value reported described a system distinct from the one in operation. The two policies came to belong to the same family without becoming identical. Table 5.6 brings the three policies together with what each one produces.

Table 5.6: The three decision policies considered for the gate, and what each produces over the same twelve companies. The second was never deployed: it was simulated over the logged decisions, and it is that simulation which rules it out. The concentration values refer to the news alerts actually delivered; the figure of 874 is the number of alerts the simulation assigns to a single company.

Policy	**What it does**	**What it produces**
Fixed threshold *(the one in place)*	The model vetoes: a news item passes if its score exceeds a single value	7 of the 12 companies get to alert; concentration 0.713
Per-company threshold *(simulated, rejected)*	The value is no longer single and becomes a percentile within each company	In companies whose score barely varies the percentile falls on the constant: one of them would generate 874 alerts
Ranking with a budget *(the one deployed)*	The model stops vetoing and starts ranking; volume is cut by a budget of five alerts per day	9 of the 12 companies get to alert; concentration 0.480

The change was measured over the alerts actually delivered, and the intended effect is observed. The quantity observed is concentration, defined as the fraction of the alerts that belongs to the three most alerted companies, and its advantage is being dimensionless, which allows windows of distinct sizes to be compared. In news alerts, which are the ones the budget governs, concentration falls from 0.713 to 0.480, with the confidence interval of the difference excluding zero, and the number of companies that come to produce an alert goes from seven to nine of the twelve monitored.

What supports the reading is not, however, the fall on its own, but the comparison with a series the change does not govern. The price movement alerts run through the same period, the same companies and the same market conditions, and the daily budget does not count them. In that series concentration goes from 0.449 to 0.485, with the interval containing zero. The governed quantity shifted and the ungoverned quantity did not, which rules out the alternative explanation that the market would have distributed itself more evenly in the second period.

Five days are excluded from the later window, and the reason is recorded because it is itself a result. Between 25 and 29 August 2026 the daily counter resided on ephemeral disk and returned to its initial value at every start of the process, which produced runs of exactly five alerts at the moments of the restarts and one day with twenty. On those days the budget was not in force, and including them would raise the later concentration to 0.514. The resampling is done by day, which is the unit on which the budget acts.

The comparison is not a randomised trial. The control series shares the period but was not assigned at random, and the two windows differ in duration and in market conditions. What is established is that the governed quantity shifted and the ungoverned one did not, and not that no other cause exists.

The metric evaluated ranks all the candidates of the test block and only then fills five slots on each day, which requires knowing the complete day before deciding and is therefore a deferred policy. The deployed system decides online, every sixty seconds, and spends the first slot on the first news item that crosses the gates, because at the moment of the decision the afternoon news item does not yet exist. The precision within the budget reported in this chapter describes that deferred selection. Knowing more candidates does not guarantee higher precision when scores can be wrong; the relationship with the online policy's precision remains to be measured.

5.7.2 The performance of the model over the decisions it took

The system logs every triage decision and, once the days needed for the price to respond have passed, confronts them with the actual outcome. Two measurements result from that log: the fraction of cases that proved material on each side of the gate, and the ranking ability of the model, measured by Receiver Operating Characteristic Area Under the Curve (ROC-AUC). Figure 5.16 presents both, over 825 decisions already matured.

Over the population it actually observes, this sample does not distinguish the model from chance on either measure. The news items it let through proved material in 58.9% of cases, against 61.7% of those it stopped, over a base rate of 60.2%; the intervals of the two fractions overlap, so the apparent ordering is not established. What is measured is the absence of a detectable benefit at this sample size, and not the absence of benefit.

These values require a considerable reservation about precision: the 825 decisions are not 825 independent observations, since the label is assigned per pair formed by company and day, and the rows are distributed across only 239 distinct pairs. Measured over those units, the ranking ability stands at 0.486 with a confidence interval between 0.403 and 0.571, an interval in which chance is exactly 0.500. The correct reading is not that the model is worse than no model, but that this sample does not distinguish it from chance.

The measurement was repeated when the log had 825 matured decisions against the 530 of the first reading, and almost nothing changed: the difference between kept and suppressed narrowed, the ranking ability went from 0.494 to 0.486 and the interval shrank. With half as much evidence again, the verdict stands and becomes firmer.

The most likely explanation is not that the model is broken, but that it is redundant. When it is invoked, the news has already crossed the relevance filter and the freshness filter, and

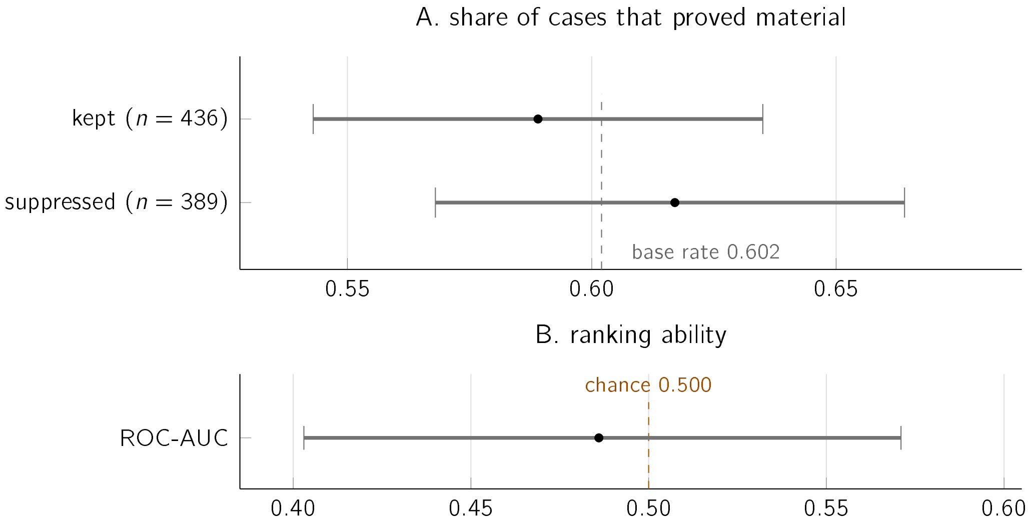

Figure 5.16: In A, the fraction of cases that proved material among the decisions the model let through and among those it stopped, with ninety-five per cent confidence intervals and the base rate of the set. The suppressed ones show a higher fraction than the kept ones, but the two intervals overlap over almost their whole extent and do not separate them. In B, the ranking ability measured over the 239 independent units, with the interval comfortably containing the chance value.

those elementary filters removed a large part of what the model was trained to remove, leaving little to separate. A model evaluated in isolation and deployed behind filters was never evaluated on the distribution it actually observes, and this is the finding of the work with the greatest capacity to transfer.

Observing that distribution has been instrumented. On 4 September 2026 the decision log began to retain, for every candidate assessed and not only for the survivor of each cycle, the value of each input and the date of the last price bar used, as described in Section 4.7. That change makes observable, for the first time, the population the model would have to rank if it were the one triaging, rather than only the fraction the elementary filters hand it. The analysis protocol was fixed before any measurement existed: the unit is the pair formed by company and day, only decisions whose price bar is no later than the news date are usable, and below a minimum number of matured pairs no metric is published. Nothing is asserted in this document about the outcome of that observation, which did not exist at the time of writing.

Three of the twelve companies monitored, AMD, Netflix and Meta, do not appear in the training corpus, and a fourth, Microsoft, appears only in the test block. The model scores them all the same, because what it receives are numerical values and not the identity of the company, but it does so over a combination of inputs it has not observed. One of the absent ones, AMD, is one of the two companies that always sit above the threshold, and another, Meta, has a median immediately above it, so the determination by company described above also bears on names the model never observed. It is a second, more literal sense in which the deployed model operates outside the distribution on which it was evaluated, and the decisions of those companies enter the 825 above like any others.

5.7.3 Drift of the distributions

A second reason, independent of the previous one, can make a deployed model cease to serve: the data change. The model was trained over the period from 2018 to 2023 and operates in 2026. The measure used is the PSI, or population stability index, which here compares the distribution of each input in the training block, between 2018 and 2022, with that of the test block, of 2023. It is not, therefore, the distance between training and the present, which the following subsection discusses separately: it is the drift internal to the protocol itself. The bands are those conventional in credit risk, under which values below 0.10 indicate stability, values between 0.10 and 0.25 moderate displacement and values above 0.25 significant displacement. Figure 5.17 presents the result.

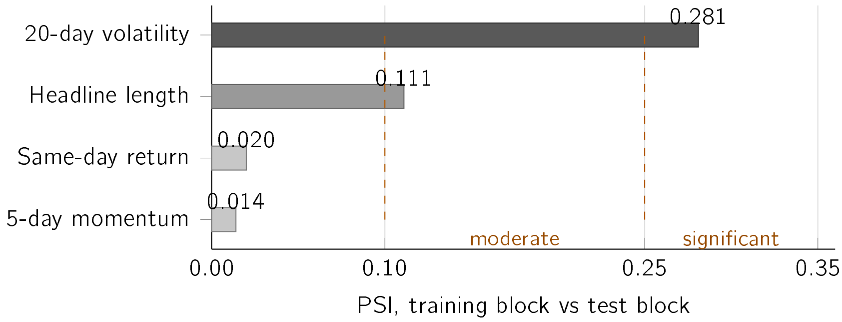

Figure 5.17: Population stability index of each input between the training block, from 2018 to 2022, and the test block, of 2023, with the conventional bands marked. The input the model most depends on is the one that shifted most. The prevalence of the label goes from 0.3854 to 0.3781 between the two ends, but it is not stable in between: in the validation block, temporally between the two, it reaches 0.4704.

The result has two halves. The input that shifted most is the volatility of the twenty preceding days, in the significant band, whereas the rest remain stable or moderate. The label shifts little between the ends, with the prevalence of positives going from 0.3854 to 0.3781, but it is not stable: the validation block, temporally between the two, reaches 0.4704. The prevalence oscillates with the volatility regime instead of drifting in one direction, which is consistent with volatility being the input that shifts most.

The correct reading of this measurement is that the drift is not external to the values of this chapter: it is the one the protocol itself runs through, since it trains on 2018–2022 and evaluates on 2023. The PR-AUC reported is already a measure under this drift, and the combination is unfavourable because the input the model most depends on is precisely the one that shifted most.

What this measurement does not establish is the distance between training and 2026, which is the period in which the system operates. A snapshot over current quotes points in the same direction, with volatility again being the input that shifts most. It rests, however, on consecutive rolling windows that share almost all their observations, which inflates the index and makes its magnitude not comparable with that of the figure. The value remains unmeasured, and is not estimated by analogy. Drift is, in either case, measured and not corrected, and Section 6.5 states what would be needed to correct it.

5.7.4 The complete system against the existing alternatives

The previous sections compare component by component. What is missing is the comparison at system level: given the news of a day, does the policy choose the five to present better than a rule available at no cost. Figure 5.18 answers. All the policies choose five per day, over the same test block and with the same label, and the tie-break is random and explicit in all of them.

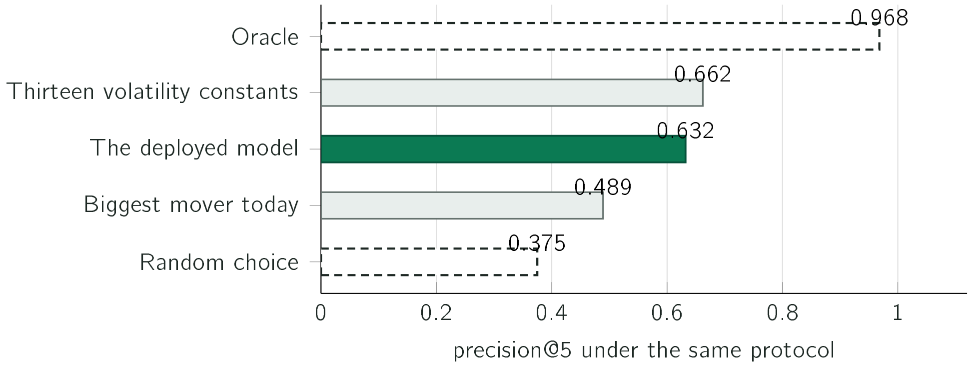

Figure 5.18: Five selection policies over the same test block. Random choice serves as the reference, and obtains 0.375 here against the 0.379 of Figure 5.11: they are two runs of the same draw by distinct procedures, and the difference lies within the spread across seeds of either of them. The oracle is a theoretical maximum that is not usable, obtained with knowledge of the answers. All of them choose knowing the complete day. The usable policies describe deferred selection; only the oracle establishes the maximum possible on this set.

Three readings follow from the figure. The first is that the system beats the realistic alternative: presenting news from the companies with the largest movement of the day is what any stock application already does at no cost, and it obtains 0.489, against 0.632 for the system, a gain of 0.143 under the same protocol. This value measures the ability to rank the approximate label on the historical block, and not the usefulness perceived by a person nor the causal value of the alerts in operation.

The second is that the system still does not beat volatility. Thirteen constants computed exclusively over the training block obtain 0.662, which is consistent with the ablation of Section 5.5.5: the model is, in practice, a table per company, and volatility is another table per company, simpler, and the one that requires no training. It is not the same quantity that Figure 5.12 labels *volatility only*, which is a regression on daily volatility and obtains 0.632 on this metric; here volatility enters as a constant per company.

Two distinct constants per company obtain, on this metric, the same value of 0.662: the mean volatility of each company, here, and the fraction of its news items that proved material, in Section 5.5.5. The coincidence is not fortuitous and follows from the shape of the measure itself. Precision within the budget depends only on which five news items are chosen on each day; and a score that is constant per company reduces that choice to an ordering of the companies that have news on that day. Two predictors that order the companies in the same way therefore select exactly the same news items, and obtain the same value, even though the quantity they assign to each company is different. That two such distinct quantities produce the same ordering is the observation of this section in another form: what separates the companies in this set is, to a large extent, their volatility.

The third, and most useful, is the location of the limitation given by the theoretical maximum. The best possible result, choosing with knowledge of the answers, would be 0.968, which means that on almost every day there are indeed five material news items in the available set. The system is not limited by scarcity of positively labelled rows, but by its selection. The margin between 0.632 and 0.968 is 0.336. Since all headlines from a company on the same day share the label, exchanging headlines within that group does not change precision. This margin measures distance from ideal selection of company-day groups; it does not locate an inability to distinguish headlines within a group or independently verify that ability.

All the policies in this comparison choose from among what the source delivered, which leaves out the question prior to them all, concerning the days on which nothing was delivered. Over a year of captured news, there was at least one relevant news item on 88.5% of the days the detector considered unusual, out of a total of 417, and on 47 of the 52 most extreme days, a size that does not allow the two fractions to be distinguished. On the remainder, no change to the system has any effect, since a news item not associated with the company by the free source never enters the path. Coverage is also very uneven, with the worst covered company standing at 50%. This value is an upper bound, and the distinction determines what it is worth: it establishes that a news item existed, and not that the right news item arrived. Section 6.4 takes it up as a limitation.

One baseline was not measured, and the reason is recorded. The most natural alternative would consist in presenting the first five news items to arrive. It is not measurable under this protocol, since the historical dataset does not keep the publication time and is sorted by date and by company, so using the order of the file would measure the alphabetical order of the companies. It would be the same artefact documented in Section 5.5, this time introduced deliberately. A wrong baseline is worse than a missing baseline.

5.7.5 Perceived usefulness of the alerts delivered

The channel now accompanies every alert delivered with two buttons, *useful* and *did not help*, and records the vote of whoever presses. The analysis rules were fixed before a single vote existed and were not changed afterwards: a minimum of twenty effective votes to report any proportion, one vote per person and per alert with the last replacing the previous one, a safeguard that repeats the computation without the dominant voter when that voter accounts for more than forty per cent of the votes, always subject to the same minimum, and Wilson confidence intervals, calculated without adjustment for clustering. Only votes on alerts present in the shared log were counted.

131 valid votes were recorded between 2026-09-02 and 2026-09-11, corresponding to 91 effective votes from 3 distinct people, on 62 alerts. 11 votes were later changed by the same person, and only the last of each pair counts. Another 29 repeat an earlier vote without changing it and count only once. A further 5 votes, not counted among the 131 above, were excluded for not corresponding to any alert in the shared log.

Of the 91 effective votes, 86 rated the alert as useful, that is 95%, with a 95% Wilson confidence interval between 88% and 98%. This binomial calculation does not adjust for dependence between votes by the same person or on the same alert; its width does not represent all the uncertainty in this sample.

A single person accounts for 58% of the effective votes, above the limit of 40% fixed in the protocol. Excluding that person, 38 votes remain, of which 34 rate the alert as useful. The proportion without that person is 89%, with an interval between 76% and 96%. This is the reading to retain.

The independence between whoever builds the system and whoever rates it is a condition of this sample: the author is not among the votes counted. The votes come from readers of the public channel, and the log keeps of each only the conversation identifier assigned by the platform.

These votes are observational feedback in a real context and do not replace the controlled study that Section 6.4 identifies as the gap of greatest weight in this work, and which remains to be carried out.

Four limitations accompany this result and none of them is resolved by more votes. The first is self-selection: whoever wants to votes, and those who find an alert indifferent tend to press nothing, which pushes the sample towards the extremes. The second is the absence of a counterfactual, since no group received the price change without the explanation that accompanies it; nothing here attributes the usefulness to the explanation itself. The third is the distance between perceived usefulness and a better decision, which is the founding hypothesis of this work and remains untested. The fourth is not knowing who votes: the channel is public and it is not known whether those who answer belong to the audience the dissertation assumes.

5.8 Síntese dos resultados

This section brings together the comparisons carried out throughout the work, including those in which the system keeps the option the measurement does not favour.

Two of the three answers are affirmative and one is negative. The negative answer is the one that required the greatest effort of verification, and it is also the one that establishes that the evaluation was built so as to be able to produce a negative answer. Table 5.7 presents each technical choice beside the alternative measured against it.

Three rows of the table deserve development. The first concerns the financial domain encoder, whose defeat has its explanation in the training objective and not in domain knowledge, as Section 5.3 establishes. The second concerns the twenty-day window, which is the hardest option to justify, since the only measurement available on this point does not support it and the responsiveness justification is not quantitative. The third concerns the volatility estimator, kept for explainability against an alternative measured as superior.

The chapter documents three occasions on which a simpler technique beat a more sophisticated one under equivalent conditions, and three occasions on which the reverse happened and the system kept the deployed option all the same. In the latter the reason differs in each case, and none of them is the result of the measurement: the equally weighted estimator was kept for explainability, the twenty-day window for responsiveness, and the smaller sentence encoder for computational cost. The distinction between the two sets is the distinction between a result and a choice, and Chapter 6 takes it up when establishing what this work demonstrates and what it merely decides.

Table 5.7: The technical decisions of the work, each with the alternative measured against it. Three rows record that the deployed option did not obtain the best result, and the reason for keeping it appears in the respective column. The last three rows are constraints fixed before any measurement, and not results: presenting them as a comparison would give them an authority they do not have.

Deployed option	**Alternative measured**	**Reading**
Rolling z-score	Fixed three per cent threshold: spread of 0.344 against 0.015	A fixed threshold measures the volatility of the company and not the rarity of the day.
Rolling z-score	*Isolation Forest* with F_1 of 0.269 and *Local Outlier Factor* with 0.280, against 0.530	The learned detectors flag almost everything and are seldom right.
Twenty-day window	Did not obtain the best result: ten days produce F_1 of 0.385, twenty days 0.516 and sixty days 0.678	The measurement favours the longer window. Kept for responsiveness, with the reservation about the circularity of the label declared.
Equally weighted standard deviation	Did not obtain the best result: exponential decay produces F_1 of 0.664 against 0.516	Kept for being explainable to a non-specialist reader, and recorded as a choice and not as a result.
Sentence embeddings	Word overlap: precision@5 of 0.196 against 0.395	Common vocabulary does not imply a common subject, and a common subject does not require common vocabulary.
MiniLM instead of MPNet	Did not obtain the best result: MPNet produces 0.422 against 0.395; a financial domain encoder produces 0.275	The heavier model is better and was set aside on cost. The domain encoder obtained the worst result of all, by measurement and not by assumption.
Logistic regression	Gradient boosted trees: PR-AUC of 0.469 against 0.538	The more expressive model did not obtain a better result, and the simple model is explainable.
Platt calibration	Isotonic regression, compared over the same validation set	Platt calibration matches or beats it on the Brier score in all five families assessed: 0.218 against 0.223 on the volatility one and 0.224 against 0.226 on the deployed one.
Cosine similarity	Euclidean distance	With normalised vectors the two produce the same ordering, and the first has a more direct reading.
Two triggers, news and price	Fusion with volume and news intensity, along the lines of panels that combine signals of distinct origins (*World Monitor* 2026): it loses in two of the three budgets, by −0.050 and −0.032, and wins in the third by +0.004	A gain that depends on the budget chosen, and an order of magnitude smaller than the losses, does not support adoption. Volume continues to be computed and presented as context, without giving rise to alerts.
Free sources only	—	Founding constraint: it defines the audience the system can serve.
No price forecasts	—	Design constraint: it is what makes everything else verifiable.
United States market	—	Scope constraint: it is where the free sources have coverage.

Chapter 6

Conclusions

Chapter 5 evaluated the four components. This one brings together what those measurements yield. It synthesises the work carried out, presents the answer obtained for each research question, identifies the limitations of the study and sets out the lines of future development. Whoever reads it will know what the work establishes, what it decides without establishing, and what remains untested.

6.1 Main conclusions

The work produced a system in operation and an evaluation of each of the techniques that compose it. InvestiGator monitors twelve companies in a continuous cycle, delivers explained alerts to a public channel and provides a page on which both the messages sent and the decisions that led to silence are recorded. Between 9 July and 13 August 2026, 367 messages were delivered; between 22 July and 20 August, 36,925 triage decisions were logged; and the reconstruction of the precedent base gathered 38,214 cases with measured impact. These numbers describe a deployed system and not a demonstration built for the purpose.

The component that gives the work the character of a dissertation is not, however, the system, but the evaluation to which it was submitted. Three of the four techniques were compared with simpler alternatives, chosen and declared before the measurement was carried out, which is the condition for the result to be able to be negative. In one of them the result was indeed negative. The fourth technique, the decomposition of the movement, does not admit a comparison of the same nature, since the split of a return has no observable external referent. What was measured of it is what remains falsifiable: the quality of the fit that supports it, and the ability to produce distinct answers for distinct companies.

The engineering contribution does not lie in the size of the learned model. The system trains a single model, with nine inputs, and that model loses to a one-variable baseline. Confidence in that value, and in the negative conclusion it supports, depends on what had to be built around it. Section 4.7 describes the seven pieces. The dataset is labelled from two independent sources. The label is declared as a decision, and nine alternative definitions are measured instead of the one adopted being assumed. The temporal separation is verified by an automatic procedure. The calibration comes with the measurement and the declaration of the bias that affects it. The artefacts are kept under version control. The deployed model is demonstrably the model evaluated. And the system in operation continues to be observed. It is in that set, and not in the model, that the work resides.

A second conclusion runs across the four case studies and belongs to none of them in particular. On four distinct occasions, a simpler technique obtained a better result than a more sophisticated one under equivalent conditions; on three others the reverse happened, and the system nonetheless kept the deployed option, for a distinct reason in each case. Figure 6.1 gathers the seven.

the simple technique obtained the better result

rolling z-score against two learned detectors
F_1 of 0.530 against 0.269 and 0.280

recent volatility against text representations
PR-AUC of 0.542 against 0.496

thirteen constants per company against the trained model
precision@5 of 0.662 against 0.632

unadjusted encoder against the one adjusted for materiality
precision@5 of 0.771 against 0.746

the alternative obtained more, and the system kept its own

exponential decay obtains more, and was not adopted
F_1 of 0.664 against 0.516 · explainability

sixty-day window obtains more, and was not adopted
F_1 of 0.678 against 0.516 · responsiveness

MPNet obtains more, and was not adopted
precision@5 of 0.422 against 0.395 · cost

Figure 6.1: On the left, the four occasions on which a simpler technique obtained a better result than a more sophisticated one under equivalent conditions; the last is the most recent, and in it the simpler technique is the absence of adjustment. On the right, the three on which the reverse happened and the system nonetheless kept the deployed option, with the reason for each. The distinction between the two columns is the distinction between a result and a choice, and Table 5.7 gathers them with the rest.

The distinction between these two cases is the distinction between a result and a choice, and the present chapter keeps it explicit: what the measurement establishes is reported as a result, and what the author decided in spite of the measurement is reported as a decision, accompanied by what that decision costs when the measurement separates the alternatives, and by the statement that it costs nothing in PR-AUC when it does not separate them.

6.2 The answers to the research questions

This section answers the four questions stated in Section 1.3 and delimits, in each case, what the answer does and does not allow to be claimed. Figure 6.2 brings the four together.

6.2.1 Detection of unusual movements

The answer to RQ1 is affirmative. A z-score over a rolling twenty-day window flags rare days at a practically identical rate in companies of very different volatilities. The spread between the most and the least flagged company is 0.015, against 0.344 for a fixed threshold rule at three per cent. Against an approximate label, the method obtains $F_1 = 0.516$ and the fixed rule 0.218. Subjected to an equivalent causal protocol, the same method further beats two detectors designed for this task, with F_1 of 0.530 against 0.269 for *Isolation Forest* (F. T. Liu, Ting, and Zhou 2008) and 0.280 for the *Local Outlier Factor* (Breunig et al. 2000).

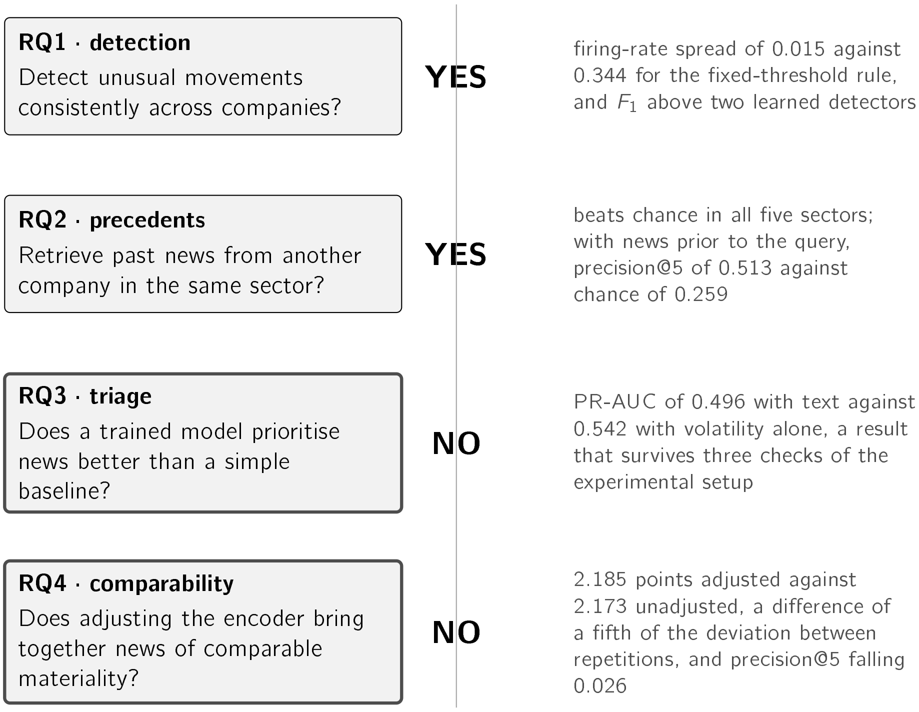

Figure 6.2: The four answers, with the two negative answers presented with the same prominence as the two affirmative ones. The answer to the second question is limited to the evaluated alternatives and the proxy criterion of sector relevance.

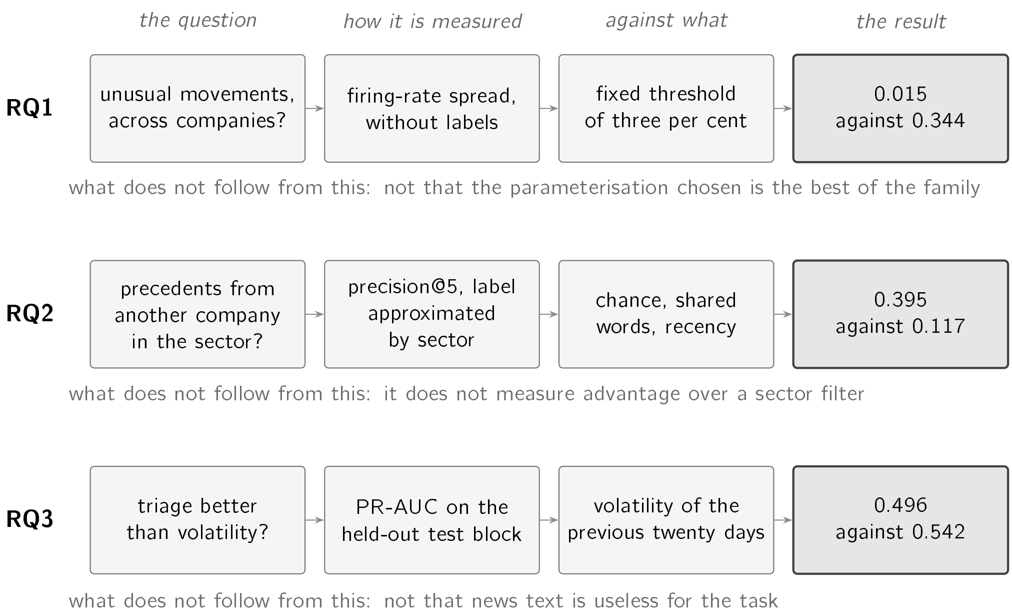

Figure 6.3: For each question, the path from the question to the result, and the line that delimits the reach of that result. The first three columns are the protocol, fixed before the measurement; the fourth is the value obtained against the alternative. The bottom line of each block is what distinguishes an evaluation from a demonstration, and is developed in the subsections that follow.

The deployed configuration is not, however, the best measured within the family itself. With the same recall, an exponentially weighted estimator obtains $F_1 = 0.664$ and a sixty-day window obtains 0.678, against the 0.516 of the configuration in production. What RQ1 establishes is that the rolling family beats the fixed threshold and the two learned detectors, and not that the parameterisation chosen is the best of that family.

This result is less surprising than the difference in values suggests, and the literature allows it to be situated. The two learned detectors define rarity from the geometry of the data, one by isolating the point and the other by comparing densities, and were designed for spaces with multiple interacting variables. The problem treated here has a single series of returns per company, in which the relevant structure is temporal and not geometric: agitated periods follow calm ones. It is the property that ARCH and GARCH models formalise (Bollerslev 1986; Engle 1982). The rolling window captures it directly; a detector trained over the complete set would have to learn it. When the assumption built into a simple method coincides with the actual structure of the problem, the margin for beating it is small.

The reach of the answer is delimited by three conditions. The comparison covers fifteen large-capitalisation companies, selected in 2026 and evaluated over the past, which survived by construction; three years of quotes; and a single market. The label used as a secondary argument is itself relative to the volatility of each company, which is why the principal argument is the consistency of the firing rate, which does not depend on labels. The three per cent threshold of the alternative rule was not optimised, so what the comparison establishes is not that no fixed threshold would obtain a better result, but that a fixed threshold rule measures a quantity distinct from the one intended.

6.2.2 Precedent retrieval

The answer to RQ2 is affirmative against the evaluated alternatives. Over the recent corpus, retrieval by sentence similarity (Reimers and Gurevych 2019) obtains precision@5 of 0.395, against 0.196 for vocabulary overlap, 0.117 for random choice and 0.204 for presenting the most recent news. Here too the deployed model is not the best measured: a larger variant obtains 0.422 under the same protocol, and was set aside on computational cost.

The aggregate value is not, however, the appropriate way of stating the result. Eighty-four per cent of the news in the complete corpus belongs to the technology sector, which makes available a strategy that dispenses with any model and does not even look at the query, consisting in invariably returning news from that sector; its precision equals the fraction of technology queries, 0.840 in the complete corpus. That strategy is useless as a product, since it is right on every query of one sector and on none of the remaining four, but its existence shows that the aggregate measures in part the composition of the corpus.

The comparison that controls for this composition is carried out within each sector, where the method beats the respective reference value in all five. The absolute margin is largest in the most abundant sector, at $+0.220$ in technology; in the sparse sectors, where the reference value is around 0.100, the method obtains several times that value with a smaller absolute margin. The scope is that of the evaluated alternatives: if selection could use the known sector of the query, filtering by that sector would suffice to satisfy the label, provided enough candidates existed. The per-sector analysis does not measure semantic advantage over that filter.

Two further checks delimit the answer. Over the complete historical corpus, with approximately eighty thousand news items from six years, precision rises to 0.595. The reference value rises too, and faster, so the margin falls from +0.278 to +0.262. The method does not degrade with scale, nor does it improve with it. Requiring the precedent to be prior to the query reduces precision to 0.513 and the reference value to 0.259, with the margin varying by 0.008. The method keeps its advantage under this chronological restriction. The test does not impose the eight-day maturation used in production and therefore does not fully validate outcome availability.

The second caveat has direct consequences for the product. Retrieval identifies topic and not direction: the agreement between the direction of the movement of the query news item and that of the precedents stands at 0.708 against a chance value of 0.688, a difference of 0.020. Two news items can deal with the same subject and the price move in opposite directions. The result is consistent with what the semi-strong form of the efficient market hypothesis would lead one to expect (Fama 1970), under which public information is already reflected in the price, without the work testing it or relying on it to establish any conclusion. The practical consequence is to present the precedents individually, with date and outcome. The historical alert reproduced in Figure 4.6 also includes an average, which summarises those cases and does not constitute a forecast for the incoming news item.

A final delimitation concerns the label. Membership of the same sector approximates relevance in a verifiable way, without coinciding with the property one intended to measure. The lexical baseline is the most elementary form of its family, without weighting by the rarity of terms and without normalisation by length. The seventeen percentage points of advantage measure the distance to a starting point, and not to a tuned ranking function (Robertson and Zaragoza 2009).

6.2.3 News triage

The answer to RQ3 is negative. No variant incorporating the text of the headline beats a baseline built only on the volatility of the twenty preceding days, with 0.496 of PR-AUC against 0.542, and the Brier score follows the same ordering. The result survives three checks. The first repeats the comparison granting the text the best conditions available, and raises its best value to 0.533, without reaching the baseline. The second resamples by groups formed by company and day, and keeps the interval excluding zero. The third repeats the measurement for nine alternative combinations of label threshold and horizon, and volatility matches or beats the model with text in all nine.

The answer gained precision over the course of the work and that precision changes what is drawn from it. The earlier comparisons place the representations side by side and therefore answer the question of which of them is better. The question of greater use to whoever builds a system is a different one, and consists in determining whether the text adds to what is already known. Measured against the per-company lookup table, which contains everything the model knows about the company and nothing about the news, the answer is affirmative: the headline adds +0.012 of PR-AUC, with an interval between +0.004 and +0.020, which excludes zero. It is the only occasion on which a text representation shows a detectable increment on this task.

The value falls below the 0.02 limit that Chapter 5 fixed before the measurements. The observed increment falls below that practical criterion; it does not establish a guaranteed bound on the effect of the text.

Three measurements prevent this increment from reopening the verdict. The sum does not beat volatility alone, and the interval of the difference contains zero. The increment does not survive the metric that describes the product: precision within the budget of five alerts per day stays at 0.662 with and without text. Ranking within each company remains at chance level. These measurements do not demonstrate an operational gain from text: the label is shared by all news items of a company on the same day, so the metric evaluates company-day selection rather than distinguishing between headlines from that day.

There is an established line of research that uses text to anticipate price movements (Ding et al. 2015; Xing, Cambria, and Welsch 2018), and this result does not contradict it. The difference does not lie in the size of the text, since Ding et al. (2015) likewise operate on headlines, but in the treatment given to it. Those authors extract event structures and train a network to predict the direction of the movement. Here headlines are compared by similarity, and direction is never the object of a prediction. The corpora and the evaluation horizons likewise differ. The claim this work supports is narrower than a refutation: under this data constraint and with this family of methods, the textual signal adds nothing to what recent volatility already indicated. The reach of the result is therefore limited to a regime of resources and to a choice of representation, and does not extend to language.

There is, even so, a theoretical reading that frames it and that is stated with the same caution. The semi-strong form of the efficient market hypothesis (Fama 1970) holds that publicly available information is already reflected in the price. If that is so, a public headline has, at the moment it is read, no information that recent prices do not contain, and the result obtained is what that hypothesis would lead one to expect. The resource condition of this work worsens the expectation rather than easing it: between the publication of a news item and its detection by free sources a median of 353 minutes passes, an interval during which any market reaction has already occurred. The system observes, by construction, information that has ceased to be new.

This reading is a framing, without the standing of a conclusion. The work does not test the efficient market hypothesis, and a negative result obtained with one family of methods over a concrete corpus does not confirm it; confirming it would require a design this one does not have. What the reading adds is the reason why the result is less surprising than the initial hypothesis assumed, and why its most likely correction lies not in the representation of the text but in the delay with which it arrives.

6.2.4 Materiality comparability

The answer is **no**, and it is negative in both halves of the question. Contrastive adjustment of the encoder does not improve the materiality comparability of the precedents retrieved: the arm trained on magnitude obtains 2.185 percentage points against 2.173 for the unadjusted model, that is a worse value, and the one trained on direction obtains 2.157. Both differences are about a fifth of the deviation between repetitions of the arm itself. And the adjustment has a cost the measurement does separate: precision@5 by sector falls by about 0.026 in both arms, with the same sign and above the dispersion.

What distinguishes this result from a setup that failed is the diagnostic that precedes it. The first attempt collapsed the encoder, and was discarded without its metrics being read; the two arms that produced the values above displaced the space substantially relative to the starting model and preserved the geometry of the similarities. They trained, therefore, and what is not confirmed is the hypothesis.

The comparison between the two arms was the part of the experiment meant to separate magnitude from direction, and it did not find the asymmetry that motivated it: the two are practically indistinguishable from one another, in the diagnostic as much as in the two measures. The conjecture that magnitude would be partially learnable from text where direction is not therefore remains without support in this setup.

A third arm, starting from a domain encoder, was trained as a control and collapsed, so its metrics were not read either. The diagnostic shows why: the space of that encoder is, before any adjustment, far less dispersed than that of the sentence encoder, and the rate the latter tolerates is enough to make it degenerate. The collapse threshold is a property of the starting model and not of the recipe.

The consequence for the system is to keep what it already did. The case base is organised by theme, and the alert states that the similarity concerns topic; this result establishes that, by this route, the guarantee that the precedent's magnitude is comparable cannot be added to it. The decision described in Section 6.2.2 to present the cases one by one rather than summarise them by a mean is thereby reinforced.

6.3 Objectives achieved

Of the four objectives stated in Section 1.3, three are fully met and one is met with the caveat described below.

The first objective, to build the complete system from data acquisition to the delivery of the alert, is met. Both triggers work end to end in production, on free sources, and the path of an event is documented in Chapter 4 with the values of a real case.

The second objective, to evaluate the components with transparent baselines, is met for the three techniques in which an external referent exists. In one of the cases it went beyond what was planned: the detector was compared not only with a fixed threshold rule, but also with two learned methods. The decomposition of the movement is the caveat: there is no ground truth against which to measure the accuracy of a split, so what was measured is what remains falsifiable in it, corresponding to the quality of the fit and to the ability to discriminate. Internal comparisons that would be possible were not carried out, such as that of the one-factor model against the two-factor model, or that of the raw sensitivities against the shrunk ones.

The third objective, to ensure that the explanations presented are faithful to what the system computed, is met by construction, without depending on subsequent verification. Every value displayed comes from the object that computed it, and the text is composed from those pieces, so it is not possible for the alert to state a value distinct from the one the system obtained. Faithfulness is, however, a property of the system, and not of the relation between the system and whoever reads it; the usefulness of the explanations for a retail investor was not measured, and Section 6.4 deals with that absence.

The fourth objective, to report the results obtained including those that do not confirm the initial expectations, is met. The negative answer to RQ3 appears in Figure 6.2 and in the body of this chapter with the same prominence as the others, accompanied by the three checks to which it was submitted and by the three occasions on which a simpler technique beat a more sophisticated one.

6.4 Limitations

This section lists the limitations of the work in order of importance. Table 6.1 presents them together, with the work that would close each one and the nature of its cost; the text that follows develops the four of greatest consequence and summarises the rest. Most are quantified. Two are not, by their nature: the absence of a controlled study is an absence and not a measure, and the distance between each label and the property it approximates would only be measurable against a human annotation that does not exist.

Table 6.1: The eleven limitations, in order of importance, with the work that would close each one and the nature of the cost of that work. The two that depend on calendar time are the ones that require human judgement, and they are also the two on which the rest most depend: until a sample annotated by people exists, neither the usefulness of the explanations nor the quality of the labels ceases to be an approximation.

Limitation	**What closes it**	**Cost**
No controlled study with people	Run the protocol designed, with six to ten participants	calendar time
The score is almost constant within each company	Inputs that vary from one headline to the next	small; retrains
The label may favour the winning baseline	Rebuild the target with estimated sensitivities	small; retrains
The labels are verifiable approximations	Annotate a sample with human judgement	calendar time
The alert claims more evidence than it has	Measure impact per news item and not per day	intraday data
Company composition varies across the three blocks	A split of constant composition	small; re-evaluates
Only the headline is read, not the body of the article	Collect and represent the full text	new data
The split is not always well estimated	Compare one factor with two; the fit is now shown	small; re-evaluates
Drift is measured and not corrected	Retrain with the production log	small; retrains
Source coverage is incomplete and uneven	Add sources, or declare the limit	outside our control
The three causes of the delay are not separated	Log when each item appears in the feed	instrumentation

No controlled study with users was carried out. This is the gap of greatest weight. The system rests on the hypothesis that an explanation accompanied by its evidence leads a non-specialist to a better decision, and that hypothesis was not submitted to an experimental design. The protocol exists, with participants, tasks and criteria defined, and was not executed. What does exist, and is reported in Section 5.7.5, is feedback collected in a real context: readers of the public channel rate the alerts they receive, under analysis rules fixed before the first vote, and the collection continues. The literature is explicit about the weight of this absence: claims of interpretability demand evaluation and are not established by declaration (Doshi-Velez and B. Kim 2018). The reason is not a formal one. The effect of an explanation on a person is not deducible from its construction: it is known that trust in automated systems fails in both directions, with the user ignoring a correct warning or accepting an incorrect one (J. D. Lee and See 2004), and that the mere presence of an explanation increases the acceptance of wrong answers (Bansal et al. 2021). The system was built to produce appropriate trust, presenting evidence rather than issuing a verdict, but that it succeeds remains a hypothesis. In the absence of the study, the two failure modes are indistinguishable in the logs: an ignored alert and an unduly believed alert produce the same absence of data.

The distinction between the two determines what each one allows to be claimed. The channel feedback is observational: whoever wants to votes, which shifts the sample towards the extremes; no group received the price change without the explanation that accompanies it, so nothing attributes the usefulness to the explanation itself; and the channel is public, so it is not known whether those who answer belong to the audience this work assumes. What those votes measure is perceived usefulness among those who chose to answer. The founding hypothesis concerns a better decision, which is another quantity, and it is that one which remains untested.

Two observations from the literature change the weight of this absence, in opposite directions, and both belong here. The first aggravates it. The founding hypothesis is not merely undemonstrated in this work: it has published measurements against it. Waa et al. (2021) found no difference between example-based explanation and no explanation, and Cau et al. (2023) found, in a stock trading task with non-experts, lower performance for the inductive style than for the abductive and deductive styles when the model is confident. Until the study is carried out, the honest position is not "still to be demonstrated" but "still to be demonstrated, with published evidence pointing the other way". Section 2.7 states the reason this work may still expect a different outcome, since it delivers the underlying computation and it is the absence of that computation those authors hold responsible for the result; but a reasoned expectation remains an expectation.

The second locates it. The closest published work in this class of system (Muntermann 2009) evaluated its artefact by simulation and likewise deferred the study with users to future research, fifteen years earlier. The review that supports Chapter 2 found, within this class of system, no study establishing causally that alerts accompanied by evidence improve a non-expert's decision. This neither excuses the absence nor makes it less of a limitation, but it changes what it means: it is not a step this work skipped, it is a step the field has not yet taken.

The deployed model could not perform the function assigned to it. The variant put into production as a decision gate receives seven company-level inputs, one day-level input and a single one that varies from one headline to the next, the length of the headline. Its score is, by arithmetic and not by accident, almost constant within each company: the mean spread within each company is 0.072 and the spread between the medians of the companies is 0.392. In 48% of the distinct headlines, the outcome was determined before the headline was read.

Two claims must be separated, since only one of them is an error. The result of RQ3 stands, because the variant with text had three hundred and eighty-four numbers per headline and still lost. The product decision was wrong, and the reason is methodological: the choice was made by a metric that interrogates the ordering of the complete set, when the product needed to select opportunities under a daily budget. The company-day label cannot evaluate the distinction between two headlines published within that same unit.

The system was changed and the model ceased to be a gate and became a ranking criterion. That change is not validated. Over the 825 decisions matured in production, grouped into the 239 independent units the label admits, the ranking ability is 0.486, with an interval between 0.403 and 0.571, which contains chance. Ranking is the least indefensible use of the model, and not a demonstrated use.

This failure belongs to a class described in the literature on machine learning systems in production. D. Sculley et al. (2015) identify a cost that appears in no quality metric, and that comes from data dependencies and from entanglement between components: the validity of one piece depends on assumptions about another that were never stated. The model was evaluated in isolation, over the distribution of the training set, and deployed behind two filters that alter it. No test failed and no log flagged an anomaly. A learned component is evaluated where it is trained and is useful where it is placed, and when those two places differ the difference does not announce itself.

The label may favour the winning baseline. The label discounts the movement of the market assuming unit sensitivity, so its error is not random: it is larger in the stocks most sensitive to the market, which are typically the most volatile. The winning baseline is precisely volatility. With what is measured it is not possible to separate the hypothesis that volatility better predicts materiality from the hypothesis that volatility better predicts the error of the label. Both the level and the ordering of the RQ3 metrics are therefore conditional on the label used.

The alert claims more evidence than it has. When it displays three precedents that agree on the direction of the movement, the alert presents that agreement as resulting from three observations. The impact is, however, measured per pair formed by company and day, so three headlines of the same company on the same day share the value by construction. Over the 247 alerts with precedents delivered, in 36.8% the cases displayed rest on fewer distinct days than the number of cases presented, and in 11.3% they all correspond to the same day; of the 120 alerts that claim unanimity, 23.3% rest on a single day. The wording of the text was corrected to count days and not headlines, which makes the claim true. The cause remains: while impact is measured per day, two news items of the same day are not two independent observations.

The remaining seven, in summary. The **labels are approximations**: relevance is approximated by membership of the same sector, materiality by the occurrence of a movement above a threshold, and rarity by a percentile of the company's own returns. All three are objective and verifiable, and none corresponds exactly to the property one intended to measure. Dependence on approximations is not peculiar to this work: in neighbouring financial domains, such as fraud detection, unsupervised methods occupy a central position because labelled anomalies are scarce (Ahmed, Mahmood, and Islam 2016).

The **company composition varies across the three data blocks**. Training contains thirteen companies and the test nine: eight are shared, five appear only in training and one only in the test. The latter accounts for 17.1% of test rows; known companies account for the remaining 82.9%. There is substantial overlap, but the frequencies change. The comparison does not isolate the effect of that change from model performance, nor demonstrate uniform generalisation to new companies.

The **news is read only at headline level**. The system represents the headline and not the body of the article, which limits what the text representation can capture and is a plausible, though unconfirmed, explanation for the negative result of RQ3. Reviews of textual analysis in finance describe works that operate over complete documents (Kearney and S. Liu 2014), and a headline is the part of the document where the most information has been discarded.

The **split of a movement is not always well estimated**. The sum of the three parts holds by construction and validates nothing. The informative measure is the coefficient of determination in the estimation window, whose median was 0.460 and which in one of the seventeen companies took a negative value, which means that in that window the fit did not describe the data. The interface now shows that coefficient alongside the three parts, compared against the median across those seventeen companies, and names in words the case in which the fit does not describe the data. The estimate does not improve by being displayed, and the limitation remains; what stops happening is the reader receiving a split with no way of knowing when it is weak. Precision-weighted shrinkage (Vasicek 1973) limits the damage of a bad estimate without turning it into a good one.

Drift is measured and not corrected. The parameters were estimated between 2018 and 2023 and were not readjusted. Between the training block and the test block, the population stability index places twenty-day volatility in the band of significant displacement, at 0.281, and that is the input on which the model most depends. The concept drift literature deals with this problem, with a taxonomy of the change and strategies for detection and adaptation (Gama et al. 2014); here detection exists and adaptation does not. The architecture that would supply it is defined in Section 4.7, with a conditional trigger, a rolling window and a promotion gate, and was not executed: the limitation is one of execution and not of design.

Source coverage is incomplete and uneven. There was at least one relevant news item on 88.5% of the 417 days the detector considered unusual, and the worst covered company stands at 50%. On the remaining days no change to the system has any effect, since a news item not associated with the company by the source never enters the path. The value is an upper bound: it establishes that a news item existed, and not that the appropriate news item arrived.

The **delay is in discovery, and its three causes are not separated**. Over the 278 alerts with a publication time and a send time, the median between publication and detection is 353 minutes, and the median between detection and delivery is 5 seconds. The move from a best-effort scheduler to a sixty-second cycle did not reduce the delay, which rules out the polling cadence as the explanation. Three causes remain that the history does not separate, and only one of them, the source listing late, would be mitigated by a paid service; the other two persist with any source.

6.5 Future work

The lines stated in this section derive from the limitations of the previous section, according to the correspondence presented in Table 6.1, and are ordered by the relation between expected effect and cost. A list of future work that does not derive from measured limitations is a justification in another form. Figure 6.4 groups them by what each one requires before it can be carried out.

1. **Run the study with users.** Six to ten participants, with sessions of approximately fifteen minutes, comparing the reading of an isolated fact with the reading of the complete alert. It closes the central claim of the work, which is that the evidence presented alongside the alert leads to a better decision. It shares with the construction of an annotated target, the third item of this list, the property that no computational

means accelerates it, since both depend on human judgement. Other claims remain to be verified independently of it, among them the quality of the three labels, the usefulness of the model over the population it actually observes and the stability of the result across periods.

2. **Give the decision model inputs that vary with the news.** It is the direct correction of the limitation of greatest technical consequence, and its cost is small, since the two most obvious quantities are already computed by the system: the similarity of the best precedent retrieved and the direction agreement among the precedents. Unlike volatility and sector, both vary from one headline to the next.
3. **Build a target with human judgement.** Annotating a sample of days, identifying which of the news items suppressed today should pass and which of those that pass add nothing, would supply for the first time a real target to optimise, in place of tuning constants against an approximation. The same sample would allow the quality of the three labels used to be estimated.
4. **Rebuild the label with estimated sensitivities.** The triage label discounts the movement of the market assuming unit sensitivity, which introduces an error correlated with the volatility of the company, which is precisely the winning baseline. Redoing the target with the shrunk sensitivities described in Section 3.5 and repeating the six families of models would determine whether the ordering obtained survives an alternative definition of materiality.
5. **Read the body of the articles and not only the headline.** It is the most direct hypothesis for explaining the negative result of RQ3 and is testable with the protocol already built, requiring only the collection of the full text.
6. **Detect repeated stories by meaning.** Suppression of repetitions rests today on vocabulary overlap, and doing it by similarity would use the technique the system already has. The point matters more than its technical size suggests: if the retail investor buys the assets that capture their attention (Barber and Odean 2008), a system that communicates the same story three times is not informing, but amplifying the same stimulus.
7. **Remove the asymmetry of reference in the same-day return.** The input corresponds, in training, to the complete movement of the day of the news, and in production to the last bar available at the moment of scoring. Replacing it with the return accumulated up to publication, or, more conservatively, with the previous day's return, removes the difference and allows performance to be reported under a single reference. The effect on the comparison between models remains to be measured: it requires rebuilding the inputs and repeating training and evaluation.
8. **Group the precedents by company-day pair before selection.** The correction applied to the text of the alert made the claim true without removing the cause. Selecting the candidates after grouping by day, displaying from each day only the closest headline, eliminates at source the possibility of one day observed three times being presented as three agreeing cases.
9. **Expose the quality of the fit in the message itself.** The split of a movement is displayed with the same confidence when the coefficient of determination of the window is high and when it is negative. Presenting it alongside the split, and adding

an explicit note below a minimum threshold, turns a limitation declared today only in the document into a property observable by whoever reads the alert.

10. **Retrain the model with the production log.** Drift is measured and the new labels derive from prices, so execution is possible. The impediment is not technical, since a retraining would change the values reported in this document and would require evaluation time of its own.

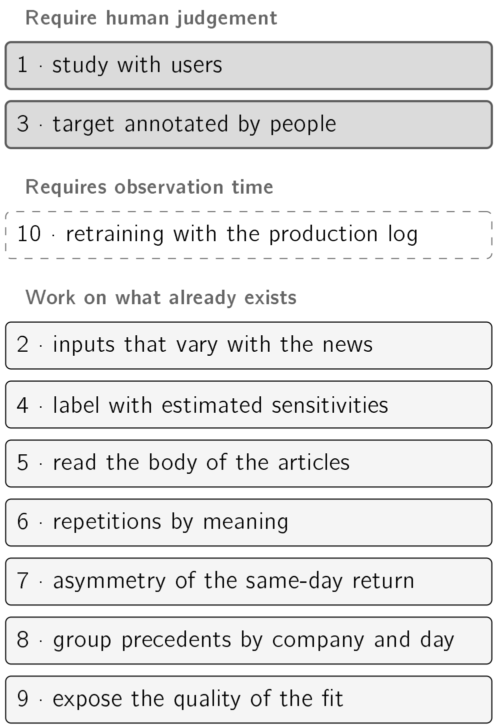

Figure 6.4: The ten lines of future work, grouped by what each one requires before it can be carried out. Seven operate on components the system already has and depend only on technical work. One awaits observation time, which began to run in September 2026. Two require human judgement and are the only ones that no computational means accelerates, which places them on the critical path despite not being the ones of greatest technical cost. The numbering is that of the list in this section, which is ordered by the relation between expected effect and cost.

One path was considered and not followed, and the reasons are recorded because they are needed by whoever takes up the work. Conformal prediction allows sets of answers to be returned with a distribution-free guarantee (Angelopoulos and Bates 2023; Vovk, Gammerman, and Shafer 2005), which suits a tool that refuses to claim more than it knows. Two reasons ruled it out. The first is that the guarantee is marginal, holding on average over cases and not case by case, which is precisely the reading a non-specialist user would make.

The second is more decisive: the guarantee requires exchangeability between calibration and test, and these data are a time series, in which the headlines arrive in order. It is the same assumption this work refuses when it splits the set by date and with an embargo. Applying the method without addressing that point would produce intervals with a guarantee the data do not support, replacing an honest value with a value that has the appearance of rigour.

Three technical directions are likewise recorded, and they are of another nature. They do not derive from measured limitations, and therefore do not belong to the list above; they follow from implementation decisions that Section 2.4 states and that the work had no reason to revisit.

The first is reranking with a cross-encoder. Retrieval compares each pair of sentences through vectors computed once, an option the scale of the case base imposes. An encoder that processes the two sentences jointly produces a finer comparison and is impracticable over tens of thousands of cases, but practicable over the first twenty results of a query. The evaluation protocol already built measures exactly that list.

The second is learning to rank under a budget constraint. The problem the product solves is not classifying each news item in isolation, but choosing five per day, and the main metric of Chapter 5 asks the first formulation. There is a family of methods dedicated to the second, and the difference between the two is measured in this work without being named: it is the distance between the PR-AUC over the complete set and the precision within the budget. Feng et al. (2021) study the ranking of news by their influence on an asset, combining the asset and event perspectives. That work offers a formulation to explore; the benefit under this system's daily budget remains to be measured.

The third is replacing the exhaustive search with an approximate index (Johnson, Douze, and Jégou 2021), mentioned in Section 2.4 as a path of growth. At the current size of the base the exact search is practicable and exact, so the replacement is not justified today; it would be justified from an order of magnitude above.

6.6 The boundary of the claim

The contribution is stated at its exact level, and the separation matters more than either half of it. What this work establishes is that the system is technically prepared to support explanations resting on verifiable historical evidence: every quantity presented comes from the component that computed it, can be checked against the record at the moment it is read, and the policy that decides the silence is documented and measured. What this work does not establish is that this form of explanation leads the retail investor to a better decision. That is the founding hypothesis, it remains untested, and its test is the first item of the previous section. Figure 6.5 presents the separation.

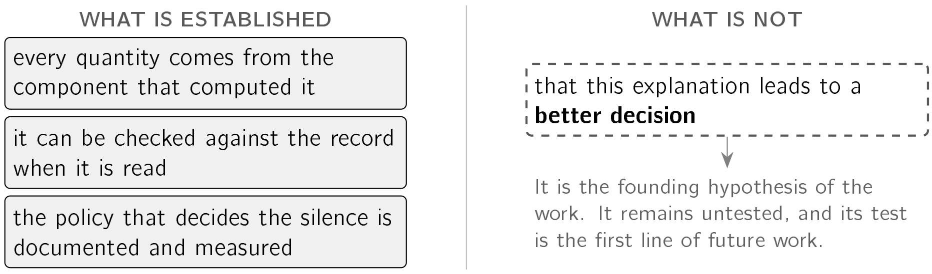

Figure 6.5: The exact limit of the claim. On the left, the three properties the system demonstrates and that are verified by inspecting what it produces. On the right, the proposition the work did not test: that presenting the evidence alongside the alert leads to a better decision. The separation is not a courtesy caveat, since the second is the hypothesis that motivated the work, and distinguishing it from what was demonstrated is what prevents the first from being read as though the second were proved.

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Appendix A

Reproducibility

This appendix documents the provenance of the results presented in the body of the dissertation. It lists the artefacts kept, establishes the correspondence between each result reported and the procedure that produces it, presents the state in which each substantive claim was left after being measured, and records the licensing conditions of the data used.

A.1 Artefacts kept

This section lists what is kept with the work and what is not, with the reason in each case. The code, the results files and the model artefacts described here accompany the dissertation and can be made available to the committee; this appendix exists so that the provenance of every value is verifiable from this document, without depending on that access.

The execution environment is described in Section 3.8, with the exact versions of the libraries that produced the values in this document. The versions are pinned, and not expressed as ranges, because a library that updates itself changes results without issuing a warning. None of the procedures requires a graphics processing unit.

Three classes of artefact are kept. The first is the learned component, comprising the trained model, the calibrator fitted on the validation block and a descriptor file of the run that produced them, with the size of each block, the prevalence of positive cases, the seed used and the metrics obtained. The second is the set of results files, one per evaluation procedure, that the text cites. The third is the reconstructed component of the precedent base, with fifty-six megabytes of vectors and eight of metadata; the base the deployed application consults is the fusion of this one with the live base, as Section 4.2.3 describes. This last one is not a sample: it is the artefact the system needs in order to start, and without it there would be no precedents to present.

The source corpora are not kept. The historical set occupies several hundred megabytes and is rebuilt by the preparation procedures described in Section 3.2. Reduced samples are kept in their place, whose purpose is to let the automatic verification run without network access, including the test that enforces the temporal separation described in Section 3.7 and that Listing 3.3 reproduces.

A.2 Origin of each result

This section establishes the correspondence between each result reported in the body of the document and the procedure that generates it.

No result in this dissertation was entered manually into the text. Each one is produced by an automatic procedure with a fixed seed, which writes its own results file, and it is that file the text cites. A second run over the same data returns the same values. Table A.1 brings together the results reported and identifies the section in which each one appears.

The section given in the third column states, for each result, the protocol, the baseline against which it was compared and the population it covered; and Section 3.8 pins the exact library versions and the seed. A reader who has the source data, whose provenance and licensing are given in Section A.4, has what is needed to rebuild any of the measurements without recourse to this work.

The determinism guarantee applies to the upper block of the table. The rows of the lower block are of another nature, and are therefore separated: they are measurements over the operation of the system, whose value depends on the moment at which the log was read and not on a seed. The section in which each one is presented declares the period and the set over which it was carried out.

One of those measurements is not regenerable, and the distinction is recorded rather than omitted. The breakdown of the funnel by gate, presented in Section 4.4, was read from the decision log on a concrete day, and that log keeps only three days, for the reason given in Section 3.2: it is republished at every cycle, so the cost is one of publication and not of storage. Running the same procedure on a later date produces a different day. The value was kept together with the date of the reading and the identification of the procedure that produced it, so that any future reading has a declared origin even though that day does not recur.

The remaining numerical values in the document belong to three categories. Values taken from the literature are cited where they appear. The configuration constants appear in Table 4.2, with the distinction between those derived from a measurement and those chosen. The third category brings together measurements that serve a passage and not a result, such as the rows discarded by the embargo, the standard deviation of the worked case of Section 3.4 or the counts per news source; each one declares, where it appears, the set over which it was carried out.

A.3 Evidence matrix

This section submits each substantive claim of the work to the evidence that supports it and records the state in which it was left.

The table of the previous section lists results. Table A.2 lists claims, a more demanding examination, since a claim may not survive the evidence that supported it. The claims withdrawn remain in the table, since a matrix that lists only the survivors is not an audit.

Two observations bear on the table as a whole and not on any single row.

The first is that withdrawn claims are compared with measurements of the work itself. The possibility of negative answers is part of evaluation; reviewing their scope also requires checking whether the interpretation matches the measured protocol.

The second is that the two rows marked as not claimed carry the same weight as the rest. One of them is the usefulness of the explanations to the user, which is not established for lack of a controlled study, despite the observational feedback reported in Section 5.7.5. Where the available evidence does not support the claim, the claim is not made.

Table A.1: The results reported in the body of the document and the section in which they appear. The upper block is deterministic and reproduces exactly over the same data; the lower block brings together measurements over the operation, whose value depends on the date the log was read.

Result	**Value**	**Where it appears**
Firing-rate spread, relative deviation against a fixed threshold	0.015 against 0.344	§5.2
F_1 against the approximate label	0.516 against 0.218	§5.2
Comparison with two learned detectors	0.530 / 0.269 / 0.280	§5.2
Ablation on the window and on the weighting	0.678 and 0.664 against 0.516	§5.2
Retrieval precision@5, four methods	0.395 / 0.196 / 0.117 / 0.204	§5.3
Reference value of the most represented sector	0.840 complete, 0.200 balanced	§5.3
Precision@5 under the symmetric protocol at scale	0.595 with chance 0.333	§5.3
Precision@5 with prior precedents only	0.513 with chance 0.259	§5.3
Direction agreement against the chance value	0.708 against 0.688	§5.3
Triage, PR-AUC of the six families	from 0.378 to 0.542	§5.5
Precision within the budget, four policies	0.163 / 0.379 / 0.662 / 0.632	§5.5
Grid of nine label definitions	9 of 9	§5.5
Identity ablation, lookup table against model	0.534 against 0.538	§5.5.5
Increment of the text over the lookup table	0.547 against 0.534	§5.5.6
Quality of the decomposition fit, median and negative cases	0.460, 1 in 17	§5.6
Gate selectivity, within and between companies	0.072 against 0.392	§5.7
Distribution drift, PSI of volatility	0.281	§5.7
Complete system against the alternatives	0.489 / 0.632 / 0.662 / 0.968	§5.7
Grounded generation check, attacks and controls	23 of 23 and 8 of 8	§4.8
Measurements over the operation: the value depends on the date the log was read		
Matured decisions, kept against suppressed	58.9% against 61.7%	§5.7
Ranking ability over the decisions taken	0.486	§5.7
News coverage on unusual days	88.5%	§5.7
Latency, discovery and delivery	353 min and 5 s	§4.6
Selectivity funnel, distinct headlines to delivered	743 → 15	§4.4
Funnel of one day, split by gate	5,060 assessments	§4.4

Table A.2: Each substantive claim, the evidence that supports it and the state in which it was left. The state *withdrawn* indicates that the work abandoned it after measuring it; the state *narrowed* indicates that it survives in a more restricted formulation than the initial one.

Claim	Evidence	State
Detection of unusual movements		
Fires at a comparable rate across companies of very distinct volatility	Spread of 0.015 against 0.344, without recourse to labels	Kept
Beats a fixed threshold on the approximate label	F_1 of 0.516 against 0.218	Kept, with the label declared as an approximation
Beats two learned detectors on the same information	F_1 of 0.530 against 0.269 and 0.280	Kept
Uses no information posterior to the day assessed	The window excludes that day, and an automatic test enforces the exclusion	Kept
The twenty-day window and the equally weighted deviation are the best options	Alternative variants obtain 0.678 and 0.664 against 0.516	**Withdrawn**: kept for explainability, not for performance
Precedent retrieval		
Beats the evaluated lexical and trivial alternatives	Precision@5 of 0.395 against 0.196, 0.117 and 0.204	Kept
Beats the best trivial reference value by a wide margin	On the balanced corpus, choosing the most represented sector obtains 0.200 and the method's margin over chance is +0.278; on the complete corpus it obtains 0.840 and overtakes the method	**Narrowed**: per sector against chance; does not assess a query-sector filter
Survives an increase in the scale of the corpus	Precision@5 of 0.595 with chance of 0.333	Kept as a scale test
Survives the prior-to-query restriction	Precision@5 of 0.513 with chance of 0.259: the margin varies by 0.008	**Narrowed**: does not test production's eight-day maturation
Captures topic and not direction	Agreement of 0.708 against chance of 0.688	Kept, and reported as a limitation
Membership of the same sector is equivalent to a relation between news items	—	**Not claimed**: it is a declared approximation
News triage		
No variant with text beats volatility	PR-AUC of 0.496 against 0.542, with nine label definitions and resampling by groups	Kept
The text adds information over the lookup table	PR-AUC of 0.547 against 0.534, with a paired increment of +0.012 and an interval between +0.004 and +0.020	**New**, positive, and with no effect on precision within the budget
The inputs incorporate no returns after the news day	Changing later sessions leaves the inputs identical and changes the label	Kept
Triage quadruples precision within the budget	The reference value of 0.163 results from alphabetical ordering; random choice obtains 0.379	**Withdrawn**: the gain is 1.67 times
The gain within the budget comes from learning	Thirteen constants per company obtain 0.662 against 0.632 for the model	**Withdrawn**
The deployed model selects better than chance	58.9% of material cases among the kept against 61.7% among the suppressed	**Withdrawn**
System		
The text of the alert cannot diverge from the computation	It is composed from the computed objects, without language generation	Kept
The free-of-charge constraint limits coverage	At least one relevant news item on 88.5% of the unusual days	Kept as an upper bound
The explanations are useful to a person	Observational feedback, no counterfactual	**Not claimed**

A.4 Data, licences and attribution

This section records the origin of the data used and the obligations their licensing imposes.

The historical set that underpins the precedent base is FNSPID (Dong, Fan, and Peng 2024), used under the Creative Commons BY-SA 4.0 licence, whose attribution is fulfilled here. Only minimal excerpts of headlines from the corpus appear in the document, for illustration.

That licence carries a second obligation, which is triggered in this case. Since files derived from FNSPID are distributed, the share-alike clause applies to those files. The licence of the code is a decision that remains open in this document, and this is the constraint that conditions it.

Prices come from free market data services, through the chain of alternatives described in Section 4.2; real-time news comes from three free interfaces, within their respective usage limits, compared in Section 4.2.2.

As to personal data, the exact formulation appears in Section 3.9.1: the system does not know the users' positions and does not ask for them, and the public channel does not identify recipients. The exception is the optional subscription to personalised alerts, in which the conversation identifier assigned by the platform and the list of companies chosen are associated, and those are the only elements stored.

A.5 Infrastructure: what enters the system, and at what cost

Figure A.1 brings together the external parts the system depends on, by the point of the path at which each one enters, and the text that follows presents the actual cost of operation. The sentence encoder is executed in the Open Neural Network Exchange (ONNX) format, which allows it to be run without the training library. No part was chosen out of preference: all of them follow from the founding constraint of using only free sources, and the two exceptions to that constraint, which are the hosting and the messaging channel, are costs of operation and not data sources.

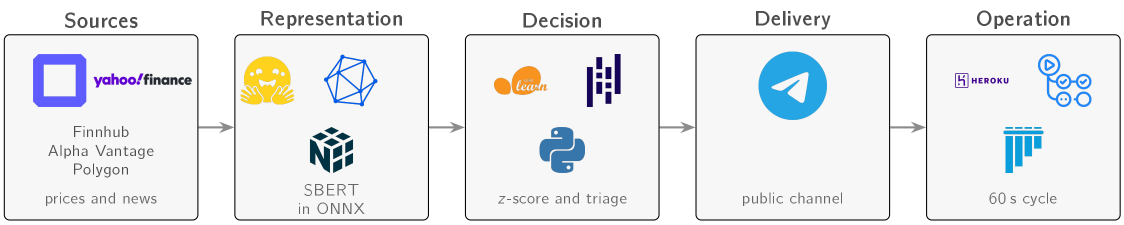

Figure A.1: The external parts the system depends on, in the order in which they enter. The marks belong to their respective holders and their use here is nominative, intended to identify the tool employed. The provenance of each file, with its corresponding cryptographic digest, is recorded in the manifest kept with the work.

Cost of operation, and the reason for the tier chosen

The system ran for most of the documented period in containers of the lowest tier, at a cost of seven dollars per month per process. On 4 September 2026 the collection process began persistently exceeding the memory quota of 512 MB, with over-quota warnings at a frequency of approximately two per minute and readings between 518 and 970 MB.

Four causes were measured and corrected, which together reduced the peak from 970 to 663 MB and none of which was sufficient: the working set of the application genuinely exceeds a 512 MB container, since the numerical libraries and the inference engine occupy 158 MB on their own and the precedent matrix becomes resident as soon as the first query walks its pages.

The technical alternative would be to remove the base of 38,214 precedents from the process, which would degrade the quality of retrieval, that is, would trade an infrastructure limitation for a limitation of result. The option taken was the reverse. The provider does not allow tiers to be mixed within the same application, so the web process, which never exceeded the quota, was dragged up with the collection process: the increase is sixty-one dollars a month and not the forty-three that changing a single process would cost.

After the change, and under the same load, not a single memory over-quota warning was recorded. The available credit balance covers approximately four months of operation at this tier, which exceeds the planned duration of the work. The constraint of using only free sources holds for the *data*, which are the subject matter of the dissertation; the hosting and the messaging channel are costs of operation and are declared as such.